Enterprise ERP

OFFICIAL DOCUMENTATION · VERSION 1.0

Enterprise ERP — User & Admin Guide

Setup, security, roles aur roz-marrah ka kaam — sab kuch step-by-step, aasaan Hinglish mein.

- Windows par installation & database setup
- Pehla admin, companies, users, roles & permissions
- HR, Payroll, Finance, Sales, Procurement, Inventory, Projects
- Employee self-service, mobile use, troubleshooting & FAQ

Version 1.0

Date 16 October 2026

Audience Business owners, administrators & staff

Table of contents

Kisi bhi heading par click karein — seedha us page par pahunch jayenge.

1 **Introduction** 5

1.1 ERP kya hai? 5
1.2 Modules ka overview 5
1.3 Multi-company concept 6
1.4 Screen ka layout 7
1.5 Yeh guide kis ke liye hai aur kaise parhein 8

2 **System requirements & Setup (Windows)** 10

2.1 System ke hisse (architecture) 10
2.2 System requirements 10
2.3 Runtimes install karna 11
2.4 Local MySQL database banana 12
2.5 Backend (Laravel API) configure karna 12
2.6 Frontend (Next.js) configure karna 15
2.7 Roz ka start: start-dev.ps1 16
2.8 Setup troubleshooting 17

3 **Database configuration** 18

3.1 Do raaste: Local ya Hosted 18
3.2 Hosted MySQL (Aiven) TLS ke sath 18
3.3 Tables dekhna: phpMyAdmin / DBeaver / System page 19
3.4 Backups aur restore 20

4 **Pehla Admin account banana** 22

4.1 Koi default ya "backdoor" account kyun nahi? 22
4.2 erp:create-admin — development mein 22
4.3 Production mein pehla admin 23
4.4 Two-factor authentication (2FA) on karna 24
4.5 Platform panel: nayi company aur us ka owner 25
4.6 Demo data (sirf testing ke liye) 29

5 **Users, Roles & Permissions** 31

5.1 Bunyadi concepts 31
5.2 Das (10) default roles 31
5.3 Permission matrix (module × role) 33
5.4 Naya user invite karna 34
5.5 Branch restriction 36
5.6 Role ki permissions badalna aur custom role banana 37
5.7 User ko edit, deactivate ya remove karna 38
5.8 Best practices 38

6 Security & access control 40

6.1 Asal hifazat server par: "button chhupana kaafi nahi" 40
6.2 403 aur 404 ka farq 41
6.3 Company data isolation 41
6.4 Login protections 41
6.5 Audit log ka istemal 42
6.6 Production security checklist 44

7 Daily use — module by module 45

7.1 Dashboard 45
7.2 Approvals inbox 46
7.3 Notifications 46
7.4 Company, Branches, Departments & Map 47
7.5 HR 51
7.6 Payroll 57
7.7 Finance / Accounting 61
7.8 Receivables & Payables 66
7.9 Sales & CRM 70
7.10 Procurement 74
7.11 Inventory 77
7.12 Projects & Tasks 80
7.13 Documents, Reports & Exports, Search 82
7.14 Settings & Themes 84

8 Employee self-service 87

8.1 Employee ko kya nazar aata hai 87
8.2 Attendance: Check in / Check out 88
8.3 Leave request karna 88
8.4 Payslips dekhna 89
8.5 Expense claim 90
8.6 Tasks aur time 91
8.7 Jo employee nahi kar sakta 92

9 Mobile use 94

9.1 Phone par ERP kholna 94
9.2 Mobile tips 96

10 Troubleshooting, FAQ & Glossary 98

10.1 Login aur account 98
10.2 Access aur permissions 98
10.3 Finance, payroll aur stock 99
10.4 IT / server ke masail 100
10.5 Glossary (istilahat) 101

A Appendix — Commands, demo accounts & permission catalogue 103
A.1 Command cheat-sheet 103
A.2 Demo accounts 104
A.3 Permission catalogue 105

CHAPTER 1

Introduction

ERP kya hai, is system mein kya kya hai, aur yeh guide kaise use karein.

1.1 ERP kya hai?

ERP (Enterprise Resource Planning) aik aisa software hai jis mein company ke saare departments — HR, payroll, accounts, sales, purchasing, stock, projects — aik hi jagah, aik hi database par kaam karte hain. Har department ka alag Excel sheet ya alag software rakhne ke bajaye, sab log aik hi system mein data enter karte hain, aur reports hamesha up-to-date rehti hain.

Misaal ke taur par: jab sales team aik **invoice** approve karti hai, to woh khud-ba-khud accounts ke **general ledger** mein post ho jati hai. Jab warehouse mein maal receive hota hai (goods receipt), to stock levels foran update ho jate hain. Payroll finalize hota hai to salary ka journal entry ban jata hai. Isi ko hum "one ledger of truth" kehte hain — aik hi sach, sab ke liye.

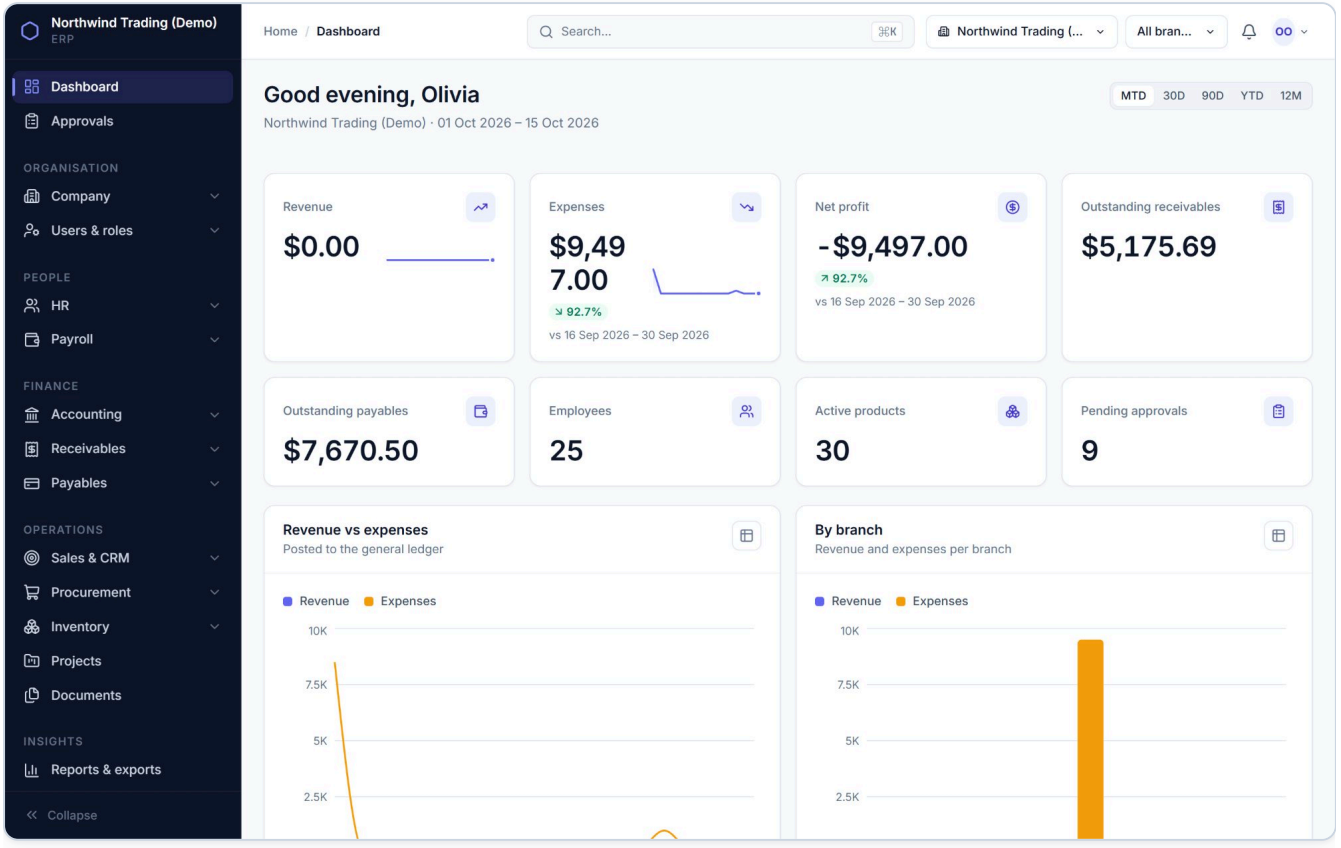

Figure 1.1 — Company Owner ka Dashboard — revenue, expenses, receivables, payables, employees aur pending approvals aik nazar mein.

1.2 Modules ka overview

Enterprise ERP mein yeh modules shamil hain. Har module sidebar (left side ka menu) mein apne section ke neeche nazar aata hai — lekin sirf tab jab aap ke role mein us ki permission ho.

Dashboard & Approvals

Executive KPIs, charts, aur aik inbox jahan har woh request aati hai jo aap ke faisle ka intezar kar rahi hai.

Company & Branches

Company profile, branding, branches, departments aur locations map.

Users & Roles

Users ko invite karna, roles assign karna, branch restriction, custom roles aur permissions.

HR

Employees, org chart, attendance, leave, shifts & holidays, performance reviews, HR reports.

Payroll

Pay components, salary structures, tax tables, monthly payroll runs aur payslips.

Finance / Accounting

Chart of accounts, double-entry journals, bank & cash, tax rates, financial periods, financial statements.

Receivables & Payables

Customer invoices, receipts, supplier bills, payments, expense claims aur aging.

Sales & CRM

Customers, leads, pipeline board, quotations, sales orders, deliveries, sales reports.

Procurement

Suppliers, purchase requisitions, purchase orders, goods receipts, procurement reports.

Inventory

Products, stock levels, movements, transfers, adjustments, warehouses, valuation reports.

Projects & Tasks

Projects portfolio, Kanban board, milestones, members, time logging aur project expenses.

Documents, Reports, Audit

Company files with versions/retention, reports & exports (CSV, Excel, PDF), aur append-only audit log.

In ke ilawa **Platform** section bhi hai (`Platform → Companies` , `Platform users` , `System`) jo sirf platform owner/admin ko nazar aata hai — is se poori installation ki companies manage hoti hain.

1.3 Multi-company concept

Yeh ERP **multi-company** hai: aik hi installation mein kai companies (tenants) chal sakti hain, aur har company ka data dusri se **bilkul alag (isolated)** rehta hai. Demo data mein do companies hain: **Northwind Trading (Demo)** (USD) aur **Indus Manufacturing (Demo)** (PKR).

- **Company** — sab se upar ka level. Har company ki apni currency, time zone, chart of accounts, roles aur users hote hain.
- **Branch** — company ke andar offices/sites (jaise Head Office, Austin Office). Users ko specific branches tak mehdood kiya ja sakta hai.
- **Department** — organisational units (Finance, HR, Sales...), jo HR, approvals aur reporting mein use hote hain.
- **Membership** — aik user aik se zyada companies ka member ho sakta hai, har company mein alag role ke sath.

Topbar mein company ka naam (company switcher) aur **All branches** (branch filter) nazar aata hai. Agar aap kai companies ke member hain to company switcher se dusri company select kar sakte hain; branch filter se dashboard aur lists sirf aik branch ka data dikhati hain.

> **NOTE**
> Platform staff (jaise `platform@erp.test`) companies **bana** aur **manage** kar sakte hain, lekin kisi company ka business data (employees, invoices waghera) tab tak **nahi dekh sakte** jab tak woh us company ke member na hon. Yeh jaan-boojh kar rakha gaya security design hai.

1.4 Screen ka layout

Login ke baad har page ka layout aik jaisa hai:

Table 1.1 — Har page par nazar aane wale hisse

Hissa	Kya hai
Sidebar (left)	Modules ka menu, sections mein: **Organisation**, **People**, **Finance**, **Operations**, **Insights**. Neeche **Collapse** button se sidebar chhota (sirf icons) ho jata hai.
Breadcrumbs	Topbar ke left mein — maslan `Home / Accounting / Journal entries` — taake pata rahe aap kahan hain.
Search…	Global search box — click karein to Search page khulta hai: employees, customers, suppliers, products, documents aur transactions.
Company switcher	Current company ka naam; kai companies hon to yahan se switch karein.
Branch filter	**All branches** ya koi aik branch.
Bell icon	Notifications — unread count ke sath.
Account menu	Aap ke initials (jaise `OO`). Is mein **My profile**, **Security**, **Notifications**, **Appearance**, theme (Light/Dark/System) aur sign out.

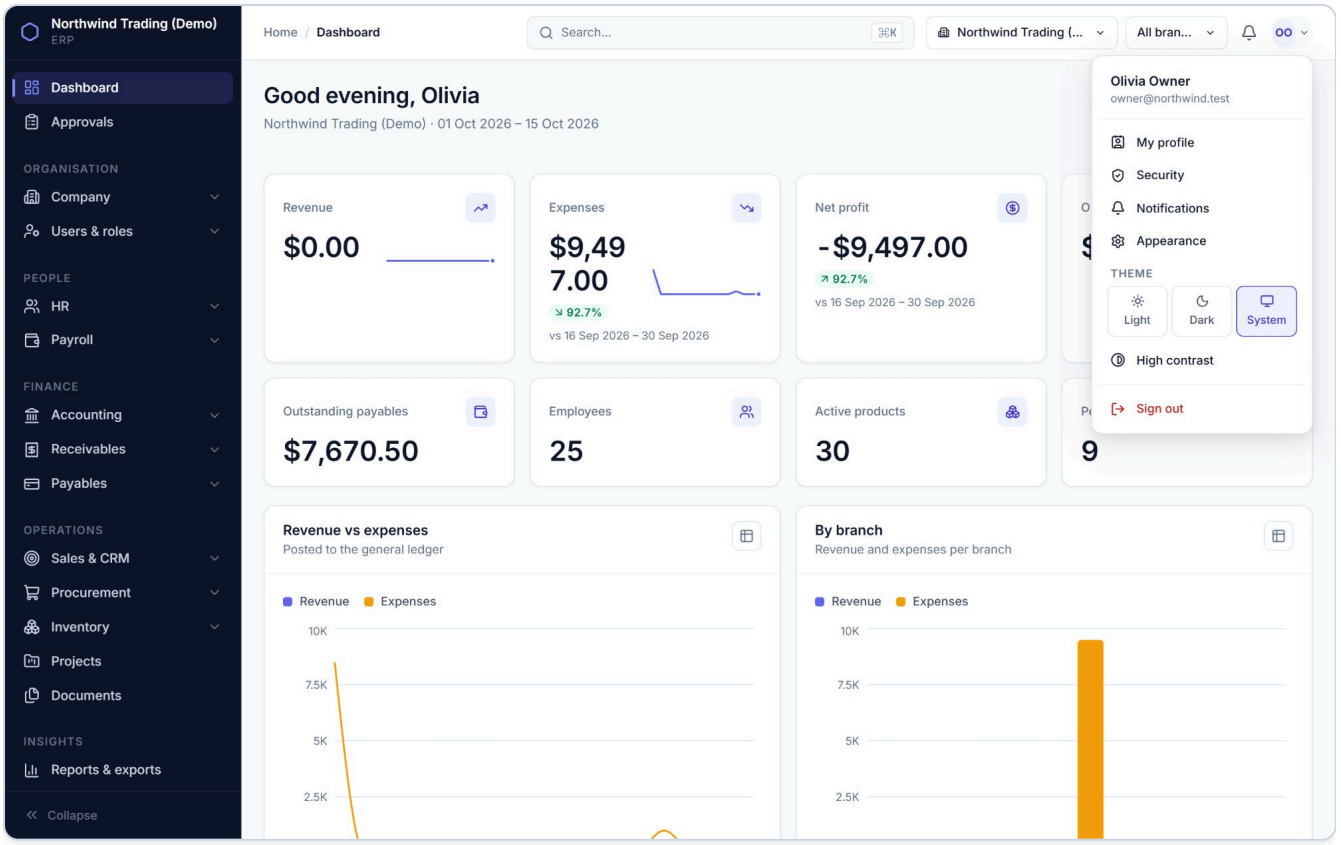

Figure 1.2 — Account menu — profile, security, notification preferences, appearance aur theme toggle.

1.5 Yeh guide kis ke liye hai aur kaise parhein

Table 1.2 — Reading guide

Aap kaun hain	Kaun se chapters parhein
IT person / jo system install karega	Chapter 2 (Setup), 3 (Database), 4 (Pehla admin), 6 (Security), Appendix A (commands).
Business owner / Company administrator	Chapter 4 (company aur owner), 5 (Users, roles & permissions), 6 (Security), 7 (Daily use), 10 (FAQ).
Department staff (HR, accounts, sales, store, purchase)	Chapter 7 mein apne module ka section, phir Chapter 10.
Aam employee	Chapter 8 (Employee self-service) aur Chapter 9 (Mobile).

Is guide mein in conventions ka khayal rakha gaya hai:

- Screen par jo label, button ya menu likha hai, woh **bilkul waisa hi English mein** highlight kiya gaya hai, jaise **New employee** ya **HR → Employees**.
- Commands, file names aur settings `is tarah` code style mein hain.
- Har kaam numbered steps mein likha hai — order mein follow karein.
- Screenshots demo company "Northwind Trading (Demo)" ke hain; aap ki company mein data alag hoga, lekin screens wohi hongi.

TIP

Rang-birangi boxes ka matlab: **Tip** = kaam aasaan karne ka mashwara; **Dhyan dein** = ghalti se bachne ki warning; **Security** = data aur access ki hifazat se mutaliq zaroori baat; **Note** = izafi maloomat.

DHYAN DEIN

Demo accounts aur demo companies (is guide mein screenshots wale) **sirf development/testing ke liye** hain. Production (asli business) server par demo data kabhi load na karein — Chapter 4 dekhein.

CHAPTER 2

System requirements & Setup (Windows)

Naye Windows computer par ERP ko zero se chalane ka mukammal tareeqa — step-by-step.

Yeh chapter technical hai aur us shakhs ke liye hai jo ERP install karega. Har step order mein karein. Aakhir mein aap ke paas teen cheezein chal rahi hongi: **MySQL database** (port 3306), **Laravel API** (`http://localhost:8000`) aur **Next.js frontend** (`http://localhost:3000`).

2.1 System ke hisse (architecture)

Table 2.1 — Project ke folders

Hissa	Technology	Folder	Address (development)
Frontend (jo browser mein nazar aata hai)	Next.js (React, TypeScript)	`frontend\`	`http://localhost:3000`
Backend API (business logic, security)	Laravel (PHP 8.3)	`backend\`	`http://localhost:8000/api/v1`
Database	MySQL 8.4	`database\local\`	`127.0.0.1:3306`
Documentation	—	`docs\`	—
Production templates	nginx, supervisor, cron	`deployment\`	—

Browser sirf frontend se baat karta hai, frontend sirf API se, aur sirf API database se. Database ka password kabhi browser tak nahi jata.

2.2 System requirements

Table 2.2 — Hardware aur software

Cheez	Kam az kam	Mashwara
Operating system	Windows 10 / 11 (64-bit)	Windows 11, latest updates
RAM	8 GB	16 GB (frontend dev server + MySQL + browser)
Disk	5 GB free	SSD, 20 GB free
Browser	Microsoft Edge / Chrome / Firefox (latest)	Edge ya Chrome
Internet	Installation ke waqt zaroori	Map tiles (OpenStreetMap) ke liye bhi

Table 2.3 — Runtimes — XAMPP ki zaroorat nahi

Software	Version	Install command (PowerShell)	Check
PHP (x64)	8.3+	`winget install PHP.PHP.8.3`	`php -v`
Composer	2.x	`winget install Composer.Composer`	`composer -V`
MySQL Server	8.4 LTS	`winget install Oracle.MySQL`	`mysqld --version`
Node.js	20+	`winget install OpenJS.NodeJS.LTS`	`node -v`

> **NOTE**
>
> **XAMPP pehle se installed hai?** Us ka bundled PHP 8.0 aur MariaDB 10.4 is ERP ke liye purane hain. Aap XAMPP ka sirf **phpMyAdmin** database dekhne ke liye use kar sakte hain (Chapter 3), aur XAMPP ki MariaDB ko port **3307** par shift kar dein (`C:\xampp\mysql\bin\my.ini`) taake dono servers aapas mein na takrayein.

2.3 Runtimes install karna

1. **PowerShell** kholein (Start menu → "PowerShell"). Normal user kaafi hai.
2. Har software ke liye upar wali table ka `winget install …` command chalayein. Har install ke baad PowerShell window **band karke dobara kholein** taake PATH update ho jaye.
3. Check commands chala kar versions confirm karein: `php -v` , `composer -V` , `node -v` . MySQL ke binaries `C:\Program Files\MySQL\MySQL Server 8.4\bin` mein hote hain.
4. **PHP extensions on karein.** `php.exe` wale folder mein `php.ini-development` ko copy karke `php.ini` naam rakhein, phir neeche wali lines set karein (shuru ka `;` hata dein).
5. `php -m` chala kar confirm karein ke saari extensions list mein hain — khaas taur par `pdo_mysql` .

php.ini

```
extension_dir = "C:/path/to/php/ext"     ; absolute path, forward slashes
extension=bcmath
extension=curl
extension=fileinfo
extension=gd
extension=intl
extension=mbstring
extension=openssl
extension=pdo_mysql
extension=sodium
extension=zip
memory_limit = 512M
upload_max_filesize = 20M
post_max_size = 25M
date.timezone = UTC
```

> ⚠ **DHYAN DEIN**
>
> Agar baad mein `db:check` "could not find driver" bataye to iska matlab hai `pdo_mysql` extension on nahi hui. `php --ini` se dekhein ke PHP kaunsi `php.ini` file parh raha hai.

2.4 Local MySQL database banana

Project apne andar aik chhota, alag MySQL data folder use karta hai (`database\local\data`) — koi Windows service nahi, admin rights nahi. Yeh sirf **development** ke liye hai. Project ke root folder (`C:\Users\…\ERP`) mein PowerShell kholein:

1. **Pehli dafa initialise karein:** `.\database\local\init.ps1` — data directory banti hai, aur `root@localhost` account (khaali password, sirf localhost) ban jata hai.
2. **Server start karein:** `.\database\local\start.ps1 -Background` — MySQL background mein `127.0.0.1:3306` par chalne lagta hai. (`-Background` ke baghair chalayein to window khuli rakhni hogi; `Ctrl+C` se band.)
3. **App database aur user banayein:** `.\database\local\create-app-db.ps1` — databases `erp` aur `erp_test` aur least-privilege user `erp_app@localhost` (random password ke sath) banta hai, aur `.env` mein paste karne wali lines print hoti hain.
4. Print hone wali `DB_*` lines kahin copy kar lein — agle step mein chahiye hongi.
5. Server band karna ho to: `.\database\local\stop.ps1` .

Expected output

```
PS> .\database\local\create-app-db.ps1

Database 'erp' and 'erp_test' ready. Paste into backend/.env:

DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=erp
DB_USERNAME=erp_app
DB_PASSWORD=<24 characters, randomly generated>
DB_SSL_CA=
```

> 💡 **TIP**
>
> `create-app-db.ps1` dobara chalana safe hai (idempotent) — lekin har dafa user ka **naya password** set karta hai, is liye phir `.env` bhi update karein.

2.5 Backend (Laravel API) configure karna

1. `cd backend`
2. Dependencies install karein: `composer install`

3. Environment file banayein: `copy .env.example .env`
4. `.env` ko Notepad/VS Code mein kholein aur `DB_*` lines (pichle step wali) paste karein.
5. Application key generate karein: `php artisan key:generate` — yeh `APP_KEY` set karta hai (encryption ke liye; kabhi share na karein).
6. Connection check karein: `php artisan db:check`
7. Database schema aur permission catalogue banayein: `php artisan migrate --seed`
8. Database self-test chalayein: `php artisan db:selftest`
9. *(Sirf testing ke liye, optional)* demo companies: `php artisan db:seed --class=DemoSeeder` — Chapter 4.5 dekhein.
10. API start karein: `php artisan serve` → `http://localhost:8000`

Important .env variables

Table 2.4 — backend\.env ke aham variables

Variable	Matlab	Development	Production
`APP_ENV`	Mahol (environment)	`local`	`production`
`APP_DEBUG`	Errors mein technical detail dikhana	`true`	`false` (lazmi)
`APP_KEY`	Encryption key	`key:generate`	Unique, secret
`APP_URL`	API ka address	`http://localhost:8000`	`https://api.example.com`
`FRONTEND_URL`	Frontend ka address (reset links, CORS)	`http://localhost:3000`	`https://erp.example.com`
`CORS_ALLOWED_ORIGINS`	Kaun se origins API call kar sakte hain (comma list, wildcard nahi)	`http://localhost:3000`	`https://erp.example.com`
`SANCTUM_STATEFUL_DOMAINS`	Kin hosts ko cookie login milta hai	`localhost:3000,127.0.0.1:3000`	`erp.example.com`
`SESSION_DOMAIN`	Cookie ka domain (frontend + API dono ka)	`localhost`	`.example.com`
`SESSION_SECURE_COOKIE`	Cookie sirf HTTPS par	`false`	`true`
`SESSION_LIFETIME`	Kitne minute idle rehne par session khatam	`120`	`60-120`
`AUTH_PASSWORD_TIMEOUT`	Password re-confirm kitni der valid (seconds)	`900`	`900`
`DB_HOST` … `DB_PASSWORD`	Database connection	create-app-db.ps1 se	Hosted DB ke credentials
`DB_SSL_CA`	Hosted DB ka TLS CA certificate path	(khaali)	`storage/certs/…pem`
`MAIL_MAILER`	Emails (password reset, invitations)	`log`	Real SMTP / API
`QUEUE_CONNECTION`	Background jobs (exports, emails)	`database`	`database` / `redis` + worker
`ERP_ALLOW_ADMIN_BOOTSTRAP`	Production mein pehla admin banane ki ijazat	`false`	`true` sirf aik dafa

> **NOTE**
> `MAIL_MAILER=log` hone par password-reset aur invitation emails bheji nahi jatin — balke `backend\storage\logs\laravel.log` mein likhi jati hain. Development mein link wahin se copy karein.

db:check aur db:selftest ka expected output

```
php artisan db:check
```

```
PS> php artisan db:check

  Connection ..................................... mysql
  Host ........................................ 127.0.0.1:3306
  Database ....................................... erp
  User ........................................ erp_app
  TLS CA ..................................... not configured
  Server version ................................. 8.4.x
  TLS cipher ............................. none (plaintext)
  Latency ....................................... 5.6 ms
  Tables ........................................ 104

  INFO  Database connection OK.
```

```
php artisan db:selftest
```

```
PS> php artisan db:selftest

  1. Schema: tables created with foreign key ............... PASS
  2. Create: inserted row #1 ............................... PASS
  3. Read: row read back with exact decimal (1234.5678) ..... PASS
  4. Update: value changed ................................. PASS
  5. Foreign key: orphan insert rejected ................... PASS
  6. Transaction: rollback restored previous value ......... PASS
  7. Delete: cascade removed child rows .................... PASS
  8. Strict SQL mode active (...) .......................... PASS

  INFO  All database self-tests passed.
```

`db:selftest` sirf temporary tables use karta hai — asli data ko haath nahi lagata. Koi bhi line **FAIL** ho to aage na barhein; troubleshooting table (2.8) dekhein.

2.6 Frontend (Next.js) configure karna

1. Nayi PowerShell window kholein aur `cd frontend`
2. Packages install karein: `npm install`
3. Environment file: `copy .env.example .env.local`
4. `.env.local` mein `NEXT_PUBLIC_API_URL=http://localhost:8000` confirm karein.
5. Dev server start karein: `npm run dev` → browser mein `http://localhost:3000` kholein. Login page nazar aana chahiye.

Table 2.5 — frontend\.env.local

Variable	Matlab
`NEXT_PUBLIC_API_URL`	API ka origin, maslan `http://localhost:8000`
`NEXT_PUBLIC_MAP_TILE_URL`	Optional — apna map tile server (default OpenStreetMap, sirf halka use, attribution zaroori)
`NEXT_PUBLIC_MAP_ATTRIBUTION`	Custom tile server ki attribution HTML

SECURITY

`NEXT_PUBLIC_*` wali har value **browser tak pohanchti hai**. In mein kabhi password, API key ya koi secret na rakhein.

2.7 Roz ka start: start-dev.ps1

Pehli dafa setup ke baad, har roz sirf aik command kaafi hai. Project root mein:

```
Project root

PS> .\start-dev.ps1
Starting…
  API       http://localhost:8000/api/v1/health
  Frontend  http://localhost:3000
  DB GUI    http://localhost/phpmyadmin (XAMPP Apache) or DBeaver → 127.0.0.1:3306
```

- MySQL background mein start hota hai (agar pehle se chal raha ho to skip).
- Do nayi PowerShell windows khulti hain: aik mein API (port 8000), dusri mein frontend (port 3000).
- Band karna: dono windows band karein, phir `.\database\local\stop.ps1`.

Background workers (exports, emails, scheduled jobs)

```
backend

php artisan queue:work --tries=3     # exports, emails, notifications
php artisan schedule:work            # dev; production mein cron:  * * * * * php artisan
schedule:run
```

Scheduler roz `inventory:check-reorder` (low-stock alerts) aur `erp:prune` (purana operational data retention ke mutabiq saaf) chalata hai. Bari reports ki export queue ke zariye banti hai — worker na chal raha ho to export "queued" hi reh jayega.

2.8 Setup troubleshooting

Table 2.6 — Aam masail aur un ka hal

Error / symptom	Wajah	Hal
`SQLSTATE[HY000] [2002]` connection refused	MySQL nahi chal raha, ya ghalat host/port	`.\database\local\start.ps1 -Background` chalayein; `DB_HOST` / `DB_PORT` check karein.
`[1045] Access denied`	Ghalat DB_USERNAME / DB_PASSWORD	`create-app-db.ps1` dobara chala kar nayi lines `.env` mein paste karein, phir `php artisan config:clear`.
`[1044] Access denied … to database`	User ko is DB par privileges nahi	GRANT dobara chalayein (create-app-db.ps1).
`[1049] Unknown database`	Database bana hi nahi	`create-app-db.ps1` ya `CREATE DATABASE erp`.
`[2026] SSL connection error`	Hosted provider TLS maangta hai	`DB_SSL_CA` mein downloaded CA file ka path dein (Chapter 3).
`could not find driver`	`pdo_mysql` extension off	`php.ini` mein `extension=pdo_mysql` on karein; `php -m` se confirm.
Login page khulta hai lekin sign in par error / 419	CORS / cookie domain mismatch	`FRONTEND_URL`, `SANCTUM_STATEFUL_DOMAINS`, `SESSION_DOMAIN` check karein; browser mein `localhost` hi use karein (127.0.0.1 aur localhost mix na karein).
"Too many attempts" login par	Rate limit (5 tries/minute)	1 minute intezar karein.
Port 3000 / 8000 already in use	Pichla server abhi chal raha hai	Purani window band karein ya process end karein.

> **TIP**
> Kisi bhi DB masle mein sab se pehle `php artisan db:check` chalayein — yeh error ke sath **Hint** line bhi print karta hai, aur password kabhi print nahi karta.

CHAPTER 3

Database configuration

Local MySQL ya hosted (Aiven) — connection, TLS, data dekhna, backup aur restore.

3.1 Do raaste: Local ya Hosted

Table 3.1 — Database options

	Option A — Hosted (Aiven)	Option B — Local MySQL 8.4
Kuch install karna?	Nahi	MySQL Community Server (≈ 300 MB)
Offline chalta hai?	Nahi	Haan
Kharcha	Free tier (1 GB) / paid	Free
TLS (encryption)	Lazmi	Optional (localhost)
Kis ke liye behtar	Demo kahin se bhi, team ka shared DB	Roz ki development, tests

Dono soorton mein configuration sirf `backend\.env` mein hoti hai. Local setup Chapter 2.4 mein ho chuka hai; neeche hosted option aur baqi operational kaam hain.

3.2 Hosted MySQL (Aiven) TLS ke sath

> **NOTE**
> Aiven ka free MySQL plan (1 node, 1 GB storage) development/demo ke liye theek hai, production ke liye nahi — free services lambi inactivity par band ho jati hain aur free tier par automatic backups nahi hote. Shuru karne se pehle Aiven ki current terms check kar lein.

1. console.aiven.io par sign up karein → **Create service** → **MySQL** → **Free plan**, qareeb tareen region chunein. Status **Running** hone ka intezar karein (2–4 minute).
2. App ko `avnadmin` superuser se **na chalayein**. Alag database aur user banayein (Users tab ya MySQL shell se) — neeche SQL dekhein.
3. Overview tab → **CA certificate** → **Download**. File ko `backend\storage\certs\aiven-ca.pem` par save karein (yeh folder git mein nahi jata).
4. `backend\.env` mein host, port, user, password aur `DB_SSL_CA` set karein.
5. **Network access band karein:** Overview → **Allowed IP addresses** mein `0.0.0.0/0` hata kar sirf apna public IP likhein. IP badle to yahan update karein.
6. Verify: `php artisan db:check` (TLS cipher nazar aana chahiye), `php artisan migrate --force`, `php artisan db:selftest`.

Aiven — dedicated database & user

```
CREATE DATABASE erp CHARACTER SET utf8mb4 COLLATE utf8mb4_0900_ai_ci;
CREATE USER 'erp_app'@'%' IDENTIFIED BY '<strong-random-password>';
GRANT SELECT, INSERT, UPDATE, DELETE, CREATE, ALTER, INDEX, DROP, REFERENCES,
      CREATE TEMPORARY TABLES, LOCK TABLES, EXECUTE, CREATE VIEW, SHOW VIEW,
      TRIGGER ON erp.* TO 'erp_app'@'%';
FLUSH PRIVILEGES;
```

backend\.env (hosted)

```
DB_CONNECTION=mysql
DB_HOST=mysql-xxxx-yyyy.aivencloud.com
DB_PORT=12345
DB_DATABASE=erp
DB_USERNAME=erp_app
DB_PASSWORD=<strong-random-password>
DB_SSL_CA=storage/certs/aiven-ca.pem
DB_SSL_VERIFY=true
```

Health check: `GET http://localhost:8000/api/v1/health/db` jawab mein sirf status, driver, version, `tls` aur latency deta hai — host, user ya password **kabhi nahi**.

> **SECURITY**
> Production mein runtime app user se `UPDATE, DELETE` on `audit_logs` revoke karein (Chapter 6) aur reporting ke liye alag SELECT-only user banayein.

3.3 Tables dekhna: phpMyAdmin / DBeaver / System page

Kabhi kabhi data ko direct dekhna zaroori hota hai (maslan support ke liye). Teen tareeqe:

A) DBeaver (recommended)

1. Install: `winget install dbeaver.dbeaver`
2. **New connection** → **MySQL**. Host `127.0.0.1`, port `3306`, database `erp`, user `erp_app` aur password (`.env` wala).
3. Hosted DB ke liye **SSL** tab mein **Use SSL** tick karein aur CA certificate = `aiven-ca.pem`.

B) XAMPP ka phpMyAdmin

1. XAMPP ki MariaDB ko port `3307` par shift karein (`C:\xampp\mysql\bin\my.ini`) taake ERP ka MySQL `3306` par rahe.
2. `C:\xampp\phpMyAdmin\config.inc.php` mein host `127.0.0.1`, port `3306` aur `auth_type = cookie` set karein.
3. XAMPP Control Panel se sirf **Apache** start karein aur `http://localhost/phpmyadmin` kholein; `erp_app` se login karein.

C) In-app System page (platform admins)

Platform → **System** page database tables ki approximate row counts aur size dikhata hai — read-only, koi SQL console nahi, koi credentials nahi.

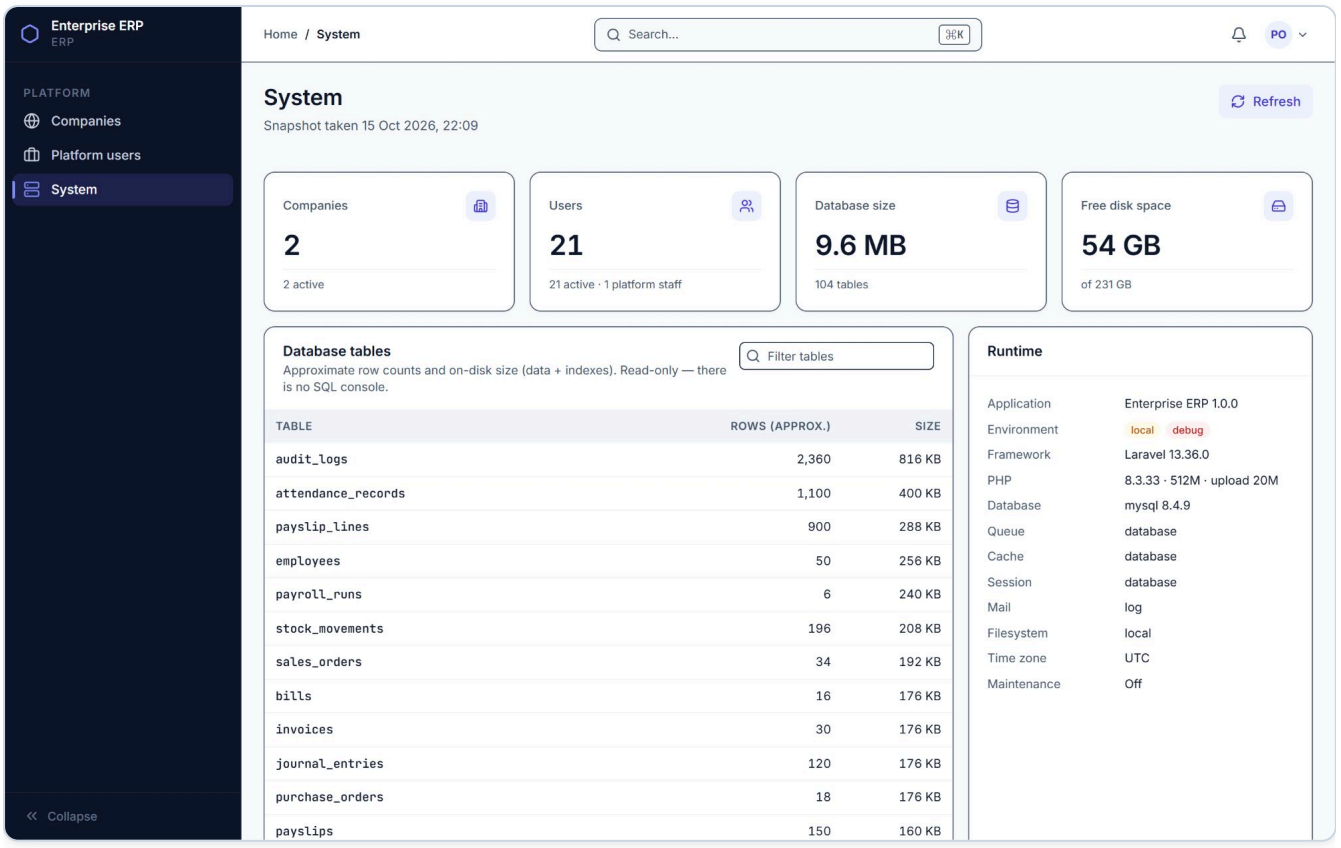

Figure 3.1 — Platform → System: companies, users, database size, free disk aur har table ki row count (read-only).

> ⚠ **DHYAN DEIN**
>
> **Read-only rahein.** phpMyAdmin/DBeaver se tables mein direct edit/delete **na karein**. ERP ke andar har posting ke sath journal entries, stock movements aur audit log jure hote hain — direct SQL se data ki consistency toot jati hai aur audit trail mein record nahi banta. Ghalti theek karni ho to app ke andar wala tareeqa (reverse, void, adjustment) use karein. phpMyAdmin ko kabhi production server par public na karein.

3.4 Backups aur restore

Backup ka matlab hai database ki mukammal copy (SQL file) jo kisi masle mein wapas laayi ja sake. Sath mein `backend\storage\app\private` folder (uploaded documents, receipts, contracts) ka backup bhi zaroori hai — yeh files database mein nahi hotin.

Backup lena (Windows, local MySQL)

PowerShell (password prompt aayega)

```
$dump = "C:\Program Files\MySQL\MySQL Server 8.4\bin\mysqldump.exe"
& $dump --single-transaction --routines --triggers -h 127.0.0.1 -P 3306 -u erp_app -p --result-file="D:\Backups\erp-$(Get-Date -Format yyyy-MM-dd).sql" erp
```

Backup lena (production Linux server, nightly)

Linux cron job

```
mysqldump --single-transaction --routines --triggers --ssl-ca=/srv/erp/backend/storage/certs/ca.pem \
  -h "$DB_HOST" -P "$DB_PORT" -u "$DB_BACKUP_USER" -p"$DB_BACKUP_PASSWORD" erp \
  | gzip > /backups/erp-$(date +%F).sql.gz
```

- Kam az kam **14 din** ke backups rakhein, aik copy **off-site** (dusri jagah) rakhein, aur encrypt karein.
- Local par Windows **Task Scheduler** se roz chala sakte hain.
- Hosted Aiven free tier par automatic backup nahi hota — khud lein.

Restore karna

1. Restore hamesha pehle aik **scratch (test) database** mein karke check karein.
2. Restore: `mysql -h 127.0.0.1 -u erp_app -p erp_restore < erp-2026-10-15.sql` (hosted ke liye `--ssl-ca=…` bhi dein).
3. Staging `.env` ko is database par point karein, phir `php artisan migrate --force` (agar backup ke baad nayi migrations aayi hon) aur `php artisan db:check`.
4. Sab theek ho to hi production par restore ka faisla karein.

> **TIP**
> Har teen mahine (quarterly) aik **restore drill** karein — jo backup kabhi restore karke na dekha ho, us par bharosa na karein.

> **⚠ DHYAN DEIN — KABHI NAHI**
> **Production database ko kabhi reset na karein.** `php artisan migrate:fresh`, `migrate:reset`, `db:wipe` ya rollback production par **sab data mita dete hain**. Migration mein masla ho to aik nayi corrective migration likhwayein. Demo seeder bhi production par refuse karta hai — aur aisa hi hona chahiye.

CHAPTER 4

Pehla Admin account banana

Platform owner, two-factor authentication, pehli company aur us ka owner.

4.1 Koi default ya "backdoor" account kyun nahi?

Bohat se software `admin / admin123` jaisa default account ke sath aate hain — aur hackers sab se pehle yahi try karte hain. Is ERP mein **koi default username/password nahi**, koi chhupa hua master password nahi, aur koi "backdoor" nahi. Pehla administrator aap khud, server par aik command chala kar, apne email aur mazboot password ke sath banate hain.

Table 4.1 — Admin accounts ki qismein

Account type	Kaun	Kya kar sakta hai
Platform owner	Poori installation ka malik (aam taur par IT head / software owner)	Companies banana/band karna, platform users, system health. Kisi company ka business data **nahi** dekh sakta jab tak member na ho.
Platform admin	Platform owner ka madadgar	Platform panel ka kaam; platform roles sirf owner de sakta hai.
Company Owner	Har company ka business owner	Us company ke andar sab kuch (Chapter 5).

4.2 erp:create-admin — development mein

Backend folder mein command chalayein aur sawalon ke jawab dein:

Interactive steps

```
PS> cd backend
PS> php artisan erp:create-admin

 Full name:
 > Adnan Ahmed

 Email:
 > adnan@yourcompany.com

 Password (min 10 chars, letters + numbers):
 >   (type karte waqt nazar nahi aayega)

 INFO  Platform owner created: adnan@yourcompany.com
 WARN  Enable two-factor authentication for this account after first login.
```

1 `Full name` — poora naam.

2. `Email` — asli, aap ka apna email (password reset isi par aayega). Pehle se mojood email accept nahi hota.
3. `Password` — kam az kam **10 characters**, letters **aur** numbers dono. Password ko known data-breach lists se bhi check kiya jata hai; leaked password reject ho jayega.
4. Kamyabi par "Platform owner created" message aata hai. Yeh event audit log mein bhi record hota hai.

> **TIP**
> Development mein `--name`, `--email`, `--password` options bhi diye ja sakte hain. Production mein `--password` **refuse** hota hai, kyun ke command line ka text shell history mein save ho jata hai.

4.3 Production mein pehla admin

Production mein yeh command ghalti se na chal jaye, is liye do tala (locks) hain: environment flag aur `--force`. Aur agar platform owner pehle se mojood ho to command chalti hi nahi.

1. `backend/.env` mein `ERP_ALLOW_ADMIN_BOOTSTRAP=true` set karein. (Agar `config:cache` use karte hain to `php artisan config:cache` dobara chalayein.)
2. Command chalayein: `php artisan erp:create-admin --force` — naam, email aur password **prompt** par dein.
3. Foran `ERP_ALLOW_ADMIN_BOOTSTRAP=false` wapas karein (aur config cache dobara banayein).
4. Login karke turant two-factor authentication on karein (4.4).
5. Mazeed platform admins ab command se nahi, balke Platform panel se manage hote hain.

```
Production
```

```
# Linux server (one-off, env var inline):
ERP_ALLOW_ADMIN_BOOTSTRAP=true php artisan erp:create-admin --force

# Agar flag/force na ho:
 ERROR  In production set ERP_ALLOW_ADMIN_BOOTSTRAP=true and pass --force.
# Agar owner pehle se ho:
 ERROR  A platform owner already exists. Manage further admins from the platform panel.
```

> **SECURITY**
> `ERP_ALLOW_ADMIN_BOOTSTRAP` ko **true** chhor dena production security checklist ki failure hai. Kaam hote hi false karein.

4.4 Two-factor authentication (2FA) on karna

2FA ka matlab: login ke waqt password ke ilawa phone ki authenticator app (Google Authenticator, Microsoft Authenticator, 1Password...) ka 6-digit code bhi chahiye. Password chori ho jaye tab bhi account mehfooz rehta hai. **Har admin aur owner ke liye lazmi samjhein.**

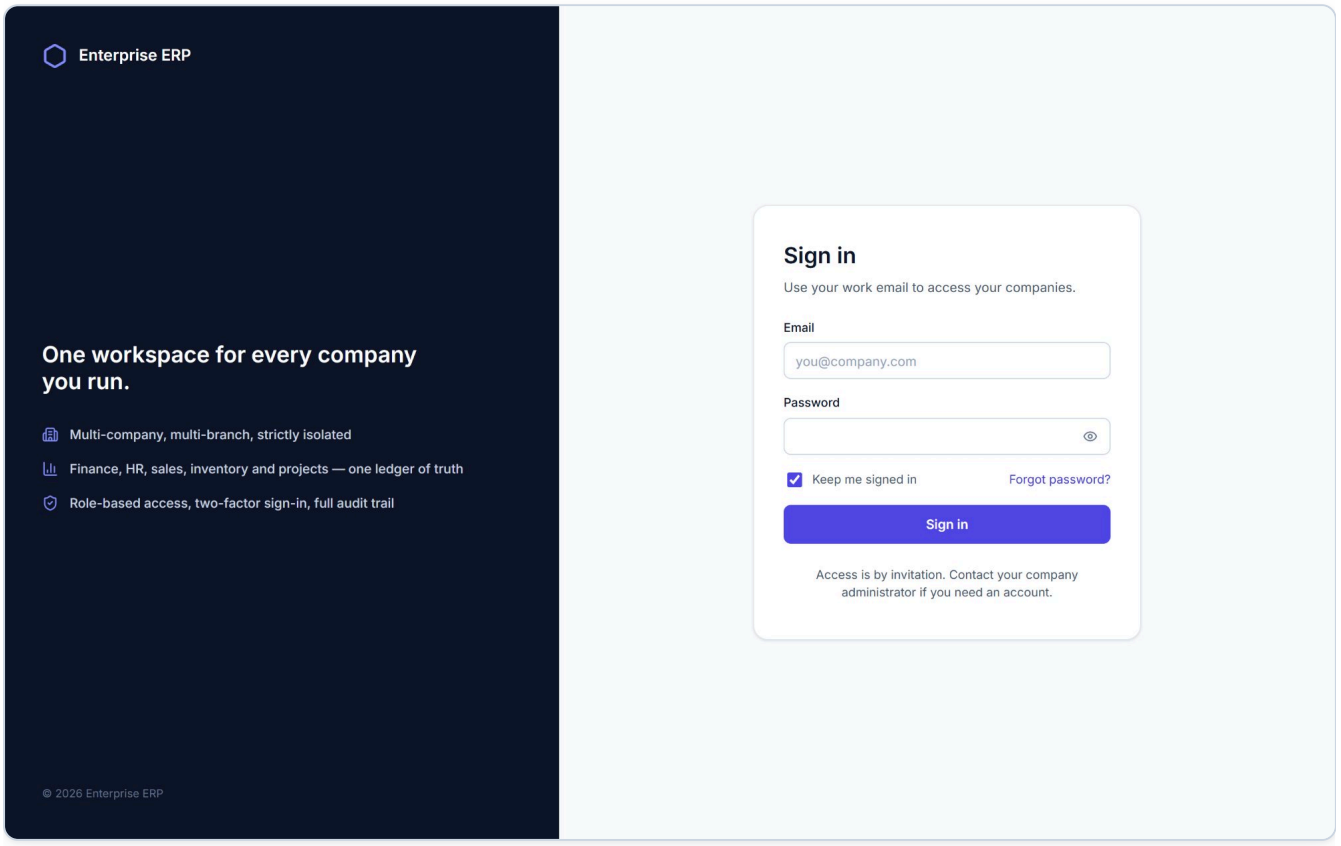

Figure 4.1 — Login page — email aur password. Access sirf invitation se milta hai; khud sign-up ka option nahi.

1. Apne email/password se login karein. Upar right mein apne initials (Account menu) par click karein → **Security**. (Ya seedha `/settings/security`.)
2. **Two-factor authentication** card mein status **Off** hoga. **Set up two-factor** par click karein.
3. Security ke liye system dobara aap ka password maang sakta hai (password confirmation, 15 minute tak valid).
4. Screen par QR code aata hai: "1. Scan the QR code with your authenticator app, or enter this key manually". Phone ki app se scan karein.
5. "2. Enter the 6-digit code it shows" — app ka code **Verification code** mein likh kar **Verify & enable** dabayein.
6. **Save your recovery codes** — recovery codes dikhaye jate hain. **Copy codes** se copy karke kisi mehfooz jagah (password manager / taala band file) rakhein. Har code sirf aik dafa chalta hai; phone gum ho jaye to inhi se login hoga.

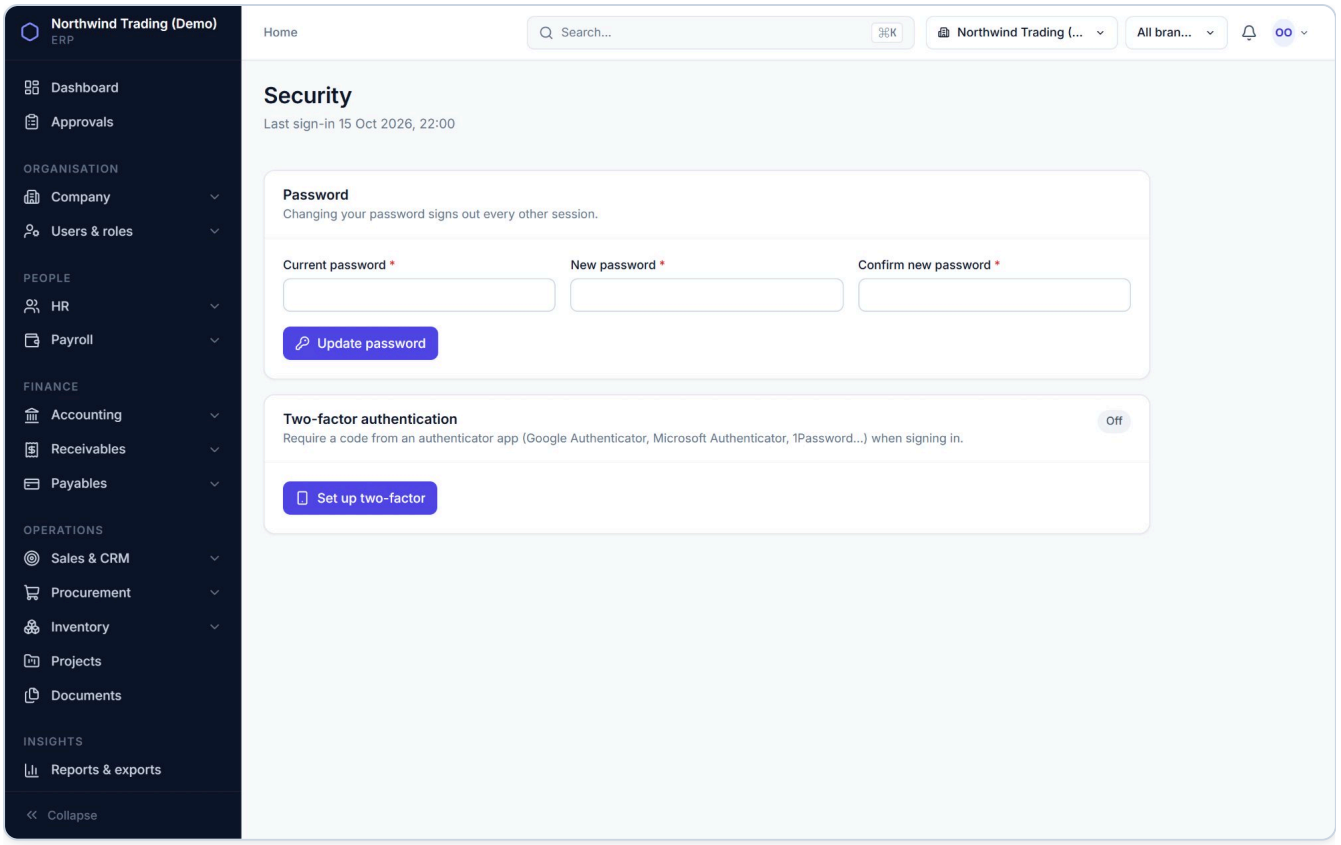

Figure 4.2 — Settings → Security: password badalna aur two-factor authentication. Status "Off" hai — Set up two-factor dabayein.

> **NOTE**
>
> Baad mein **Regenerate recovery codes** se naye codes bana sakte hain, aur 2FA band karne ke liye **Password to disable** mein password de kar **Disable** .

4.5 Platform panel: nayi company aur us ka owner

Platform owner login karta hai to sidebar mein sirf **Platform** section hota hai: **Companies** , **Platform users** , **System** .

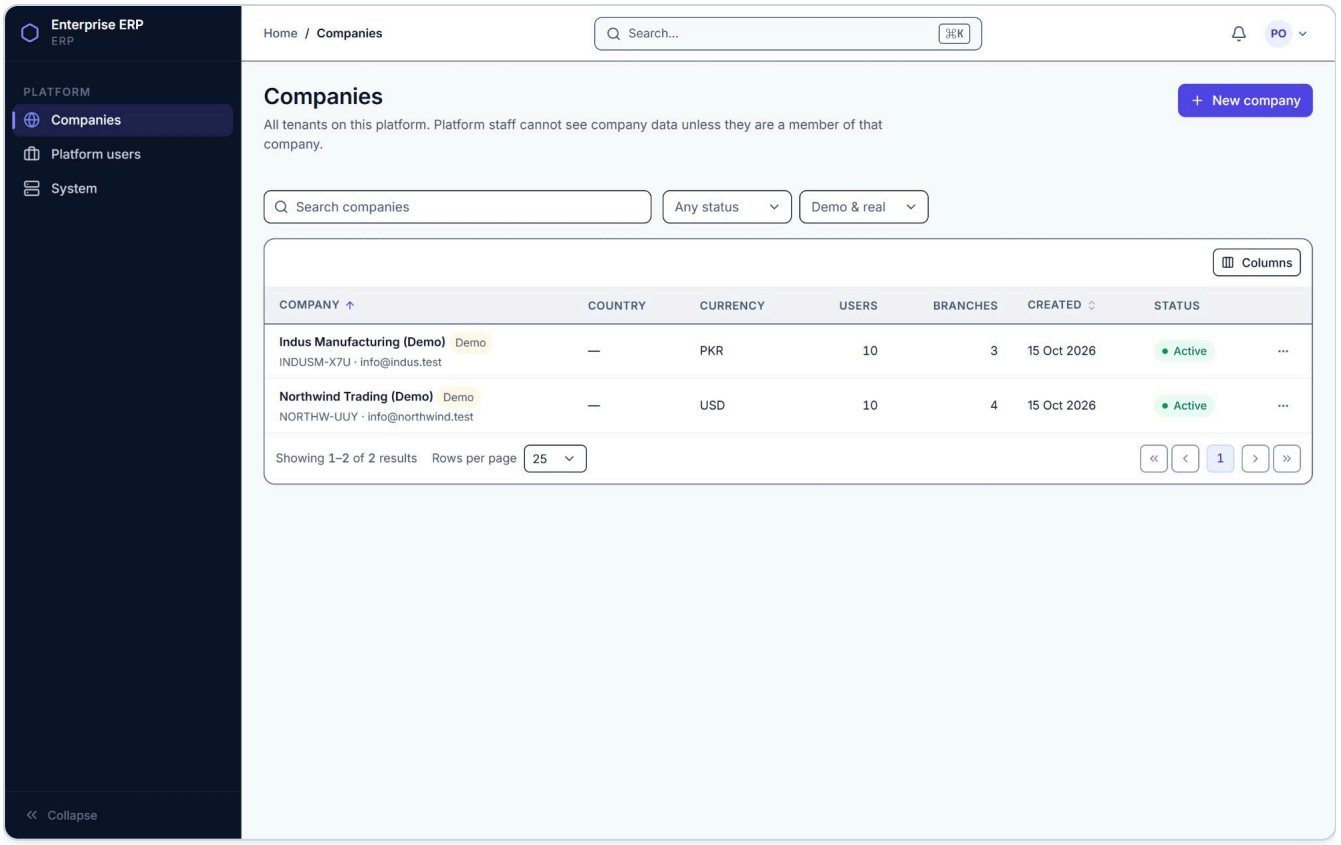

Figure 4.3 — Platform → Companies: installation ki saari companies, demo badge, users aur branches ki tadaad.

1. Sidebar mein **Platform → Companies** kholein aur upar right **New company** dabayein.
2. Drawer mein company ki details bharein: **Name**, **Legal name**, **Email**, **Country** (ISO code, jaise PK, US), **Base currency**, **Time zone**.
3. **Owner name** aur **Owner email** — yeh shakhs company ka **Company Owner** banega.
4. **Demo company** switch sirf testing ke liye on karein ("Clearly labelled as demo data throughout the app"). Asli company ke liye off rakhein.
5. Save karein. System khud-ba-khud company, aik **head office** branch, standard **10 roles**, **chart of accounts** aur defaults bana deta hai, aur owner ko invite karta hai.

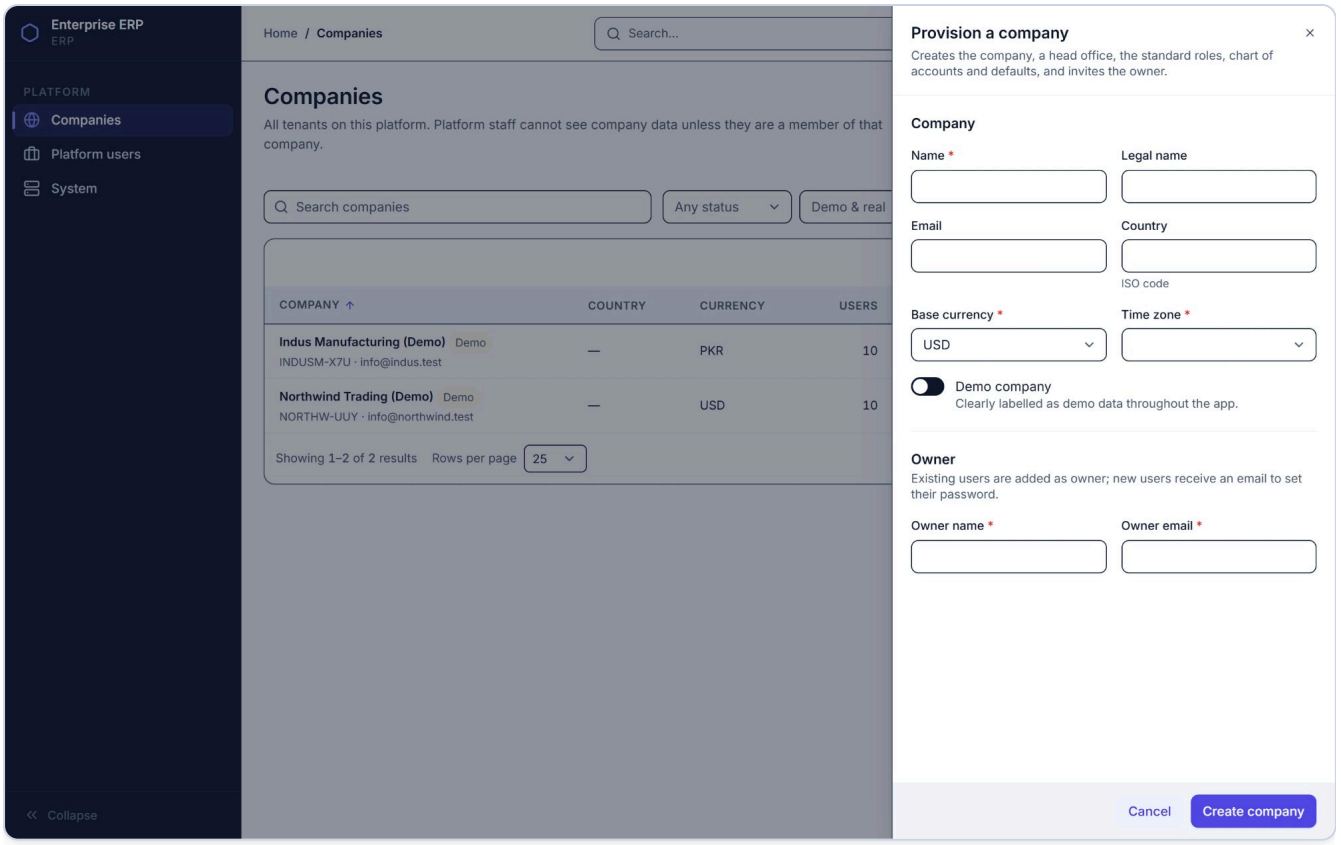

Figure 4.4 — New company drawer — company details aur owner ka naam/email. "Creates the company, a head office, the standard roles, chart of accounts and defaults, and invites the owner."

Owner ko kya milta hai?

- **Naya user** ho to: email "You've been added to <Company>" — "An account was created for you… Check your inbox for a link to set your password, then sign in." Sath hi aik alag **password-set (reset) email** aati hai jis ke link se owner apna password khud chunta hai. Kisi email mein password nahi bheja jata.
- **Pehle se mojood user** ho to: usay company mein owner ke taur par add kar diya jata hai aur email mein **Sign in** link hota hai; login ke baad company switcher se nayi company chunein.
- Development mein (`MAIL_MAILER=log`) yeh emails `backend\storage\logs\laravel.log` mein milti hain.

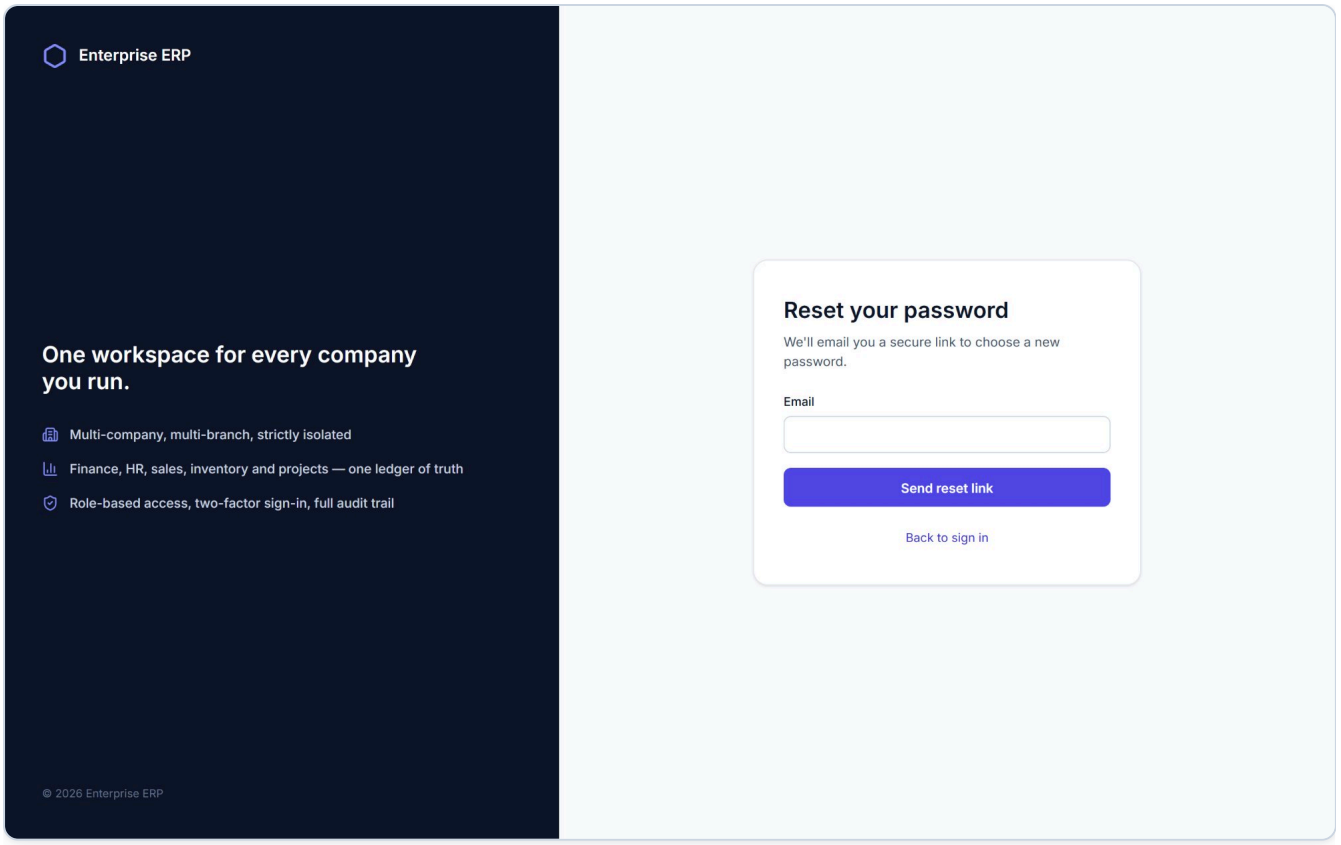

Figure 4.5 — Forgot password page — agar link expire ho jaye to owner yahan se naya password-reset link mangwa sakta hai.

Platform users

Platform → Platform users mein installation ke tamam accounts, un ki platform role, companies ki tadaad, 2FA status aur last sign-in nazar aate hain. Platform roles sirf platform owner de sakta hai, aur koi apni platform role khud nahi badal sakta.

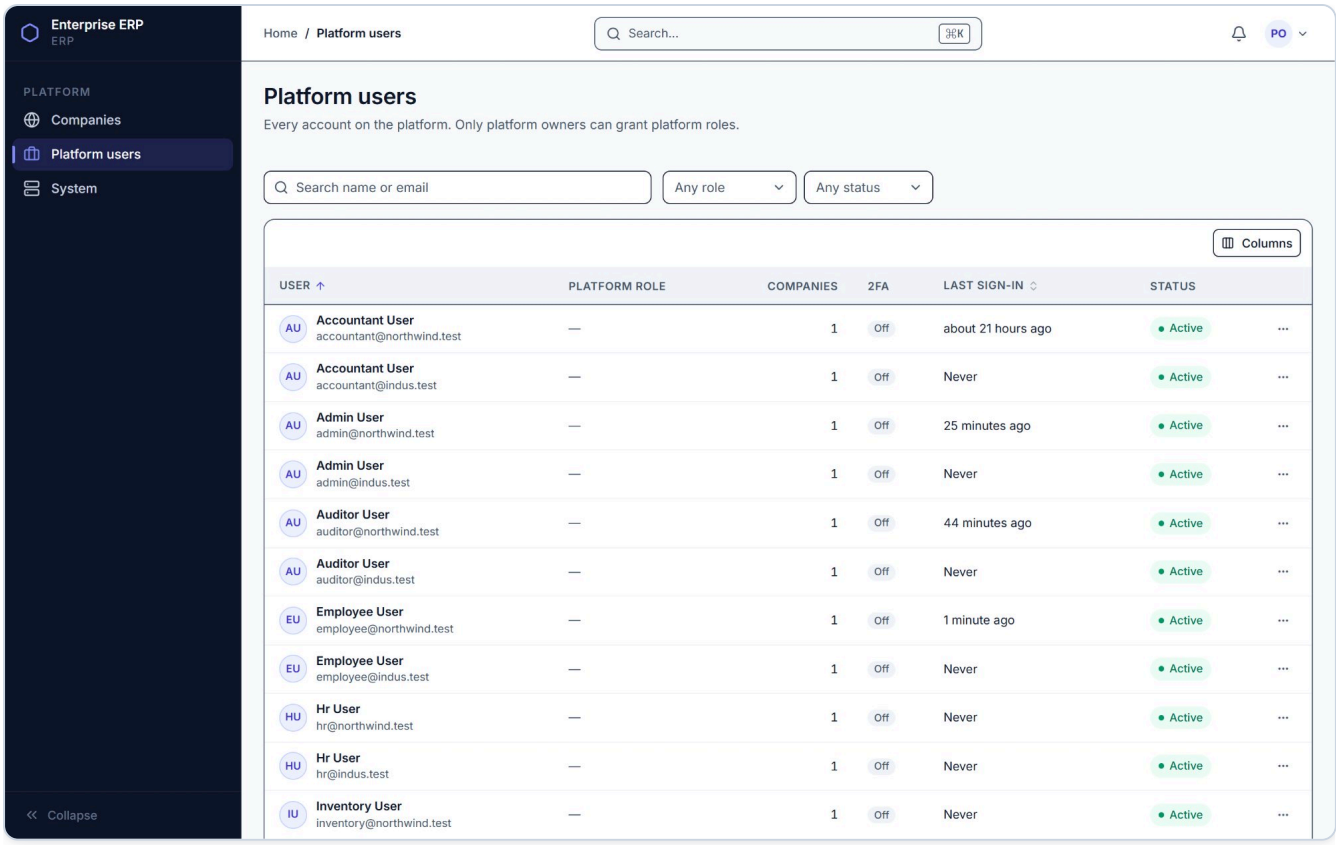

Figure 4.6 — Platform users — har account ka 2FA status aur last sign-in; yahan se dekh sakte hain kis ne 2FA on nahi kiya.

> **NOTE**
> Platform owner jab `/dashboard` kholta hai to "Dashboard unavailable — Your role does not include access to the executive dashboard" nazar aata hai. Yeh normal hai: platform staff company ka business data nahi dekhte. Seedha **Platform → Companies** use karein.

4.6 Demo data (sirf testing ke liye)

Training ya testing ke liye do tayyar demo companies load ki ja sakti hain — **Northwind Trading (Demo)** (USD) aur **Indus Manufacturing (Demo)** (PKR) — jin mein chhe mahine ki consistent history hai: employees, attendance, leave, payroll runs, products/stock, suppliers/POs, customers, quotations, orders, invoices, journals, projects aur documents.

```
backend (development only)

php artisan db:seed --class=DemoSeeder
```

- Demo users ka password `DEMO_PASSWORD` environment variable se set hota hai; set na ho to random password ban kar print hota hai.
- Demo companies `is_demo` flag ke sath hoti hain aur app mein har jagah "Demo" badge nazar aata hai.
- Seeder production environment mein chalne se **inkaar** kar deta hai.

> ⚠ **DHYAN DEIN**
>
> **Demo data kabhi production mein nahi.** Demo users ke passwords sab ko maloom hote hain (training material mein likhe hote hain). Production checklist ka aik point yahi hai: kisi company par demo flag na ho.

CHAPTER 5

Users, Roles & Permissions

Kis ko kya access dena hai — invite, roles, branch restriction, custom roles aur best practices.

5.1 Bunyadi concepts

- **Permission** — aik chhota sa haq, jaise "Approve invoices" (`invoices.approve`) ya "View employees" (`employees.view`). System mein ~90 permissions hain (Appendix A.3).
- **Role** — permissions ka aik bundle, jaise "Accountant" ya "HR Manager". User ko permissions direct nahi, roles ke zariye milti hain.
- **User ki access = us ke tamam roles ki permissions ka majmooa (union)**. Aik user ke do roles hon to dono ki permissions milti hain.
- **Company-wise** — roles har company ke andar alag hote hain. Aik shakhs Company A mein Accountant aur Company B mein Employee ho sakta hai.
- **Branch restriction** — user ko sirf kuch branches ke data tak mehdood kiya ja sakta hai.

> **TIP**
> Sidebar mein sirf wohi menus nazar aate hain jin ki permission ho. Lekin asal hifazat server par hoti hai — Chapter 6 dekhein.

5.2 Das (10) default roles

Har nayi company ko yeh 10 system roles khud-ba-khud milte hain:

Table 5.1 — Default roles (Northwind demo company se ginti)

Role	Kis ke liye	Permissions
Company Owner	Full access to everything in the company.	89
Company Administrator	Manages users, settings and all modules except payroll processing.	88
HR Manager	Employees, attendance, leave, performance and payroll.	28
Accountant	Finance, accounting, invoices, bills, payments and financial reports.	31
Sales Manager	Customers, leads, quotations, sales orders and invoices.	22
Inventory Manager	Products, warehouses and all stock operations.	20
Procurement Officer	Suppliers, requisitions, purchase orders and goods receipts.	18
Department Manager	Approves leave, requisitions and expenses for their team; manages projects.	24
Employee	Self-service: own leave, expenses, tasks, time and payslips.	10
Read-Only Auditor	Reads every module including audit logs; cannot change anything.	36

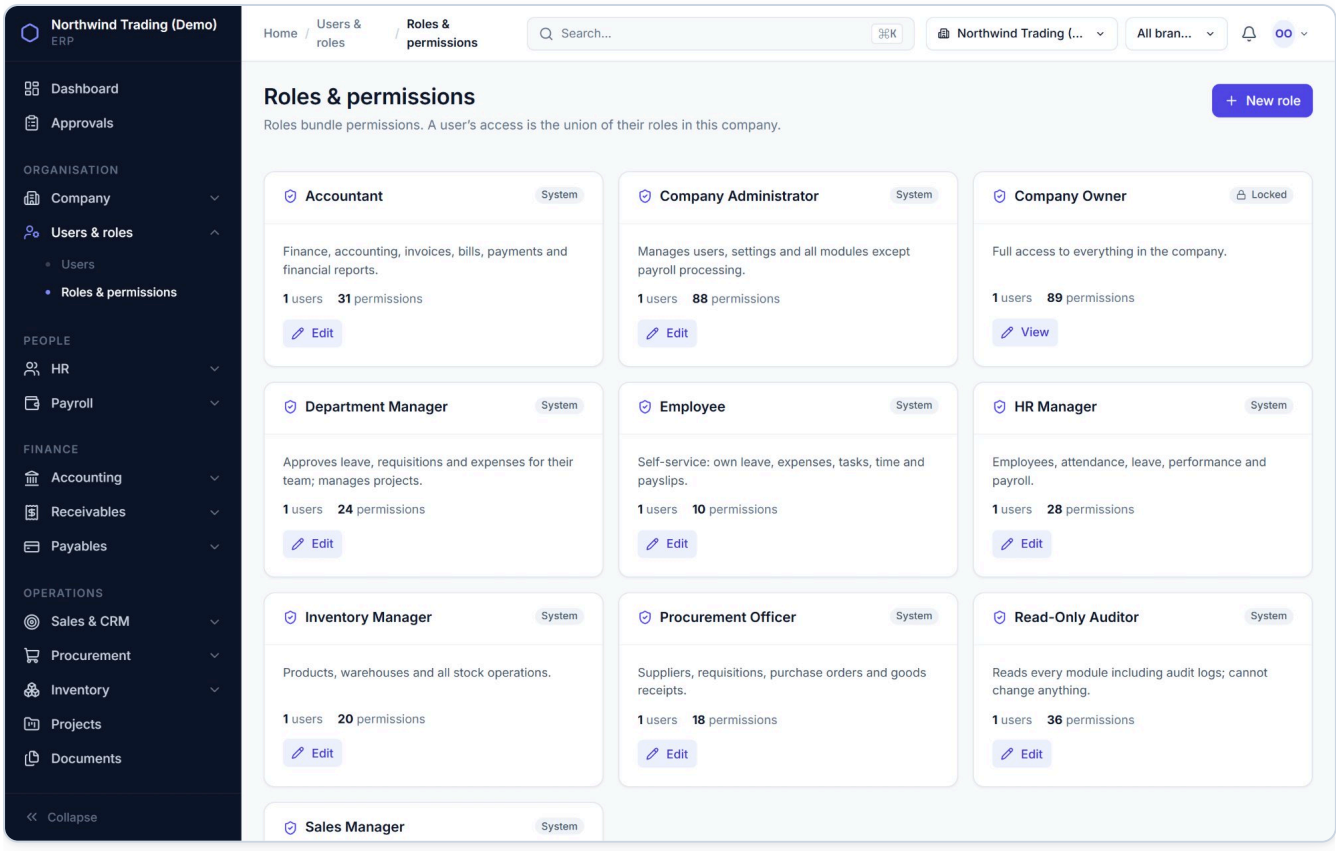

Figure 5.1 — Users & roles → Roles & permissions: har role card par users aur permissions ki tadaad. "Company Owner" par Locked badge, baqi par System badge.

5.3 Permission matrix (module × role)

Neeche wali table batati hai ke kaunsa role kis module mein kya kar sakta hai. Yeh database mein assign ki gayi asli permissions se banayi gayi hai.

✓ = poora access (dekhna + banana/badalna/approve) 👁 = sirf dekhna (view) ◐ = kuch actions (neeche detail) — = koi access nahi

Module / area	Owner	Admin	HR Mgr	Accountant	Sales Mgr	Inventory Mgr	Procurement	Dept Mgr	Employee	Auditor
Dashboard	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Company profile & settings	✓	✓	👁	👁	👁	👁	👁	👁	👁	👁
Branches & departments	✓	✓	◐	👁	👁	👁	👁	👁	—	👁
Users & roles	✓	✓	—	—	—	—	—	—	—	👁
Employees (HR records)	✓	✓	✓	—	—	—	—	👁	—	👁
Attendance & shifts	✓	✓	✓	—	—	—	—	👁	—	👁
Leave	✓	✓	✓	—	—	—	—	◐	◐	👁
Performance reviews	✓	✓	✓	—	—	—	—	—	—	👁
Payroll (runs & setup)	✓	◐	✓	👁	—	—	—	—	—	👁
Own payslips	✓	✓	✓	—	—	—	—	✓	✓	—
Chart of accounts & journals	✓	✓	—	✓	—	—	—	—	—	👁
Customer invoices	✓	✓	—	✓	◐	—	—	—	—	👁
Supplier bills	✓	✓	—	✓	—	—	◐	—	—	👁
Payments & receipts	✓	✓	—	✓	👁	—	—	—	—	👁
Expense claims	✓	✓	—	◐	—	—	—	✓	◐	👁
Bank, tax & periods	✓	✓	—	✓	—	—	—	—	—	—
Financial reports	✓	✓	—	✓	—	—	—	—	—	✓
Customers & leads (CRM)	✓	✓	—	👁	✓	—	—	—	—	👁
Quotations & sales orders	✓	✓	—	—	✓	👁	—	—	—	👁
Suppliers & requisitions	✓	✓	—	👁	—	👁	✓	◐	—	👁
Purchase orders & goods receipts	✓	✓	—	—	—	◐	◐	—	—	👁
Products & warehouses	✓	✓	—	—	👁	✓	👁	—	—	👁
Stock operations	✓	✓	—	—	👁	✓	👁	—	—	👁
Projects, tasks & time	✓	✓	—	—	—	—	—	✓	◐	👁
Documents	✓	✓	✓	👁	👁	👁	👁	👁	👁	👁
Approvals inbox	✓	✓	✓	✓	✓	✓	✓	✓	—	✓
Reports & exports	✓	✓	✓	✓	✓	✓	👁	👁	—	✓
Audit log	✓	✓	—	—	—	—	—	—	—	✓

◐ (partial) ka matlab — tafseel

Table 5.2 — Partial access ki detail

Area	Role	Yeh actions milte hain
Branches & departments	HR Manager	View branches; View departments; Manage departments
Leave	Department Manager	View leave requests; Submit own leave requests; Approve/reject leave
Leave	Employee	Submit own leave requests
Payroll (runs & setup)	Company Administrator	View payroll; Manage salary structures
Customer invoices	Sales Manager	View invoices; Create invoices; Update draft invoices
Supplier bills	Procurement Officer	View supplier bills; Create supplier bills
Expense claims	Accountant	View expense claims; Approve expense claims
Expense claims	Employee	Submit expense claims
Suppliers & requisitions	Department Manager	View purchase requisitions; Create purchase requisitions; Approve purchase requisitions
Purchase orders & goods receipts	Inventory Manager	View purchase orders; Record goods received
Purchase orders & goods receipts	Procurement Officer	View purchase orders; Manage purchase orders; Record goods received
Projects, tasks & time	Employee	View projects; View tasks; Manage tasks; Log time

> **NOTE**
> Ahem farq: **Company Administrator** ke paas sab kuch hai **siwaye** "Run & approve payroll" (`payroll.process`) ke — payroll chalana HR Manager/Owner ka kaam hai. **Procurement Officer** purchase orders bana/manage kar sakta hai lekin approve nahi — PO approval Owner/Admin karte hain. **Read-Only Auditor** har module dekh sakta hai, audit log bhi, aur reports export kar sakta hai, lekin kuch badal nahi sakta.

5.4 Naya user invite karna

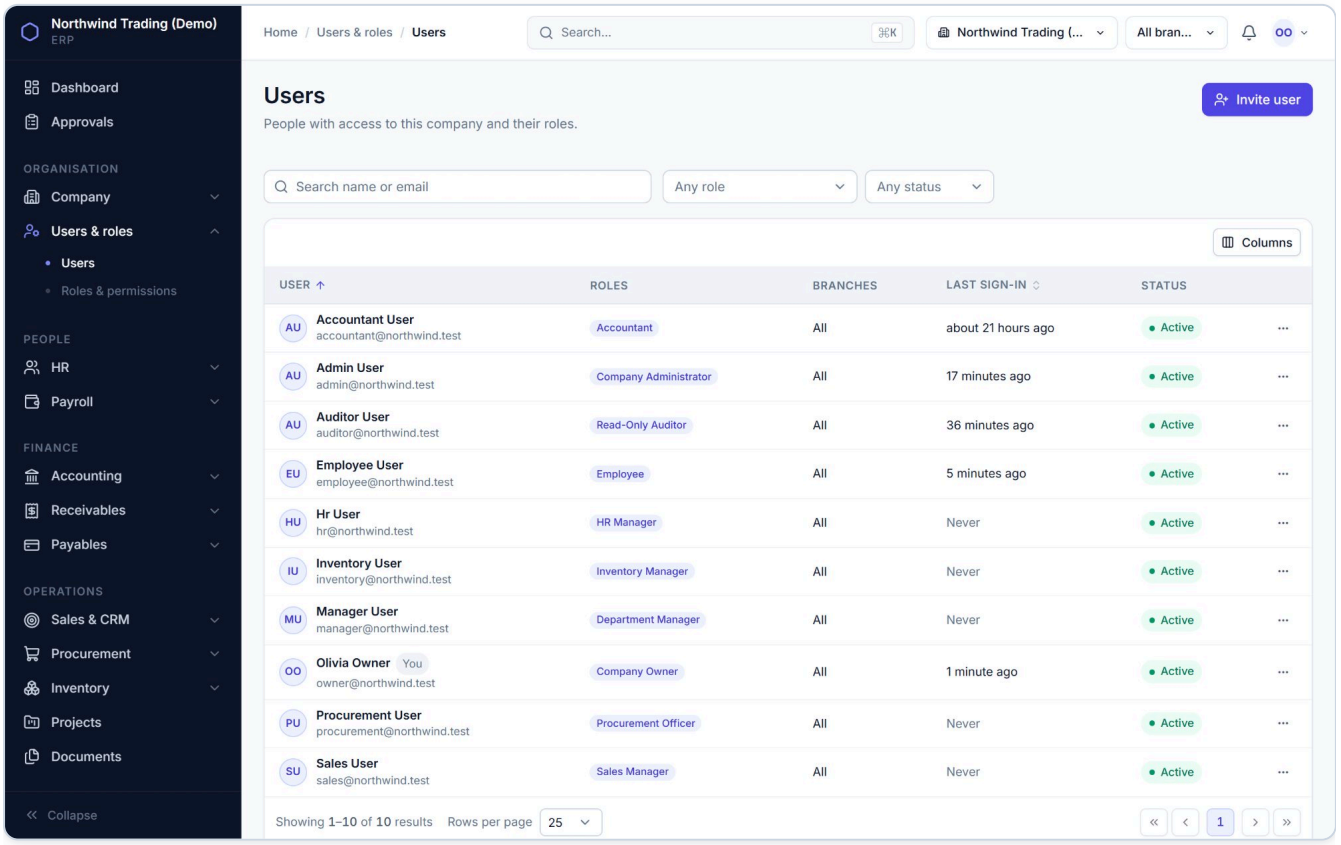

Figure 5.2 — Users list — har user ke roles, branches, last sign-in aur status. Role aur status se filter bhi kar sakte hain.

1. Sidebar mein **Users & roles → Users** kholein.
2. Upar right **Invite user** button dabayein — right side se aik drawer khulega.
3. **Full name** aur **Work email** bharein (dono lazmi).
4. **Roles** mein aik ya zyada roles tick karein. Hint: "Permissions are the union of all assigned roles."
5. **Access to all branches** — default on. Agar user ko sirf kuch branches tak rakhna hai to off karein aur **Allowed branches** mein branches chunein (5.5).
6. Invite/save karein. Naye shakhs ko email milti hai jis ke link se woh apna password khud set karta hai. Agar email pehle se kisi account ki hai, to usi account ko company mein add kar diya jata hai.

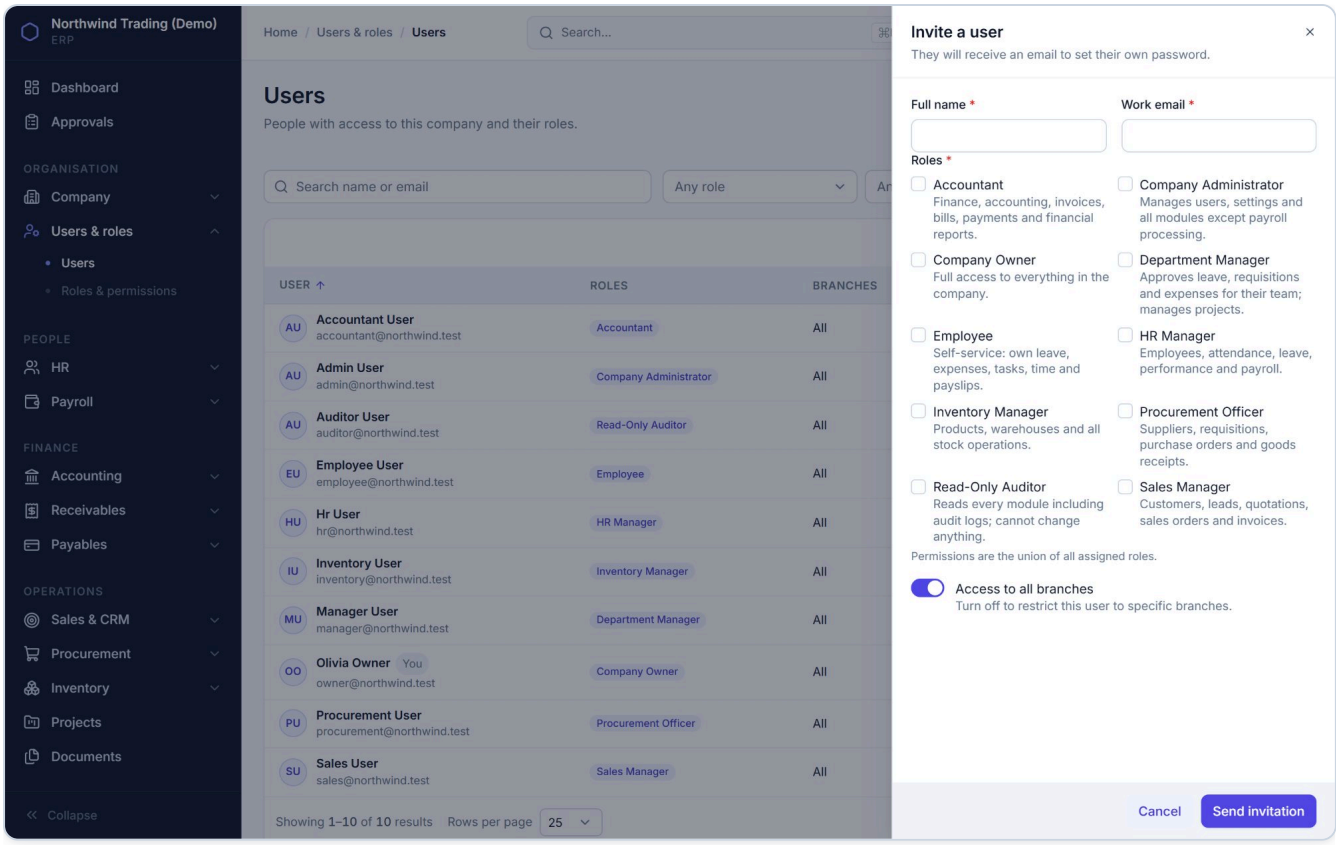

Figure 5.3 — Invite user drawer — naam, work email, roles, aur "Access to all branches" switch.

> **TIP**
> User ne abhi tak login nahi kiya to row ke actions menu mein **Resend invitation** option hota hai.

5.5 Branch restriction

Bari companies mein aksar aik branch ka manager doosri branch ka data na dekhe. Iske liye:

1. User ko invite karte waqt (ya baad mein **Edit access** se) **Access to all branches** switch **off** karein — "Turn off to restrict this user to specific branches."
2. **Allowed branches** mein sirf woh branches select karein jahan ka data user dekh sake.
3. Save karein. Users list ke **BRANCHES** column mein ab "All" ke bajaye branch names nazar aayenge.

> **NOTE**
> Is version mein branch restriction yeh records mehdood karti hai: customer invoices, supplier bills, payments/receipts, journal entries, payroll runs/payslips, warehouses, projects, dashboard aur reports/exports. Company-level settings aur jin records par branch nahi, woh role ki permission ke mutabiq dikhte hain.

> ⚠ **DHYAN DEIN**
>
> **Known limitation:** kuch lists abhi branch restriction lagu nahi kartin — maslan HR → Employees, sales orders, purchase orders, attendance, leave aur stock. Branch-restricted user in modules mein doosri branches ka data bhi dekh sakta hai (agar role mein us module ki permission ho). In modules ke liye asal hadd role ki permissions hain.

5.6 Role ki permissions badalna aur custom role banana

System role edit karna

1. **Users & roles → Roles & permissions** kholein.
2. Jis role ko badalna hai (maslan **Accountant**) us ke card par **Edit** dabayein.
3. Drawer mein note hoga: "System role: the name is fixed, but you can adjust its permissions." — naam/description nahi badlenge, permissions badal sakte hain.
4. **Filter permissions** box mein likh kar permission dhoondein, ya module ke **All** checkbox se poora module on/off karein.
5. Har permission ke neeche us ka technical code (jaise `company.view`) bhi likha hai.
6. **Save role** dabayein. Is role wale tamam users par tabdeeli foran lagu ho jati hai.

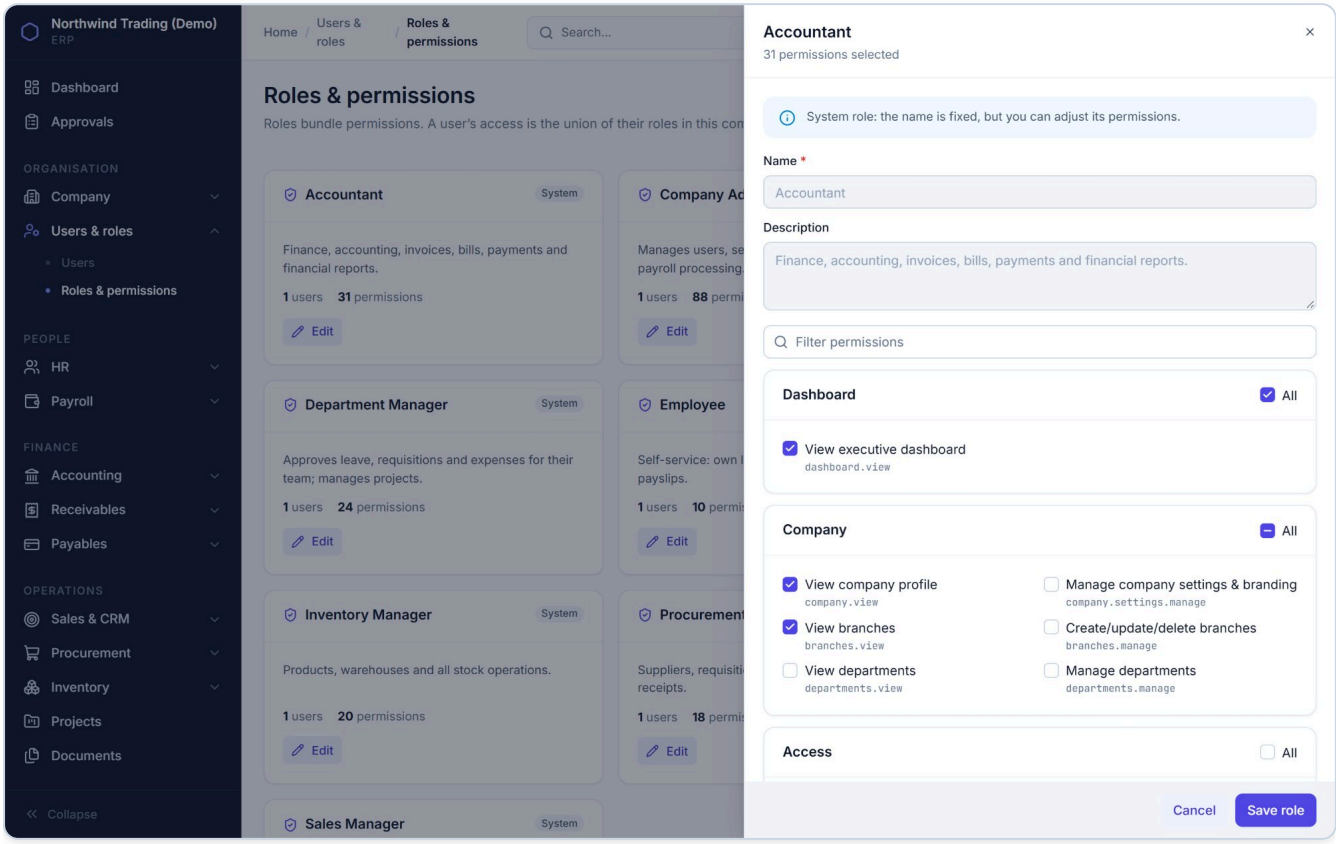

Figure 5.4 — Accountant role editor — module-wise permission checkboxes, "31 permissions selected", Filter permissions aur Save role.

Custom role banana

1. Roles page par upar right **New role** dabayein.
2. **Name** (maslan "Branch Cashier") aur **Description** likhein.
3. Sirf zaroori permissions tick karein (least privilege — 5.8).
4. **Create role** dabayein, phir user ko **Edit access** se yeh role assign karein.

> **NOTE**
> **Company Owner role locked hai:** "The Company Owner role always has every permission and cannot be edited." Is mein hamesha saari permissions rehti hain. Custom role delete karne ke liye pehle us ke tamam users se role hatayein; system roles delete nahi hote.

5.7 User ko edit, deactivate ya remove karna

Users list mein har row ke aakhir mein actions menu (···) hai:

Table 5.3 — User actions

Action	Kya hota hai
Edit access	Roles, branches aur **Active in this company** switch badlein.
Active in this company = off	"Inactive members cannot sign in to this company." User ka record aur history baqi rehti hai; agle request par woh is company se bahar ho jata hai. Wapas on karke access bahal kar sakte hain.
Resend invitation	Sirf un users ke liye jinhon ne kabhi login nahi kiya.
Remove from company	"They lose access to this company immediately. Their account and history in other companies are not affected."

> **DHYAN DEIN**
> Aap apne aap ko remove ya deactivate nahi kar sakte, aur apna woh aakhri role nahi hata sakte jis se users manage hote hain — taake company baghair admin ke na reh jaye.

> **TIP**
> Koi employee company chhore to **usi din** us ki membership inactive ya remove karein. Delete karne ke bajaye inactive karna behtar hai agar audit/history ke liye record chahiye.

5.8 Best practices

- **Least privilege** — har shakhs ko sirf utni access dein jitni us ke kaam ke liye zaroori hai. "Sab ko Admin bana do" kabhi nahi.

- **Maker–checker (separation of duties)** — jo record banaye, wohi approve na kare. System khud bhi kuch jagah yeh rok lagata hai: jis user ne **payroll run banaya** woh usay approve nahi kar sakta ("You cannot approve a payroll run you created."), aur requisition banane wala apni requisition approve nahi kar sakta. Approvals inbox mein "Your own requests never appear here."
- **Owner accounts kam rakhein** — Company Owner role 1–2 logon tak; roz ka admin kaam Company Administrator se.
- **Shared accounts nahi** — har shakhs ka apna login, taake audit log mein sahi naam aaye.
- **2FA** — owners, admins, accountants aur HR ke liye lazmi.
- **Har mahine review** — Users list ko "Last sign-in" se dekhein; purane/ghair-zaroori accounts inactive karein.
- **Bahar ke auditor** ko *Read-Only Auditor* role dein, aur audit khatam hote hi remove karein.

CHAPTER 6

Security & access control

System aap ke data ki hifazat kaise karta hai — aasaan zaban mein — aur production checklist.

6.1 Asal hifazat server par: "button chhupana kaafi nahi"

Jab kisi user ke paas permission nahi hoti to sidebar mein woh menu nazar nahi aata, aur page par woh button (jaise `Approve`) bhi nahi hota. Lekin yeh sirf suhulat ke liye hai. **Asal faisla har request par server (API) karta hai.**

- Agar koi chalak user URL khud type kar le, ya browser tools se request bhej de, tab bhi server permission check karta hai aur inkaar kar deta hai.
- Har API action par policy check aur route par permission check hota hai — dono jagah.
- Approve, payroll process, user manage, export, stock adjust aur journal posting jaise sensitive kaam audit log mein bhi likhe jate hain.

Neeche employee ka screenshot hai jis ne direct `/finance/invoices` URL khola. Sidebar mein Receivables menu tha hi nahi, aur server ne data dene se inkaar kar diya:

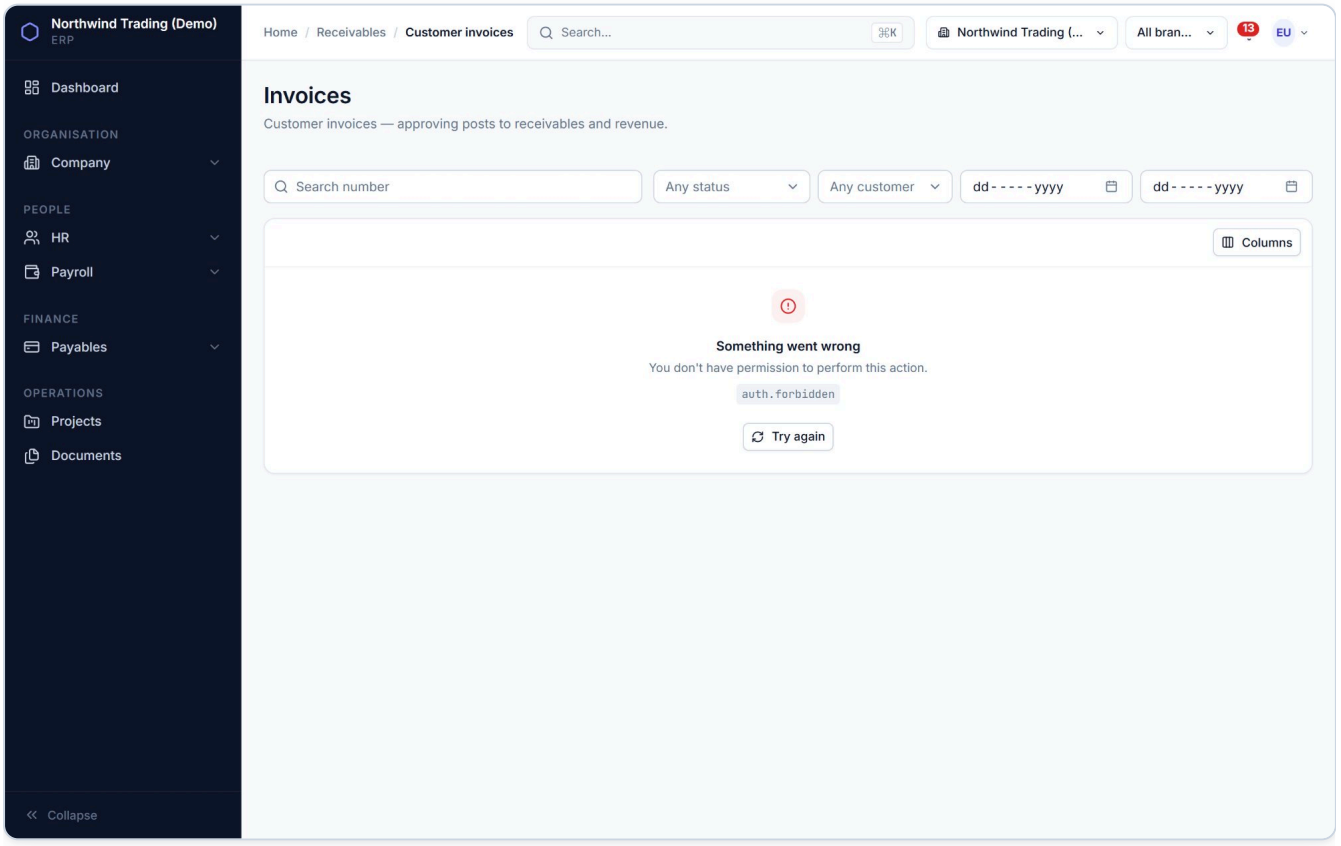

Figure 6.1 — Employee ne /finance/invoices khola — server ka jawab: "You don't have permission to perform this action." (auth.forbidden). Koi invoice data nahi dikhaya gaya.

6.2 403 aur 404 ka farq

Table 6.1 — Error codes ka matlab

Soorat	Server ka jawab	Screen par
Aap is company ke member hain lekin permission nahi	**403 Forbidden**	"Something went wrong — You don't have permission to perform this action." (`auth.forbidden`)
Kisi **dusri company** ke record ka ID try kiya	**404 Not found**	"Page not found" / record nahi milta
Platform pages (`/platform/…`) bina platform role ke	**404**	Page mojood hi nahi lagta
Session khatam / login nahi	**401**	Login page par wapas
Bohat zyada login attempts	**429**	"Too many attempts" — thori der baad try karein

404 kyun? Agar dusri company ke record par "permission nahi" (403) aaye to is se pata chal jata ke woh record mojood hai. 404 dene se kisi ko yeh bhi maloom nahi hota ke woh ID kisi aur company mein hai ya nahi.

6.3 Company data isolation

- Har request ke sath current company bheji jati hai, aur server check karta hai ke aap us company ke **active member** hain. Warna `tenant.forbidden`.
- Database ki har company-wise table mein company ka ID hota hai, aur har query khud-ba-khud sirf current company tak mehdood hoti hai.
- Form mein jo bhi related ID aaye (customer, warehouse, account…) server check karta hai ke woh isi company ki hai — client par bharosa nahi.
- Automated tests yeh confirm karte hain ke Company A, Company B ka data read, update, delete ya export nahi kar sakti.
- Platform staff bhi company data tab tak nahi dekh sakte jab tak member na hon.

6.4 Login protections

Table 6.2 — Sign-in ki hifazat

Control	Kaise kaam karta hai
Password policy	Kam az kam 10 characters, letters + numbers, aur known leaked passwords reject. Passwords bcrypt hash mein store hote hain.
Rate limiting	Aik email + IP se **5 login attempts/minute**, aik IP se 30/minute; 2FA codes 5/minute; forgot/reset 10/minute.
Aik jaisa error message	Ghalat email ho ya ghalat password — message aik jaisa, taake koi emails "guess" na kar sake. Forgot-password hamesha same jawab deta hai.
Two-factor (TOTP)	Authenticator app code + single-use recovery codes; secret encrypted store hota hai.
Session expiry	Session cookie HttpOnly hai (JavaScript nahi parh sakta). `SESSION_LIFETIME` (default 120 minute) idle rehne par logout.
Password change	"Changing your password signs out every other session." — dusre devices se khud logout.
Re-confirmation	2FA on karne jaise sensitive kaam se pehle password dobara (15 minute ke liye valid).
Deactivation	Inactive user agli request par hi logout ho jata hai.
CSRF / CORS	Har badalne wali request par CSRF token; API sirf allowed frontend origin se baat karti hai.

🔒 **SECURITY**

Agar kisi ka password leak hone ka shak ho: (1) us user se password change karwayein (dusre sessions khud band), (2) 2FA on karwayein, (3) Audit log mein "Sign-in & security" filter laga kar dekhein ke kahan se login hue.

6.5 Audit log ka istemal

Insights → Audit log aik **append-only** record hai: "Entries cannot be edited or deleted." Is mein security events, settings changes, postings aur exports — kis ne, kab, kis IP se, kya badla — sab likha hota hai. Passwords aur secrets automatically chhupa (redact) diye jate hain.

1. Sidebar mein **Audit log** kholein (permission: `audit.view` — Owner, Admin, Auditor).
2. Category filter se chunein: **Sign-in & security**, **Users**, **Roles & permissions**, **Company settings**, **Employees**, **Financial postings**, **Inventory**, **Exports**.
3. Table mein **WHEN**, **ACTOR**, **ACTION**, **RECORD**, **IP** columns hain.
4. Kisi row par click karein — drawer mein poori detail aur purani/nayi values ka farq (diff) nazar aata hai.

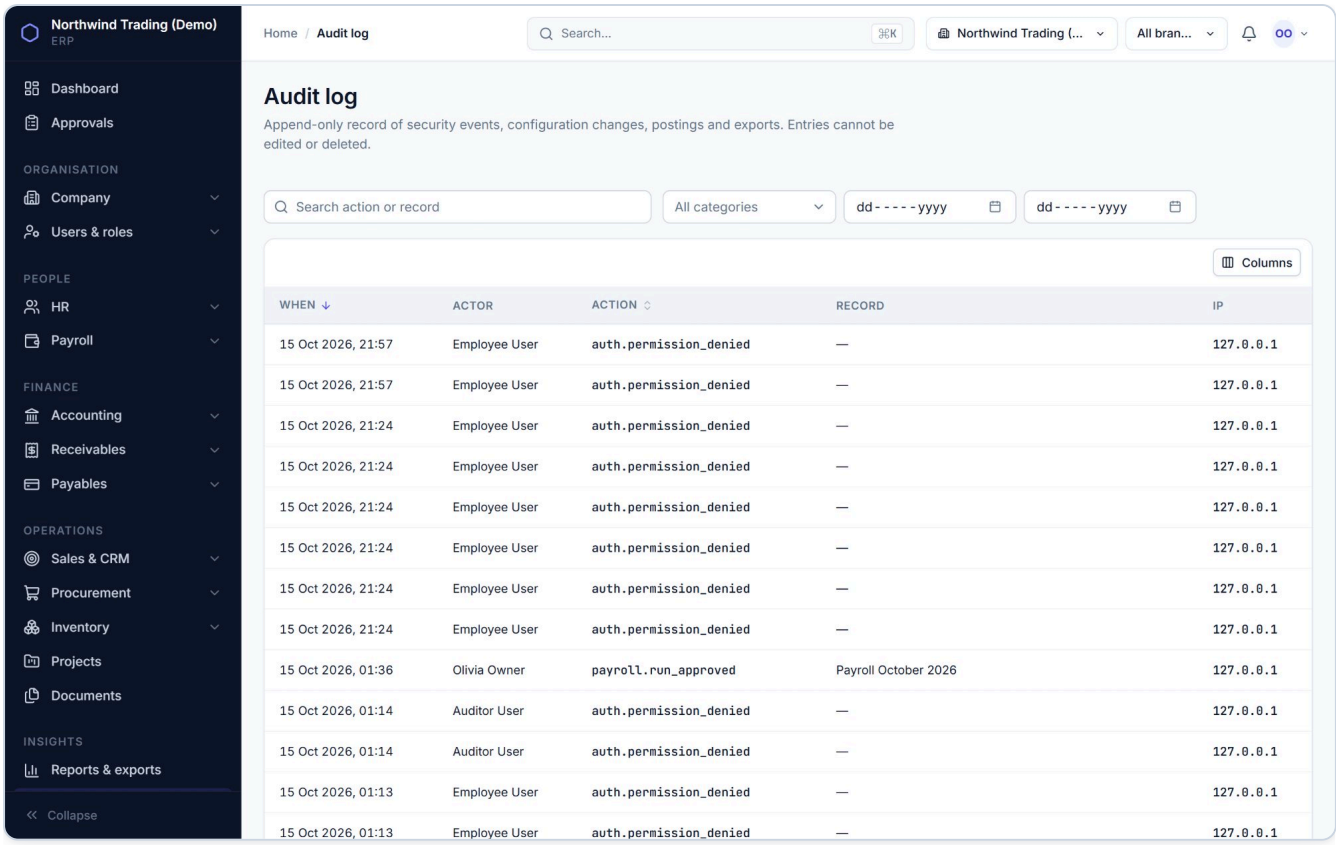

Figure 6.2 — Audit log — har action ka waqt, actor, action code, record aur IP. Upar wali entries mein employee ki denied attempts (auth.permission_denied) bhi nazar aa rahi hain.

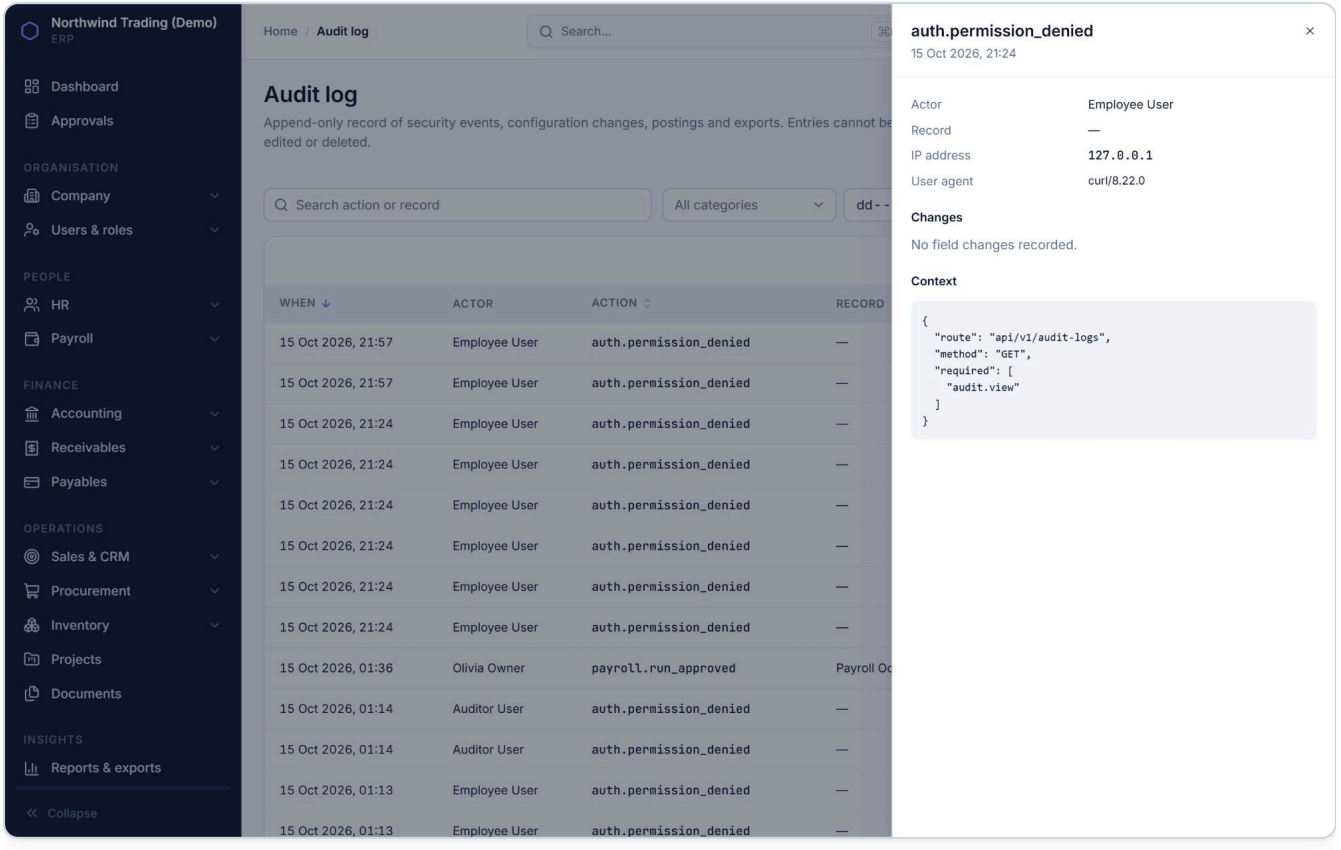

Figure 6.3 — Audit entry drawer — aik entry ki mukammal detail.

> **TIP**
> Mashkook activity dhoondne ke liye: `auth.permission_denied` (koi baar baar mana kiye gaye pages khol raha hai), failed logins, aur raat ke waqt exports dekhein. Audit log default **730 din** rakha jata hai (`ERP_RETAIN_AUDIT_DAYS`).

6.6 Production security checklist

Live (production) jane se pehle IT person har point tick kare:

- [ ] `APP_ENV=production` , `APP_DEBUG=false` , unique `APP_KEY`
- [ ] HTTPS har jagah; `SESSION_SECURE_COOKIE=true` ; proxy par HSTS
- [ ] `CORS_ALLOWED_ORIGINS` / `SANCTUM_STATEFUL_DOMAINS` / `SESSION_DOMAIN` asli domains par set
- [ ] Runtime database user ke paas `DROP` / `ALTER` nahi; migrations ke liye alag user
- [ ] Runtime user se audit log ki editing revoke: `REVOKE UPDATE, DELETE ON erp.audit_logs FROM 'erp_app'@'%';`
- [ ] Hosted DB: TLS CA configured (`DB_SSL_CA`) aur IP allow-list restricted
- [ ] Mail transport configured (password reset, invitations)
- [ ] Queue worker aur scheduler supervisor ke under chal rahe hain
- [ ] `php artisan config:cache route:cache event:cache`
- [ ] Raat ko encrypted backups, aur restore test kiya hua (Chapter 3.4)
- [ ] `composer audit` aur `npm audit --omit=dev` saaf ya review kiye hue
- [ ] Pehla admin `erp:create-admin` se bana, 2FA on, `ERP_ALLOW_ADMIN_BOOTSTRAP` wapas `false`
- [ ] Demo data mojood nahi (kisi company par `is_demo` nahi)

> **⚠ DHYAN DEIN**
> Security masle (vulnerability) ka pata chale to system owner ko **private** taur par batayein — public jagah post na karein.

CHAPTER 7

Daily use — module by module

Har module ka maqsad, aam kaam step-by-step, aur woh qawaid jo yaad rakhne zaroori hain.

Is chapter ke screenshots **Company Owner** (`owner@northwind.test`) ke hain jis ke paas har permission hai. Aap ke role ke mutabiq kuch menus ya buttons nazar nahi aayenge — yeh normal hai (Chapter 5).

Har list page par aik jaisi cheezein

- Upar right **primary button** (jaise **New employee**, **New invoice**) naya record banata hai.
- Filters (status, date, branch…) aur search box list ko chhota karte hain; **Columns** se columns chhupa/dikha sakte hain.
- Column heading par click se sorting; neeche **Rows per page** (10/25/50/100) aur page numbers.
- Row par click karne se detail page ya drawer khulta hai; row ke aakhir mein ⋯ menu mein Edit/Delete jaise actions.
- Status rang-daar badges mein dikhta hai (Draft, Pending approval, Approved, Paid…).

7.1 Dashboard

Dashboard business ki sehat aik nazar mein dikhata hai. Upar right **MTD** / **30D** / **90D** / **YTD** / **12M** se period badlein; topbar ke branch filter se aik branch ka data dekhein.

Table 7.1 — Dashboard widgets

Card / widget	Kya batata hai
Revenue, Expenses, Net profit	General ledger mein posted amounts, pichle barabar period se muqabla (% change).
Outstanding receivables / payables	Customers se lena aur suppliers ko dena baqi.
Employees, Active products, Pending approvals	Headcount, active products, aur aap ke faisle ke muntazir items.
Revenue vs expenses chart, By branch	Waqt aur branch ke hisaab se trend.
Recent invoices, Pending approvals, Low stock	Shortcut lists — "View all" / "Inbox" / "View" se poori list.
Headcount by department, Recent activity	HR distribution aur haaliya audit activity.

7.2 Approvals inbox

Approvals har us cheez ki aik jagah list hai jo **aap ke faisle** ka intezar kar rahi hai: leave requests, purchase requisitions, purchase orders, sales orders, invoices, bills aur expense claims. Page par likha hai: "Your own requests never appear here." — apni banayi hui cheez aap ko approve ke liye nahi dikhti (maker–checker).

1. Sidebar mein **Approvals** kholein. Upar tabs mein har type ki ginti hoti hai, jaise **All (9)**, **Leave request (3)**.
2. Kisi item par click karein — us ka detail page khulta hai.
3. Detail parh kar **Approve** ya **Reject** karein (reject par wajah likhein).

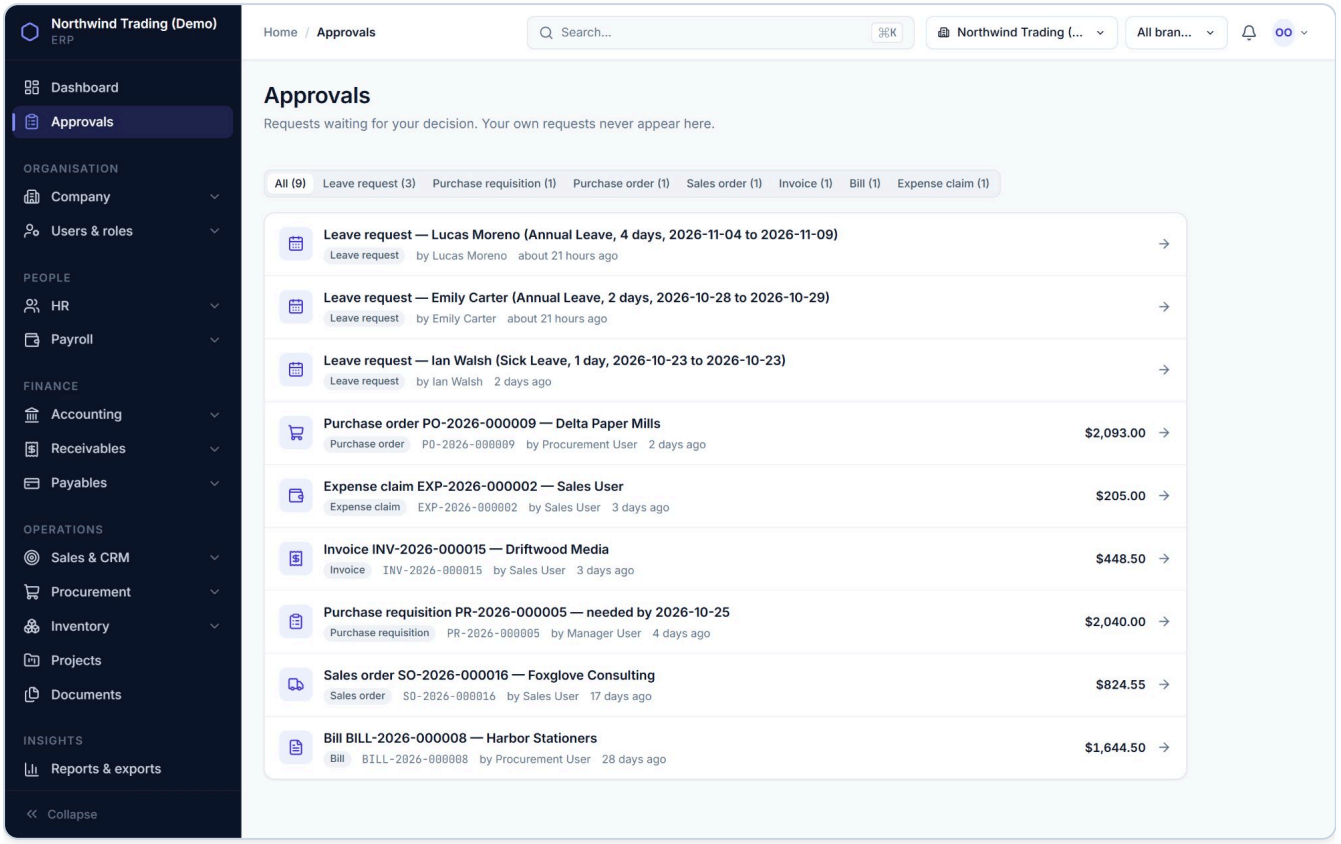

Figure 7.1 — Approvals inbox — type-wise tabs, har request ka khulasa aur kis ne kab bheja.

7.3 Notifications

Bell icon par unread ginti nazar aati hai. **Notifications** page par **All** / **Unread** filter aur **Mark all read** button hai. Notifications approvals, assignments, low-stock alerts, export tayyar hone waghera par aati hain. Kis event ki notification app mein aur kis ki email par aaye, yeh **Account menu → Notifications** (`/settings/notifications`) mein set karein; security emails hamesha aati hain.

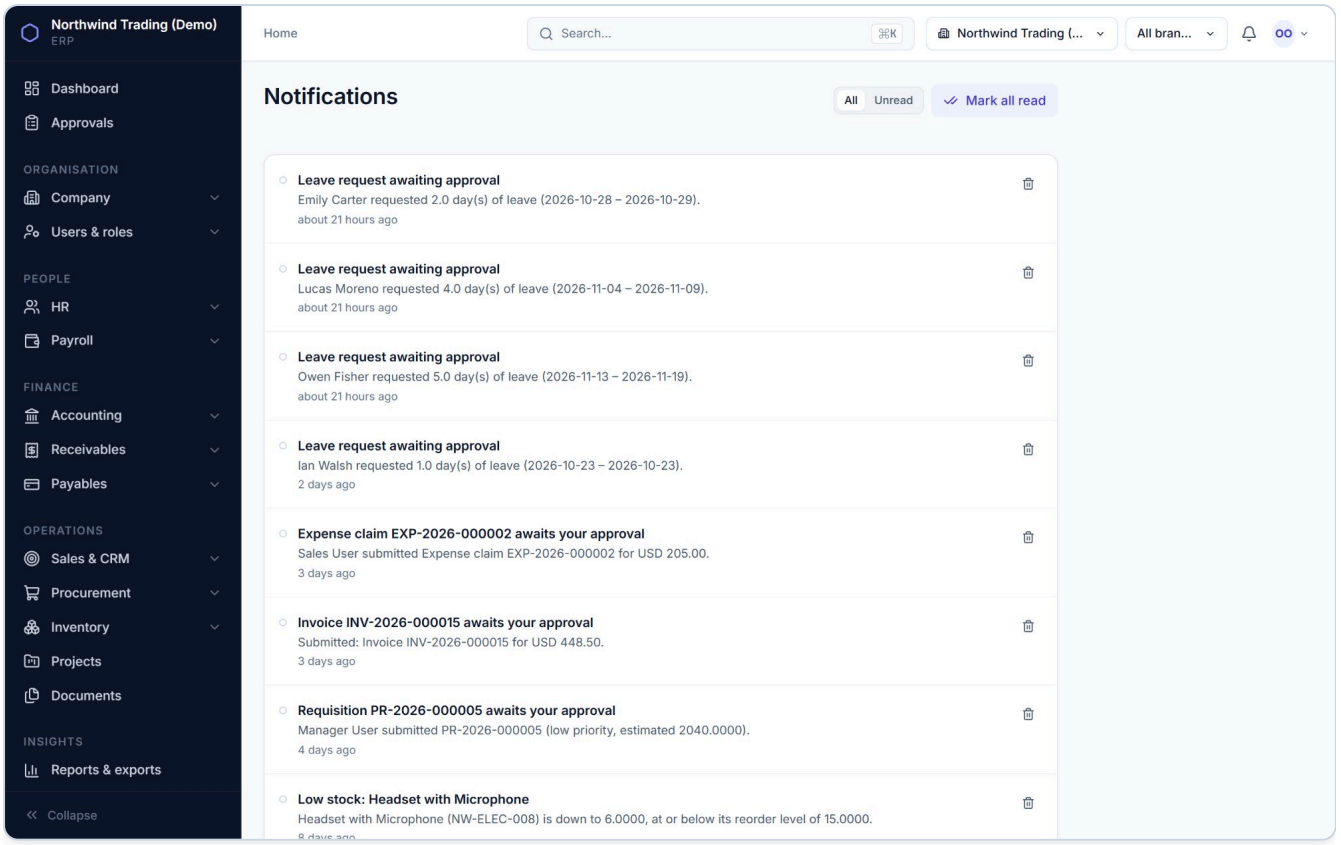

Figure 7.2 — Notifications — leave approvals ki requests aur doosre alerts.

7.4 Company, Branches, Departments & Map

Company profile

Company → Profile & settings mein logo (**Upload logo**, PNG/JPG/WebP/SVG, max 2 MB), legal name, tax/registration numbers, registered address aur localisation & finance settings hain. Badal kar **Save changes** dabayein.

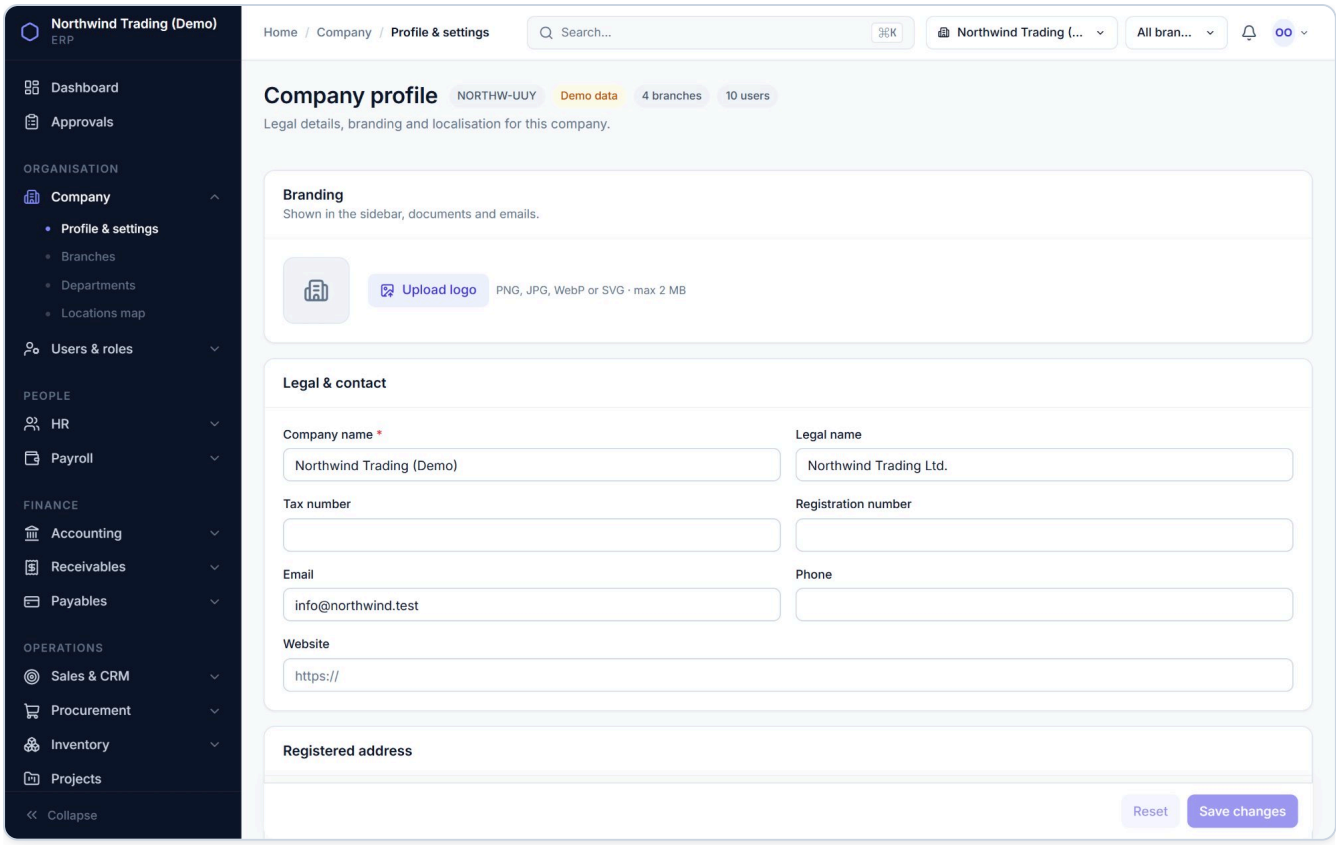

Figure 7.3 — Company profile — Branding, Legal & contact, Registered address, Localisation & finance.

Branch banana

1. Company → Branches kholein → New branch.
2. Naam, code, address aur (map ke liye) location bharein; head office ho to mark karein.
3. Save karein. Branch ab users ki branch restriction, employees, warehouses aur reports mein available hai.

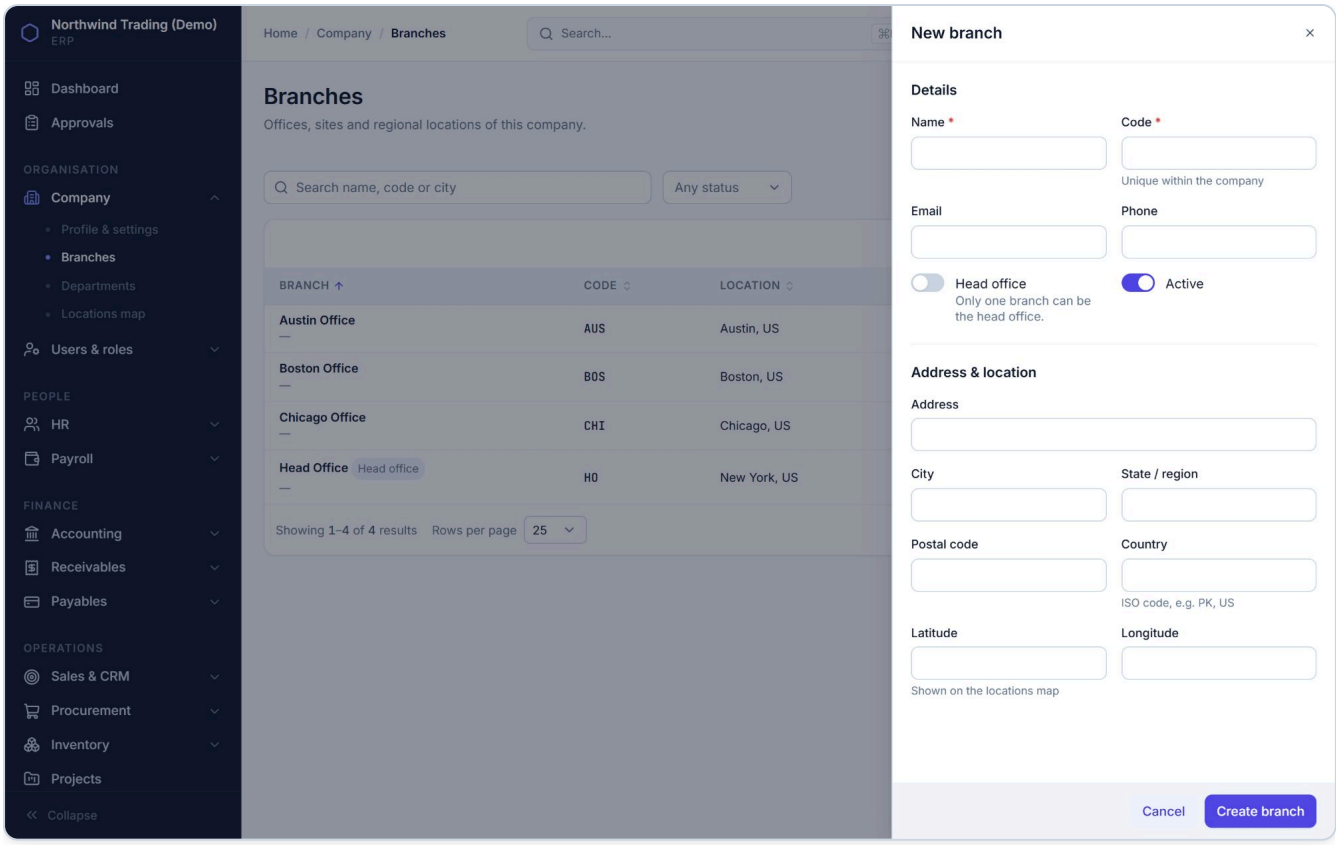

Figure 7.4 — New branch drawer — branch ki details. Save se pehle drawer band karne par kuch save nahi hota.

Departments aur map

Company → Departments mein New department se department banayein; parent department chun kar hierarchy (jaise Finance → Accounts Payable) bana sakte hain. Company → Locations map par branches, warehouses, customers aur suppliers jin ke coordinates saved hain, map par nazar aate hain (Map / List).

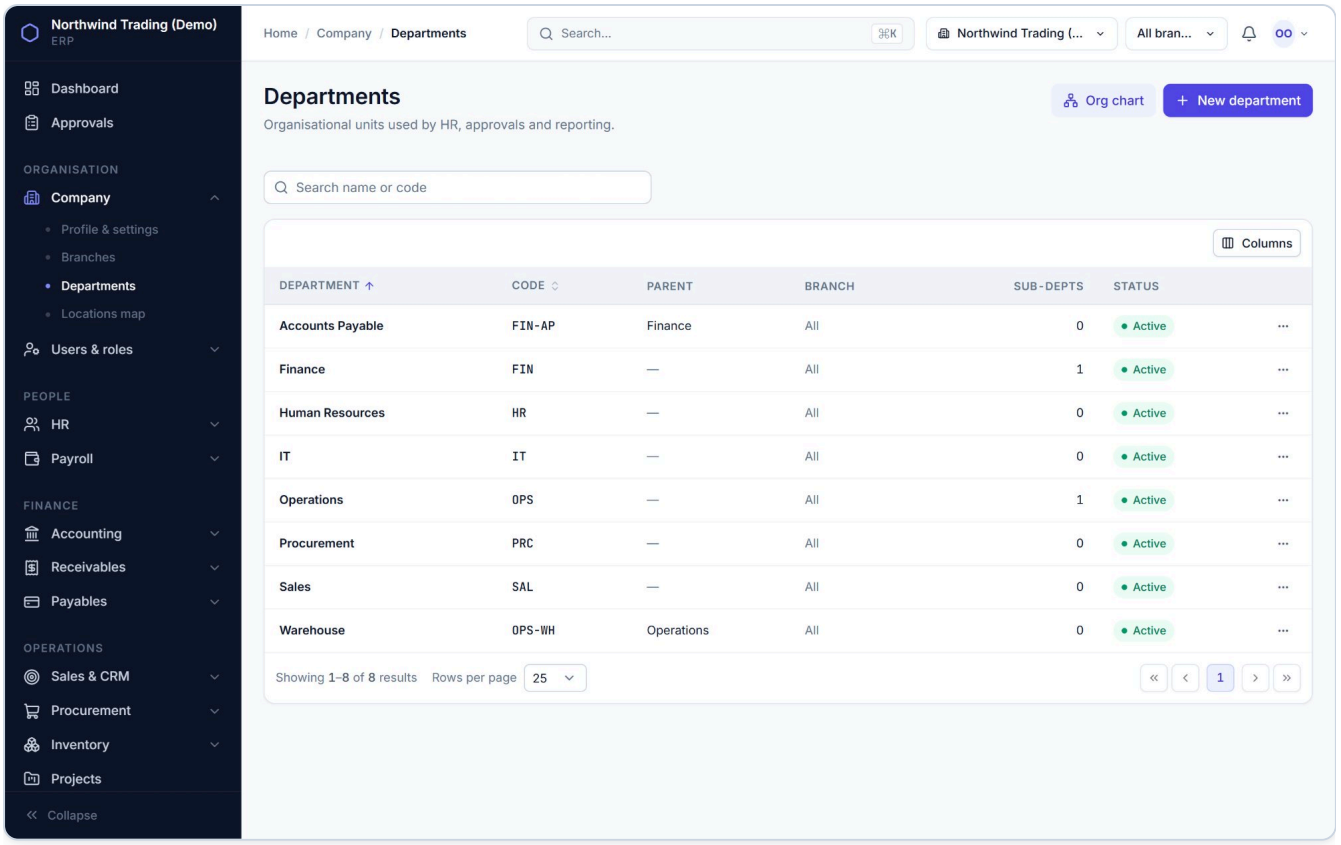

Figure 7.5 — Departments — code, parent, branch, sub-departments ki tadaad.

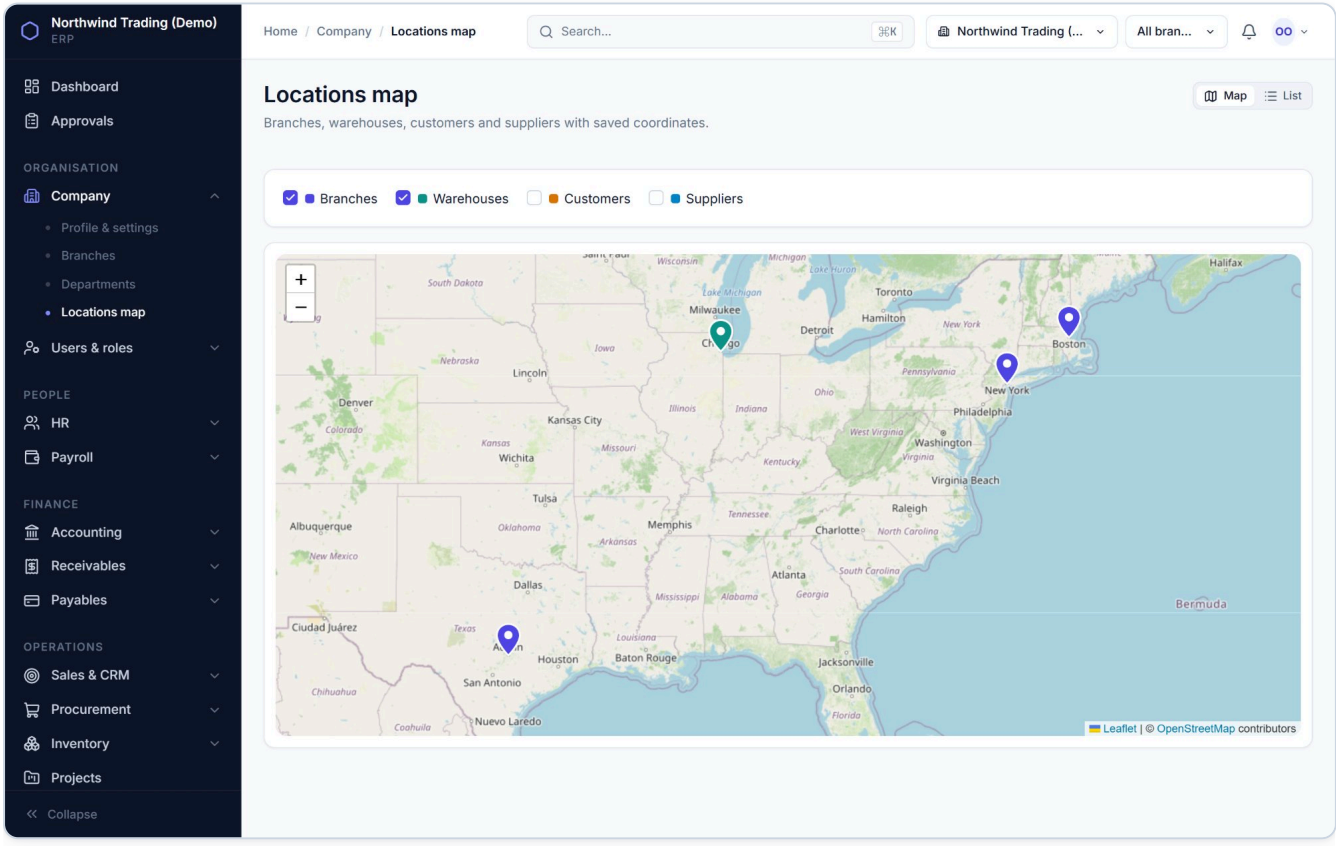

Figure 7.6 — Locations map — OpenStreetMap par branches, warehouses, customers aur suppliers ke filters.

7.5 HR

Employees

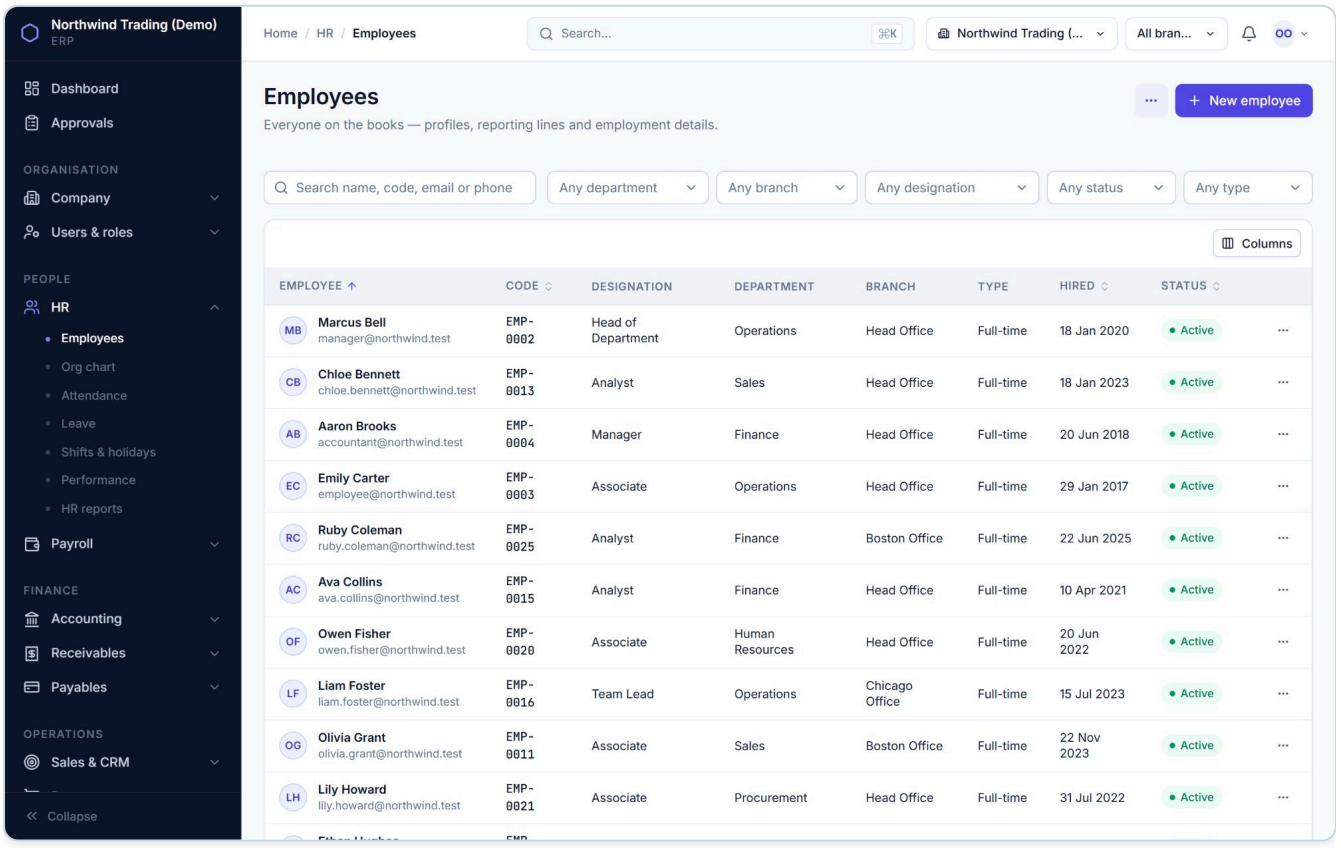

Figure 7.7 — HR → Employees — department, branch, designation, status aur type ke filters.

1. **HR → Employees** kholein → **New employee**.
2. Sections bharein: **Personal** (first/last name), **Contact & address** (work email lazmi), **Emergency contact**, **Employment** (branch, department, designation, manager, type, hire date), **Payroll basics** (base salary), **Identity, bank & tax**, **Linked user**, **Notes**. Employee code khud ban jata hai.
3. **Create employee** dabayein.
4. **Linked user** mein us shakhs ka login select karna zaroori hai — warna woh self-service (check-in, leave, payslips) use nahi kar sakega.

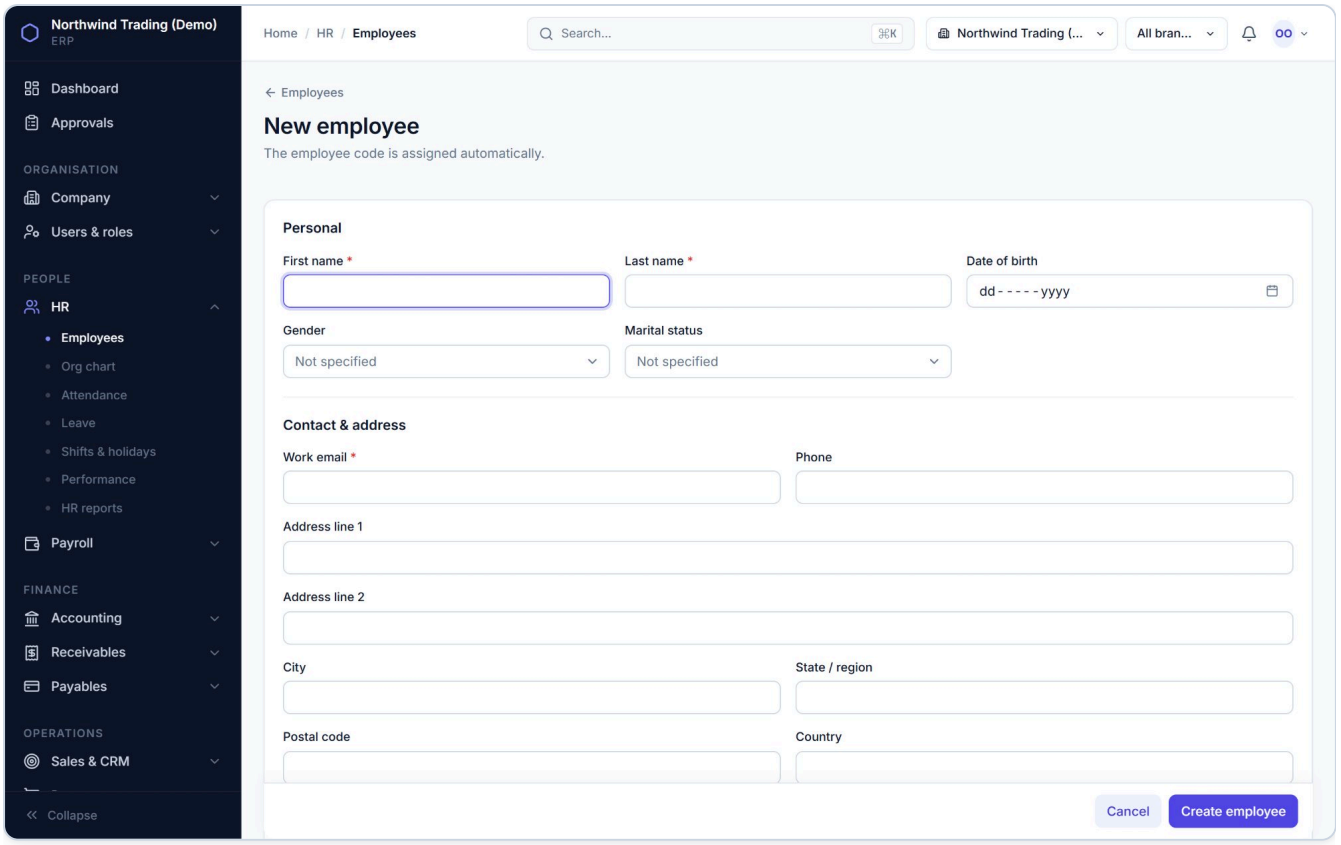

Figure 7.8 — New employee form — "The employee code is assigned automatically."

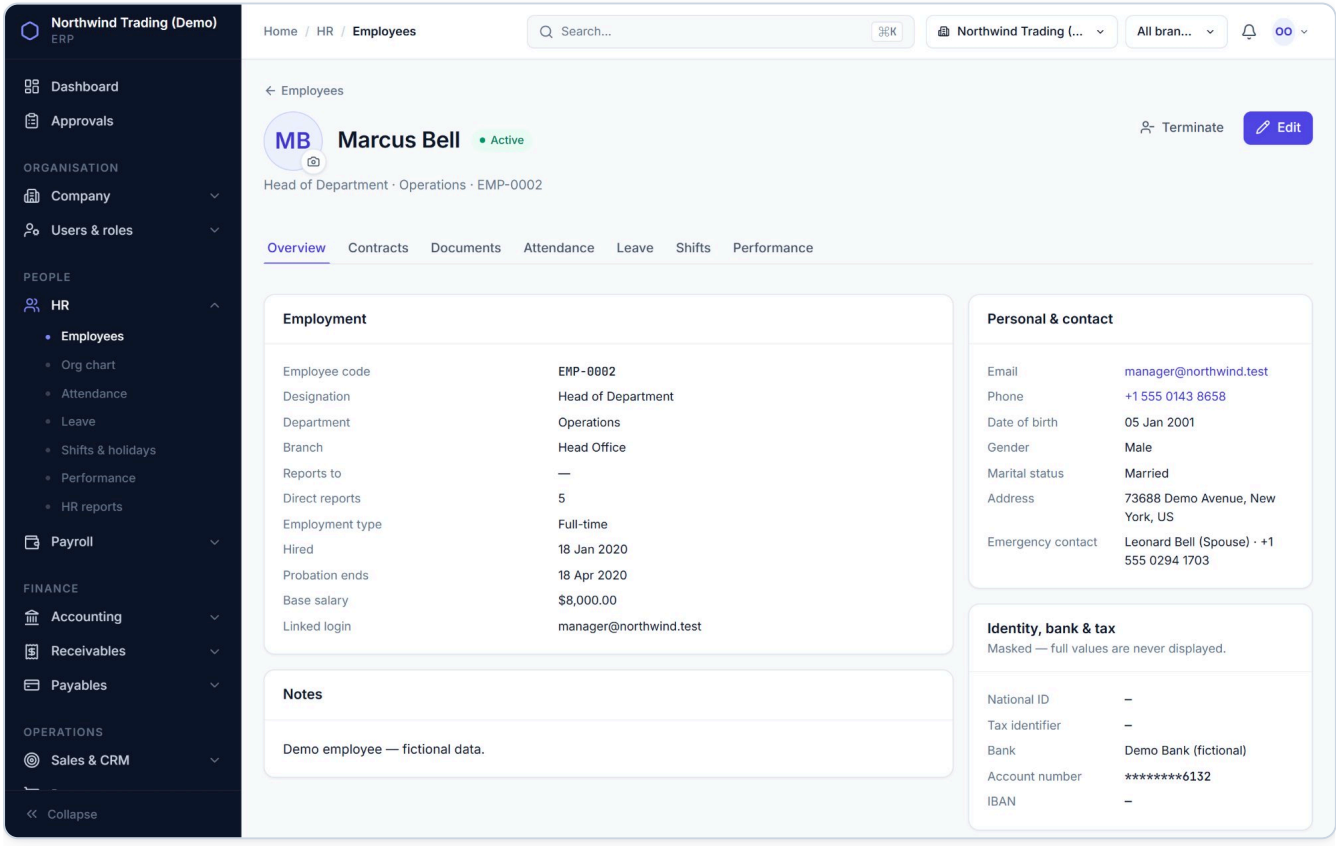

Figure 7.9 — Employee profile — tabs: Overview, Contracts, Documents, Attendance, Leave, Shifts, Performance. Linked login yahan nazar aata hai.

> **TIP**
>
> Kisi ka manager (Reports to) set karein to woh **HR → Org chart** mein sahi jagah aata hai. Employee chhore to **Terminate** use karein — record delete na karein.

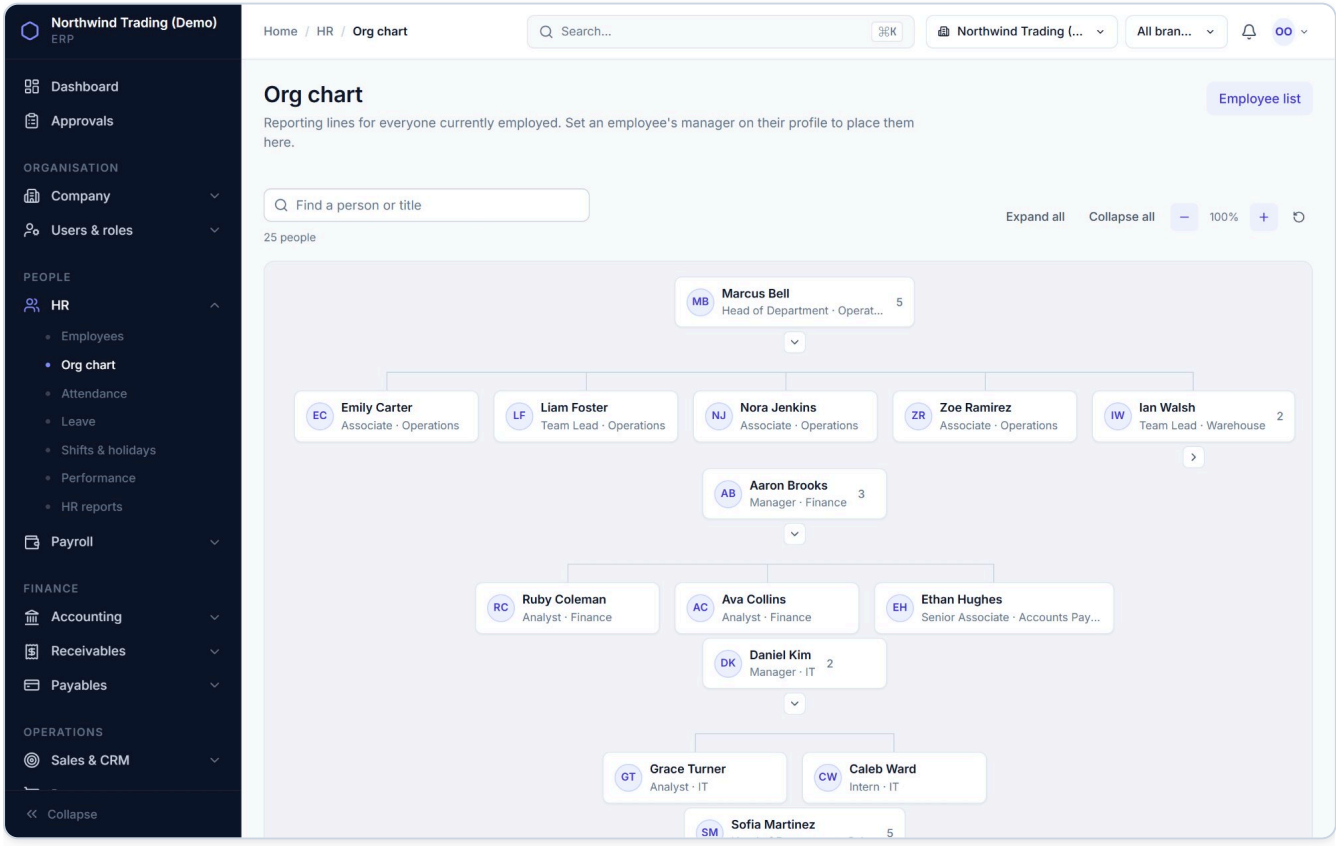

Figure 7.10 — Org chart — reporting lines; Expand all / Collapse all.

Attendance

HR → Attendance mein teen tabs hain: **Records** (rozana check-ins aur manual records), **Bulk entry** (kai employees ki attendance aik sath) aur **Summary** (period ka present/late/absent khulasa). **Record attendance** se HR manual record daal sakta hai. Employees khud **Check in** / **Check out** karte hain (Chapter 8).

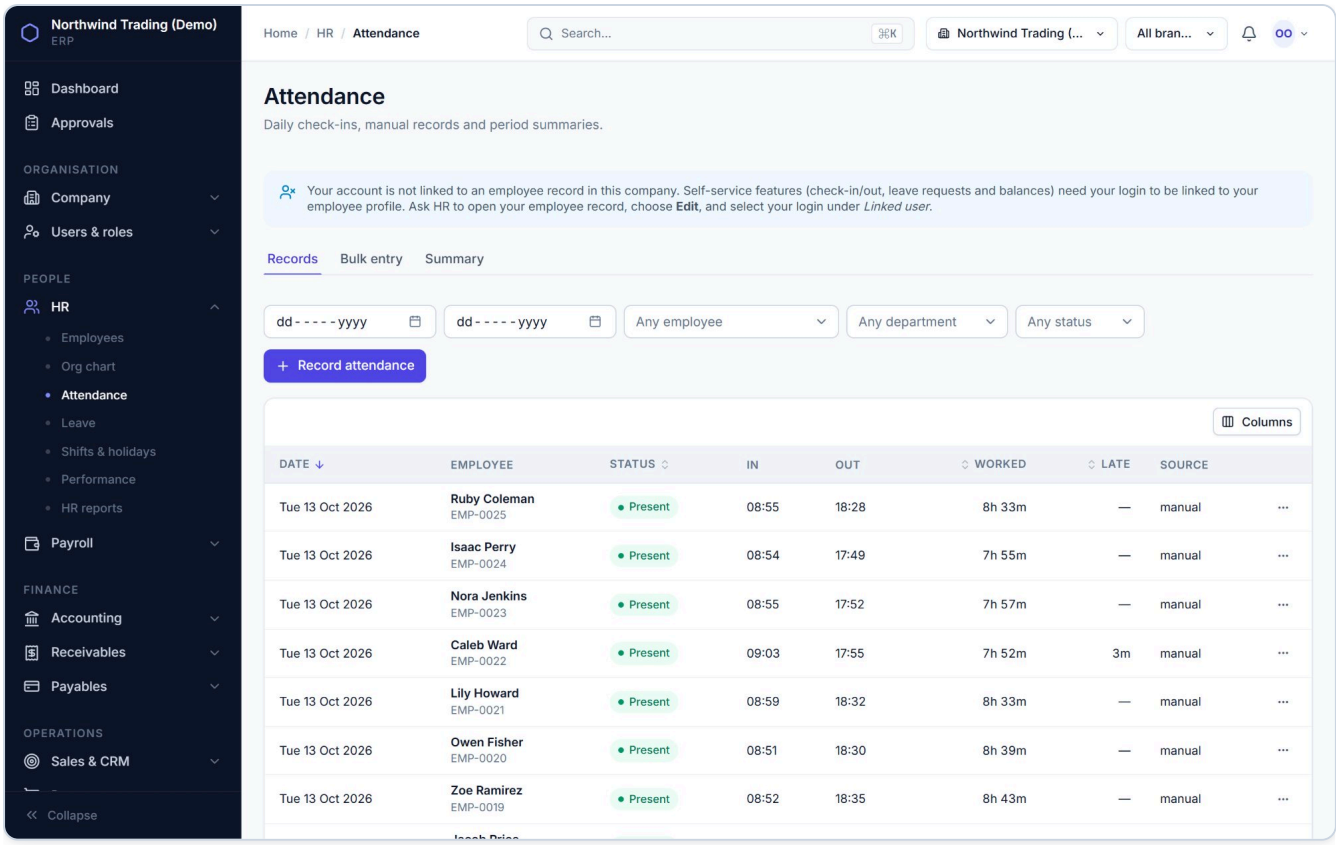

Figure 7.11 — Attendance records — status, worked time, late minutes. Owner ka login employee record se linked nahi, is liye upar self-service ka note aa raha hai.

Leave

HR → Leave tabs: **My leave**, **Approvals** (ginti ke sath), **All requests**, **Balances**, **Leave types**.

1. **Approvals** tab kholein — pending requests ki list aati hai.
2. Employee, type, dates, days aur reason dekhein.
3. **Approve** ya **Reject** dabayein. Employee ko notification chali jati hai aur balance update hota hai.

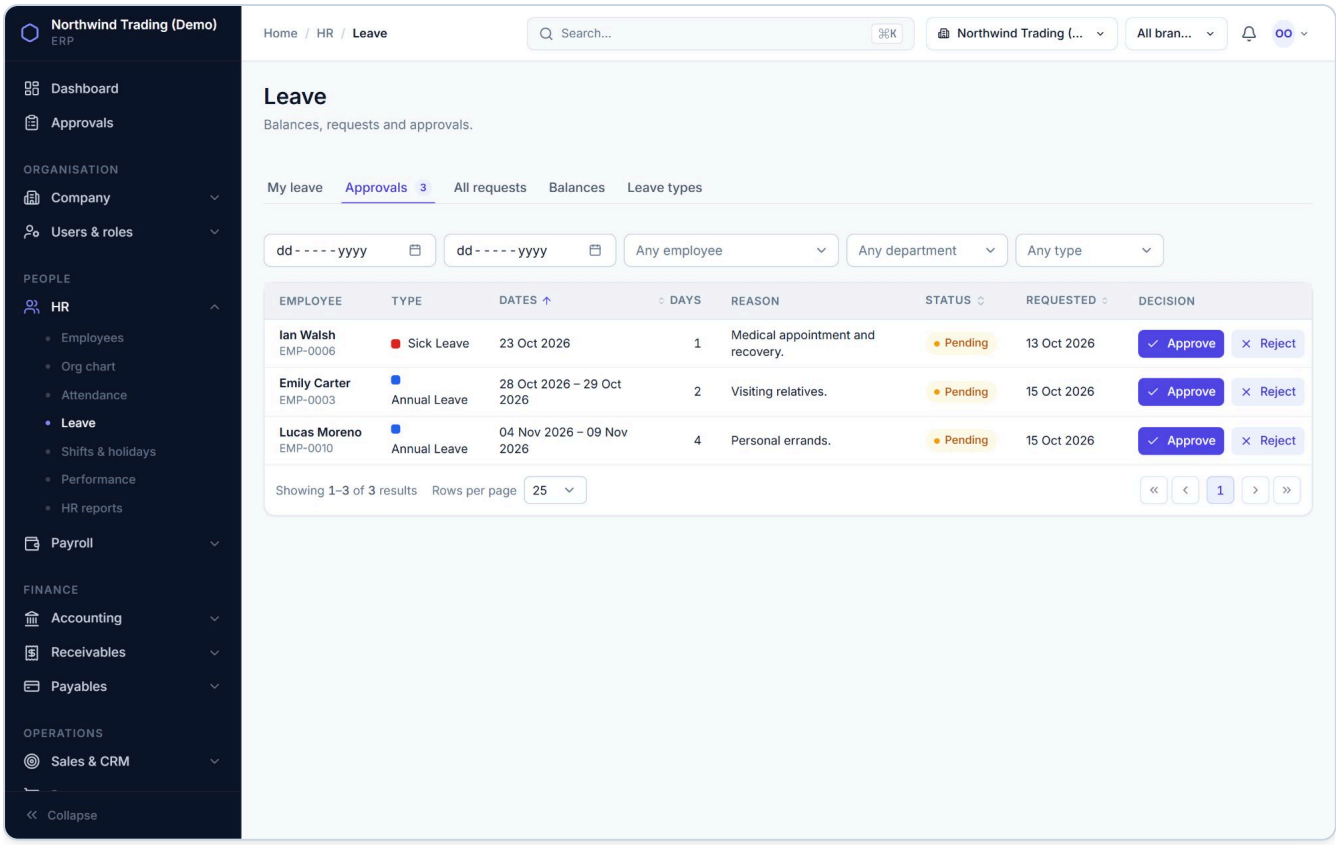

Figure 7.12 — Leave → Approvals — har request par Approve / Reject.

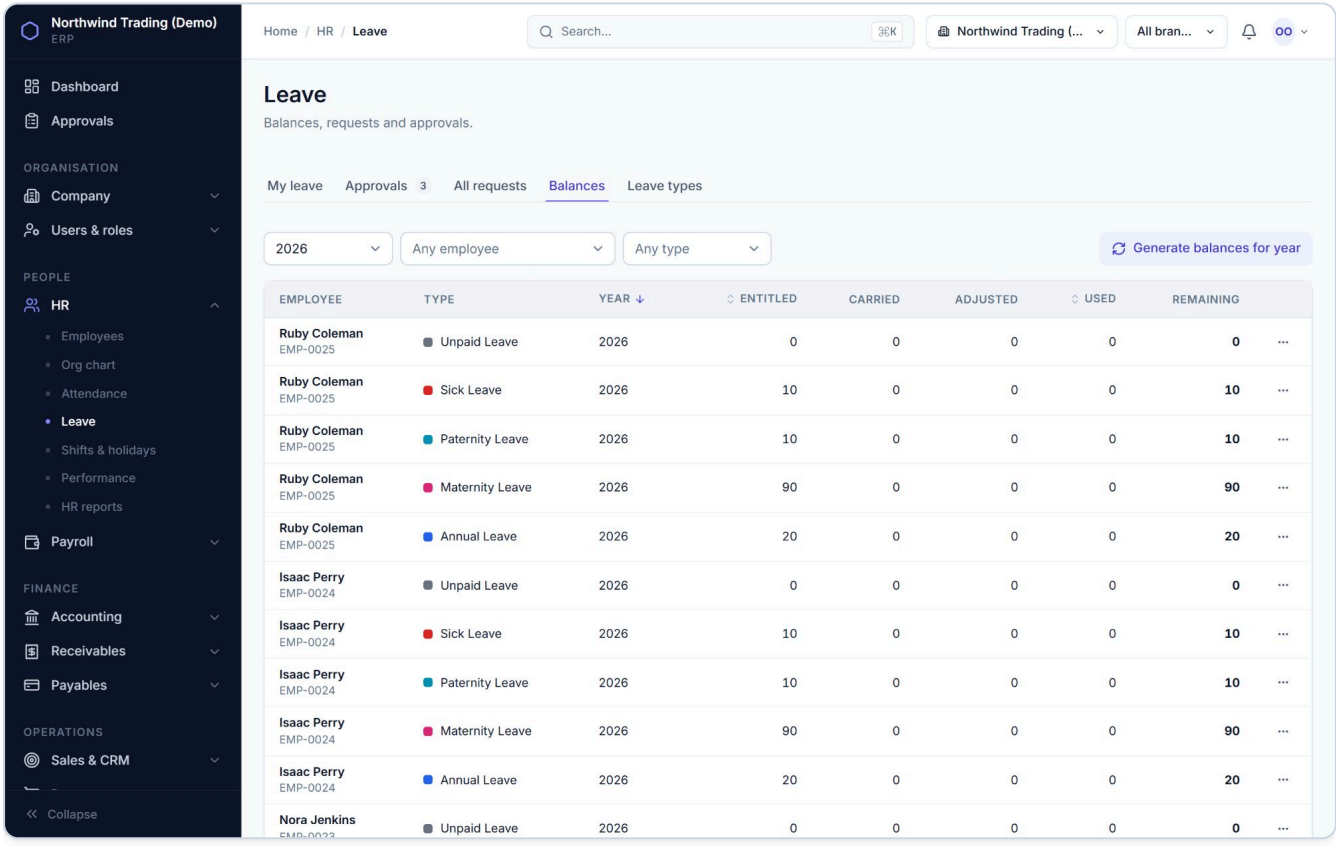

Figure 7.13 — Leave → Balances — har employee ka leave type-wise balance.

Shifts, Performance, HR reports

- **HR → Shifts & holidays**: working hours/days (shifts) late aur worked-time ka hisaab chalate hain; public holidays leave ke dinon mein shumar nahi hotin. Shift employee profile se assign hoti hai.
- **HR → Performance**: **New review** se review cycle; status Draft → Submitted → Acknowledged.
- **HR → HR reports**: **Headcount**, **Attendance**, **Leave** — joiners, leavers, department/branch-wise charts.

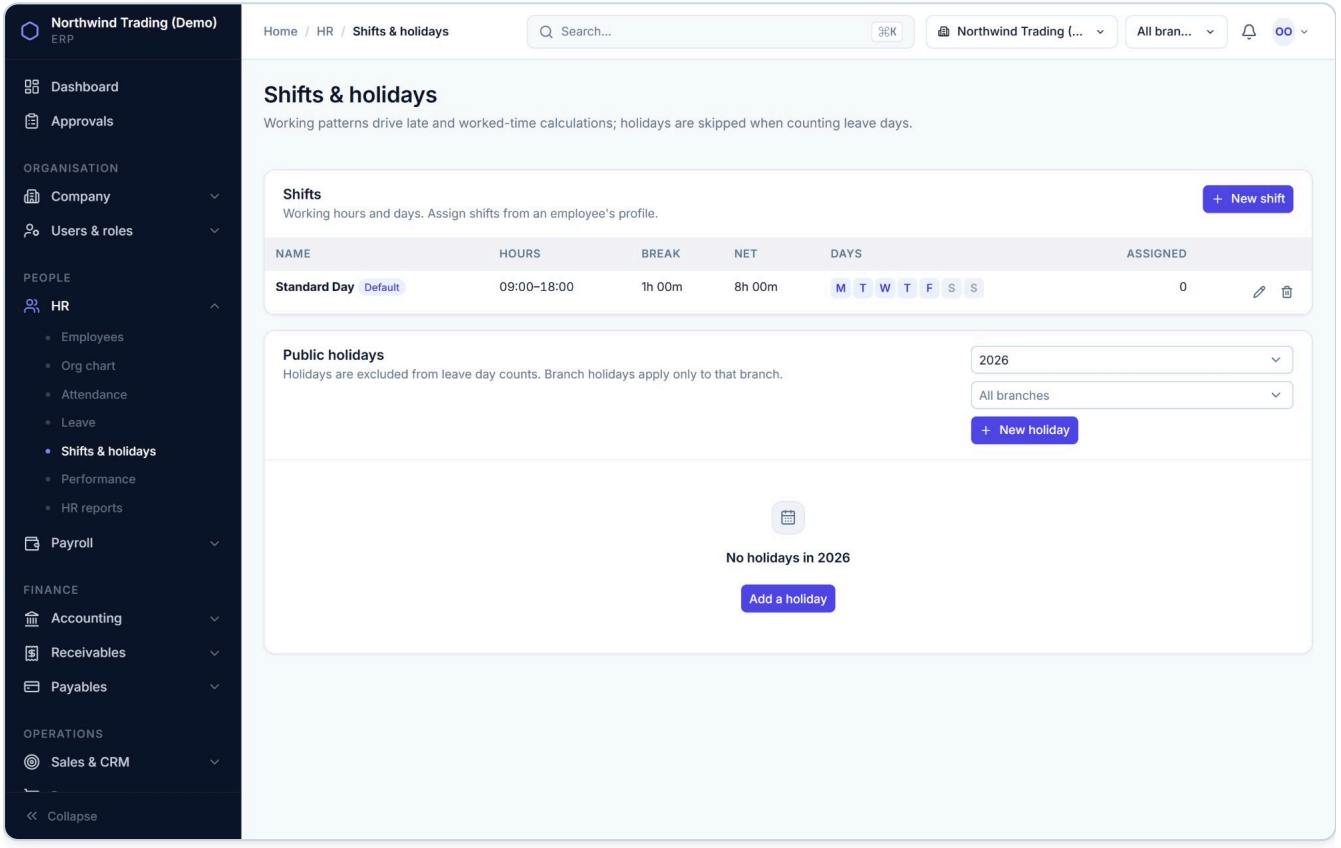

Figure 7.14 — Shifts & holidays — Standard Day shift (09:00–18:00, 1h break) aur public holidays.

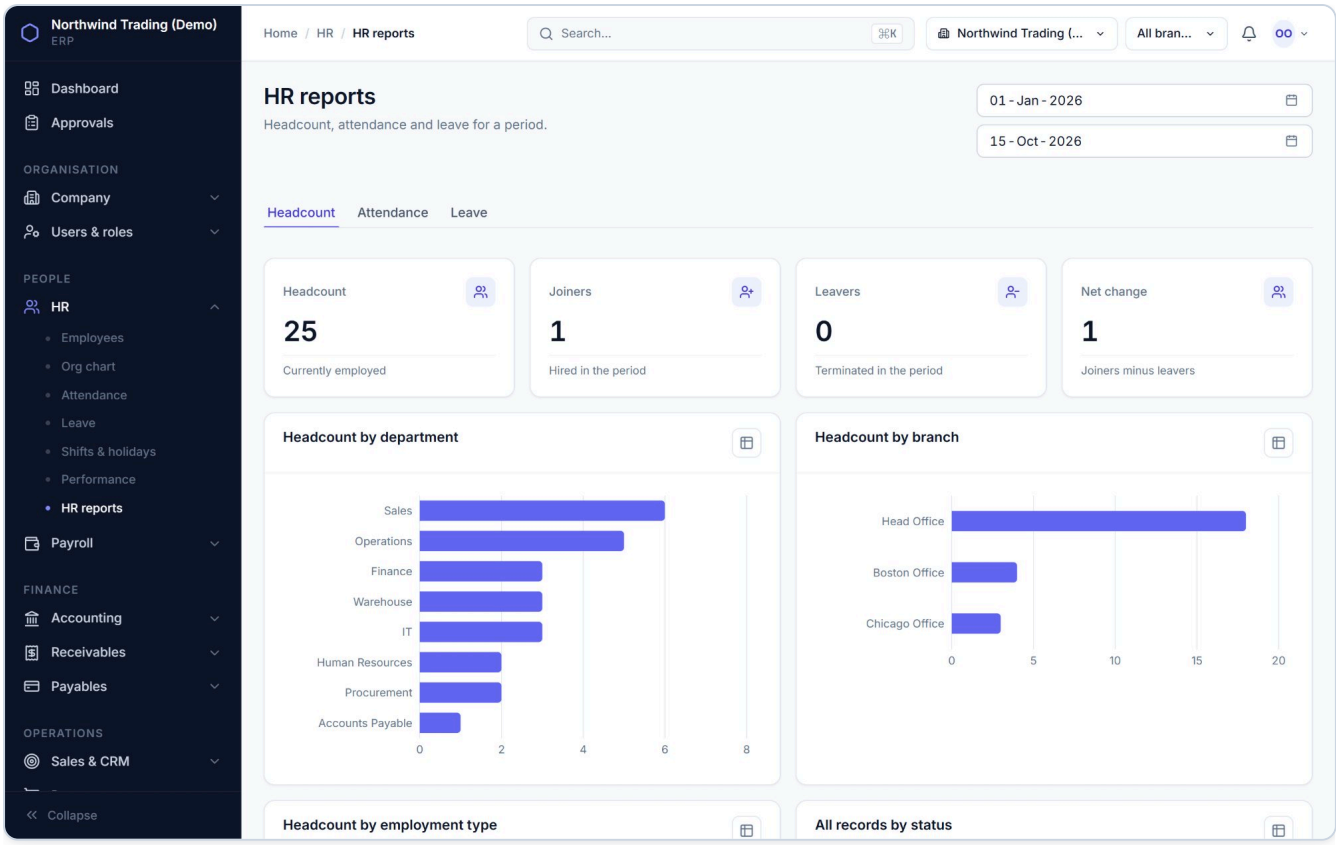

Figure 7.15 — HR reports — headcount, joiners/leavers aur department/branch charts.

7.6 Payroll

Payroll ka har mahine ka run is order mein chalta hai — har qadam ka record aur waqt run page par stepper mein nazar aata hai:

Draft → Calculated → Approved → Finalized → Paid

Table 7.2 — Payroll run ke actions

Qadam	Kaun / kya hota hai
New run	Period, pay date aur scope (company-wide / branch / department) chun kar draft run banta hai.
Calculate	Payslips current salaries, attendance/unpaid leave, pay components aur active tax table se bante hain. Status Calculated. Zaroorat ho to Recalculate .
Approve	**Kisi doosre** shakhs ka kaam jis ke paas "Run & approve payroll" ho. Run banane wala approve nahi kar sakta.
Finalize	Ek payroll journal entry general ledger mein post hoti hai; run number type karke confirm karna parta hai. "This cannot be undone." Payslips employees ko dikhne lagti hain.
Mark paid	Salary ki adaigi record hoti hai (payment journal).
Cancel run	Draft/Calculated/Approved run cancel ho sakta hai; record audit ke liye rehta hai, kuch post nahi hota.

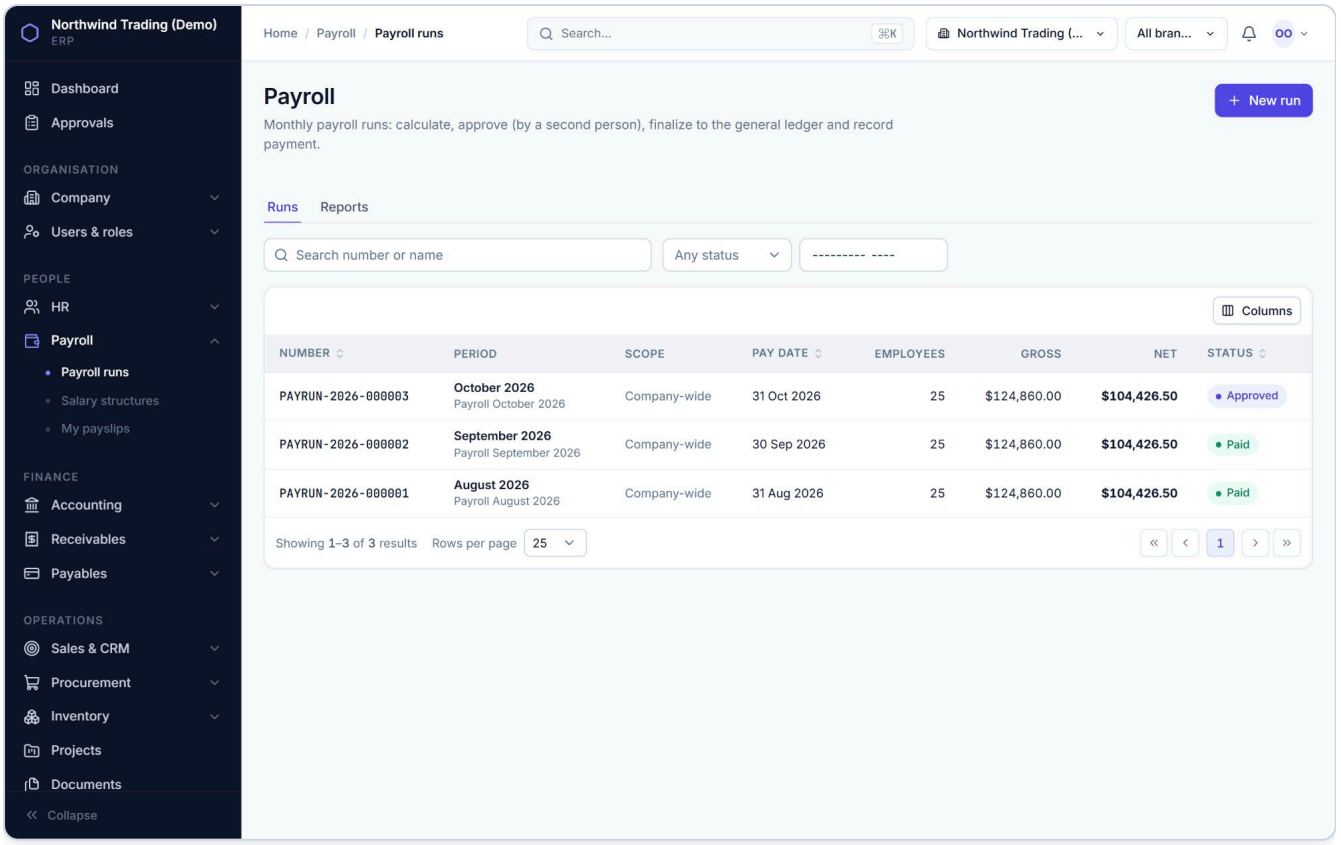

Figure 7.16 — Payroll runs — har run ka period, scope, pay date, employees, gross, net aur status.

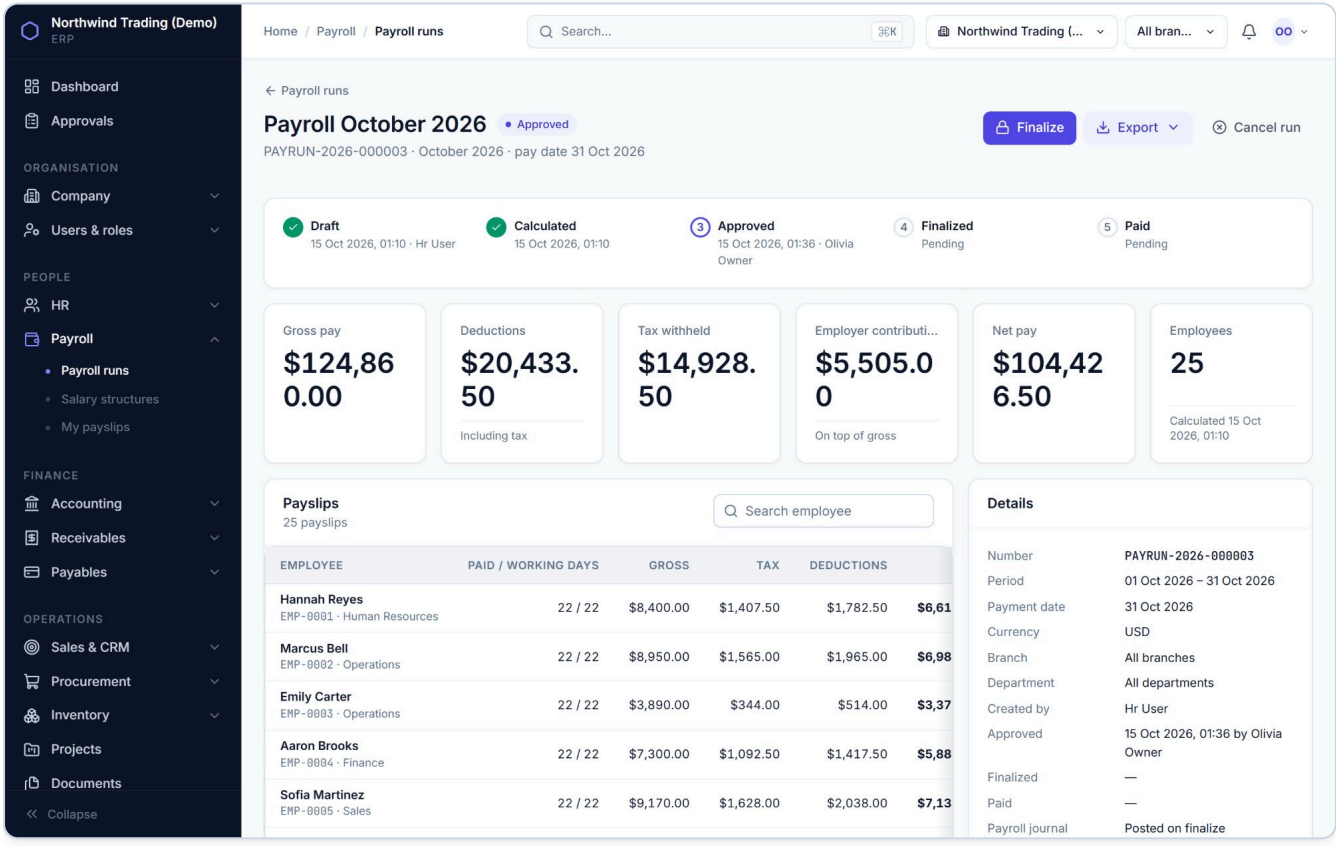

Figure 7.17 — Approved run (October 2026) — stepper, gross/deductions/tax/net cards; agla qadam Finalize.

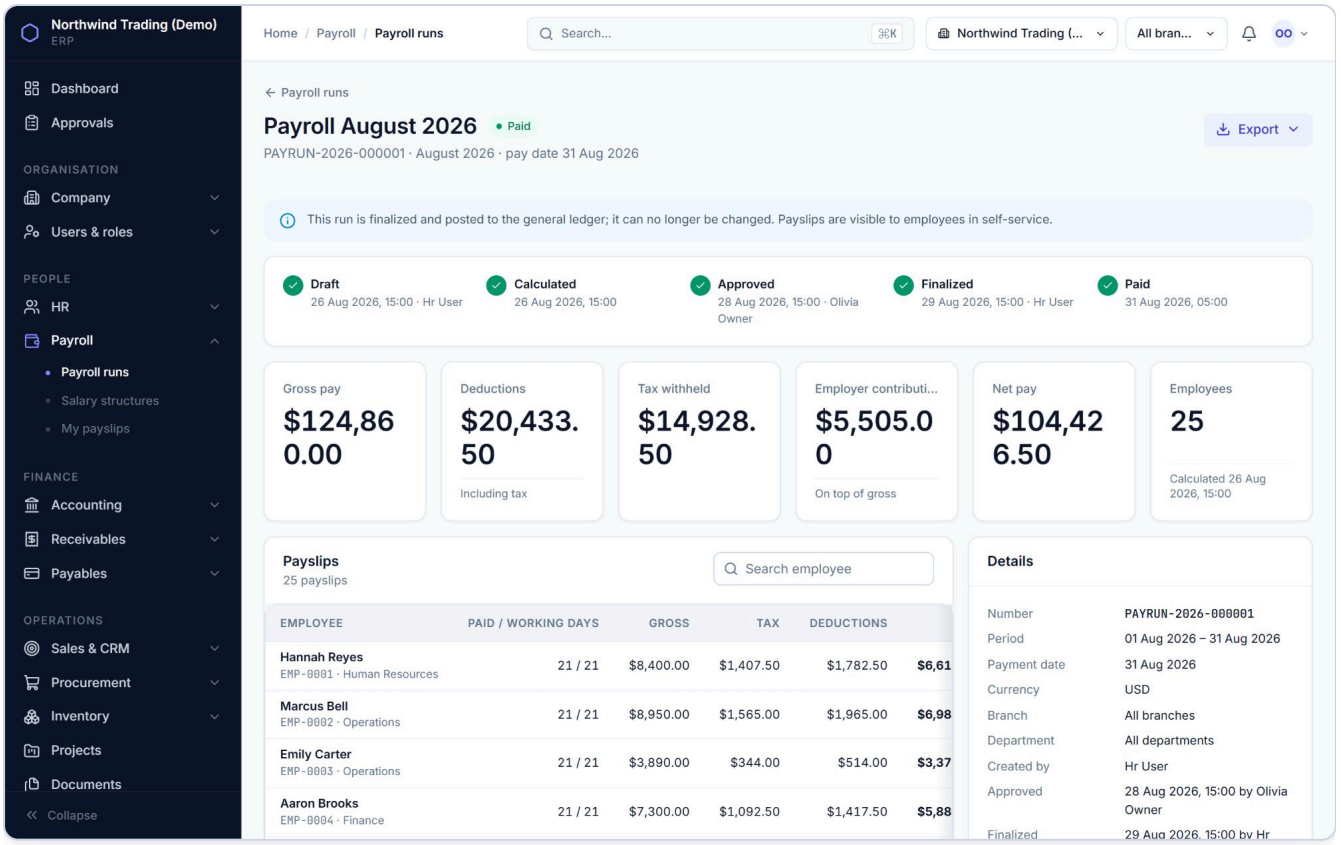

Figure 7.18 — Paid run — "This run is finalized and posted to the general ledger; it can no longer be changed."

> ⚠ **DHYAN DEIN**
>
> Finalize se pehle payslips achhi tarah check karein — finalize ke baad run badla nahi ja sakta. Ghalti ho to accounts team journal ke zariye correction karegi.

Payroll setup

Payroll → Salary structures (page ka naam "Payroll setup") mein chaar tabs: Pay components (allowances, deductions, employer contributions — fixed ya % of base), Salary structures, Tax tables aur Employee salaries.

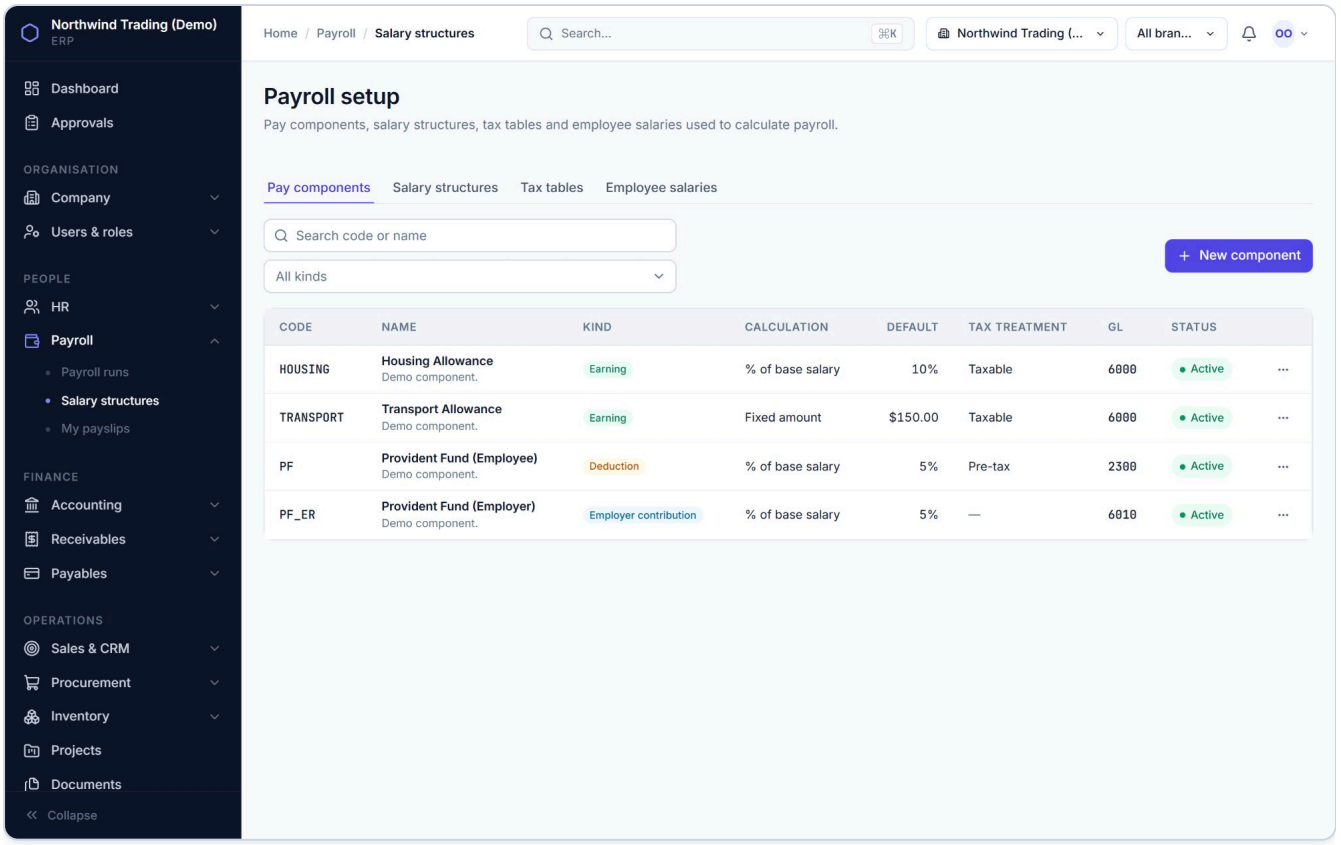

Figure 7.19 — Pay components — kind, calculation, default, tax treatment aur GL account.

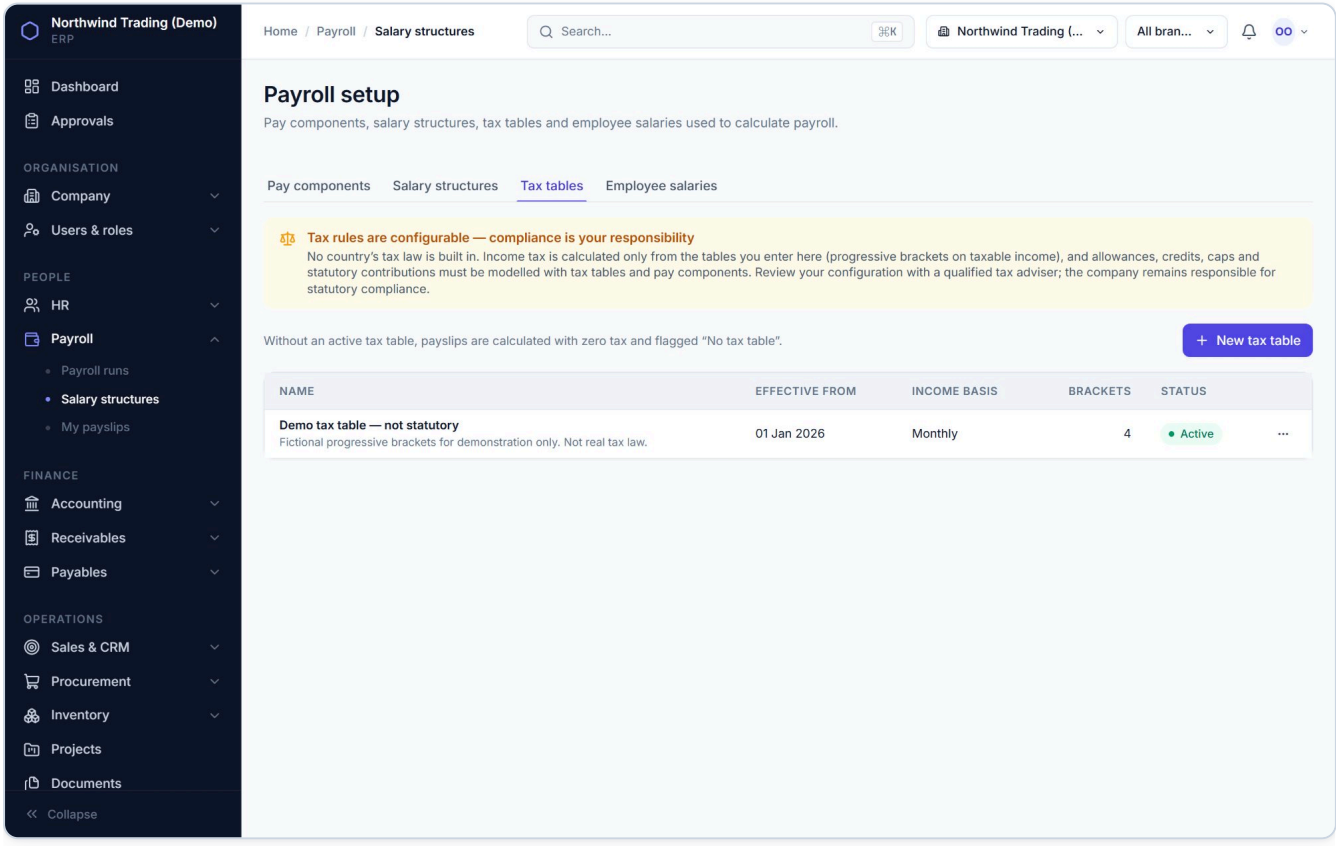

Figure 7.20 — Tax tables — "Tax rules are configurable — compliance is your responsibility." Active tax table na ho to tax zero aur payslip par "No tax table" flag.

> ⚠ **DHYAN DEIN**
>
> System mein kisi mulk ka tax law built-in **nahi**. Tax brackets aur contributions khud enter karein aur kisi qualified tax adviser se review karwayein.

7.7 Finance / Accounting

Chart of accounts

Accounting → Chart of accounts — har ledger account (code, naam, type: Asset/Liability/Equity/Revenue/Expense). **New account** se naya account. Nayi company ko default chart khud milta hai.

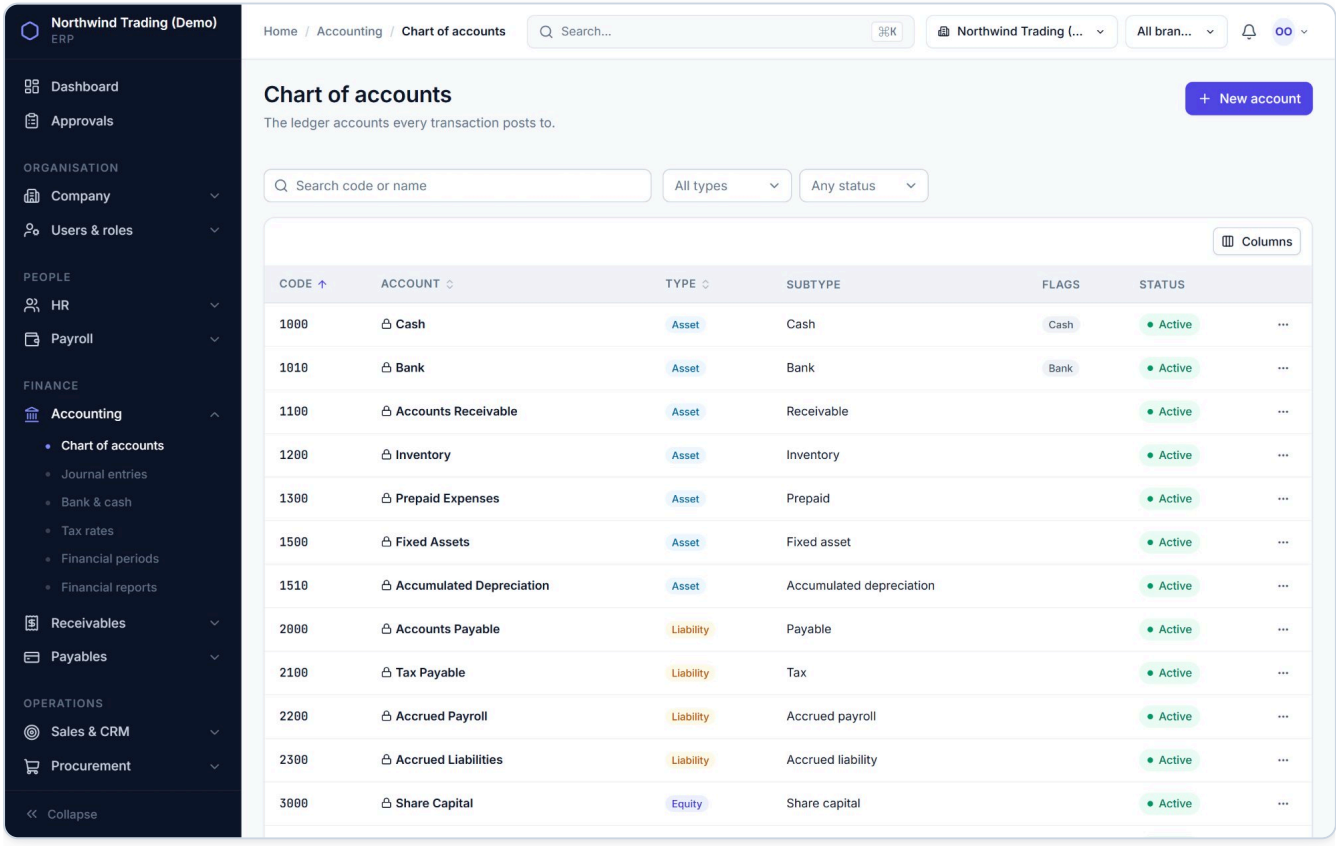

Figure 7.21 — Chart of accounts — code, type, subtype, flags (Cash, Bank, Receivable...).

Journal entry banana (double-entry)

1. **Accounting → Journal entries** → **New entry**.
2. **Date**, **Reference** aur **Memo** bharein.
3. Har line mein account chunein, description likhein, aur **ya** Debit **ya** Credit amount (aik line mein dono nahi).
4. Zaroorat ho to **Add line**. Neeche totals barabar hon to **Balanced** badge aata hai — warna save button band rehta hai.
5. **Save draft** — entry draft ban jati hai.

6 Draft khol kar **Post** → "Posting writes to the general ledger. Posted entries cannot be edited — only reversed."

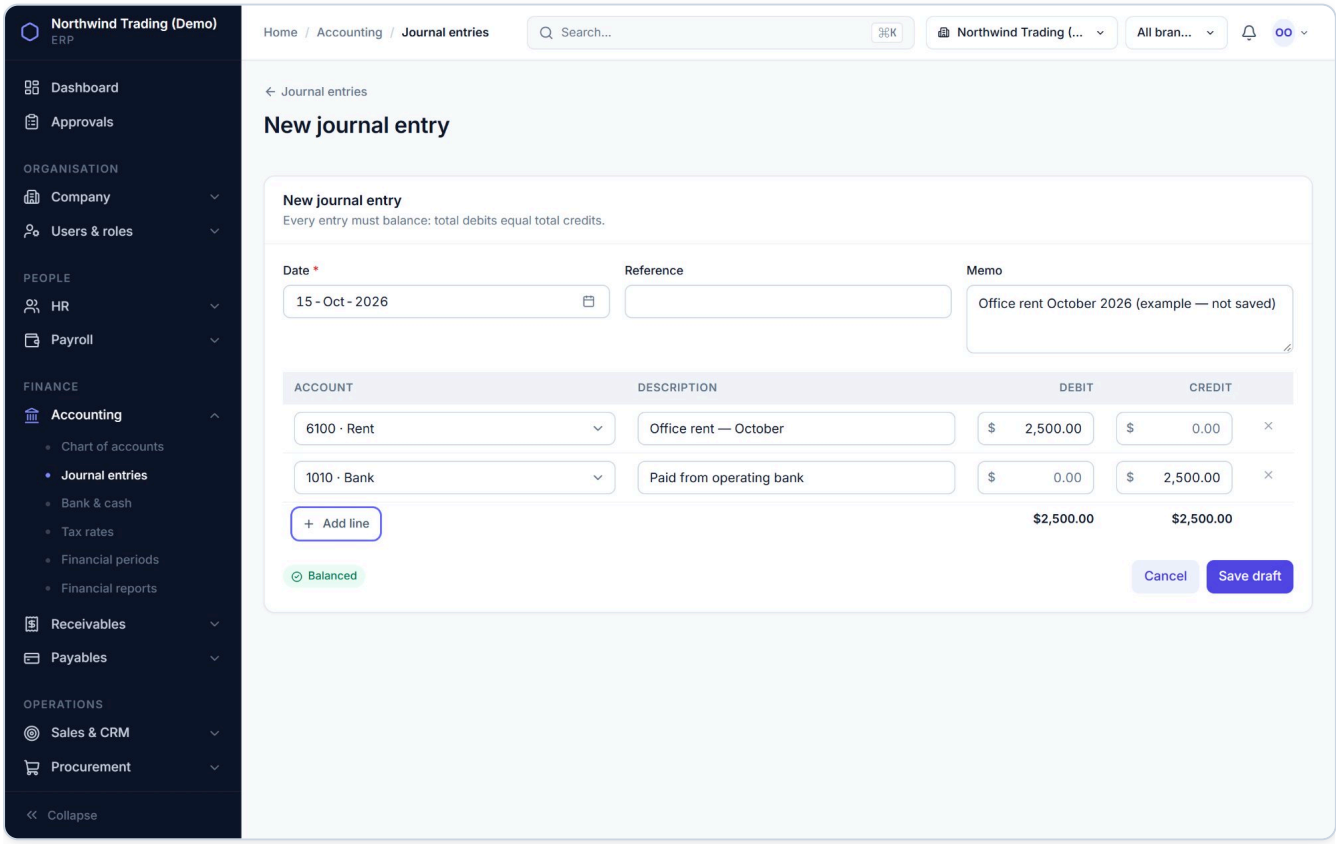

Figure 7.22 — New journal entry — Rent (6100) debit 2,500 aur Bank (1010) credit 2,500; "Balanced" badge. (Example — save nahi kiya gaya.)

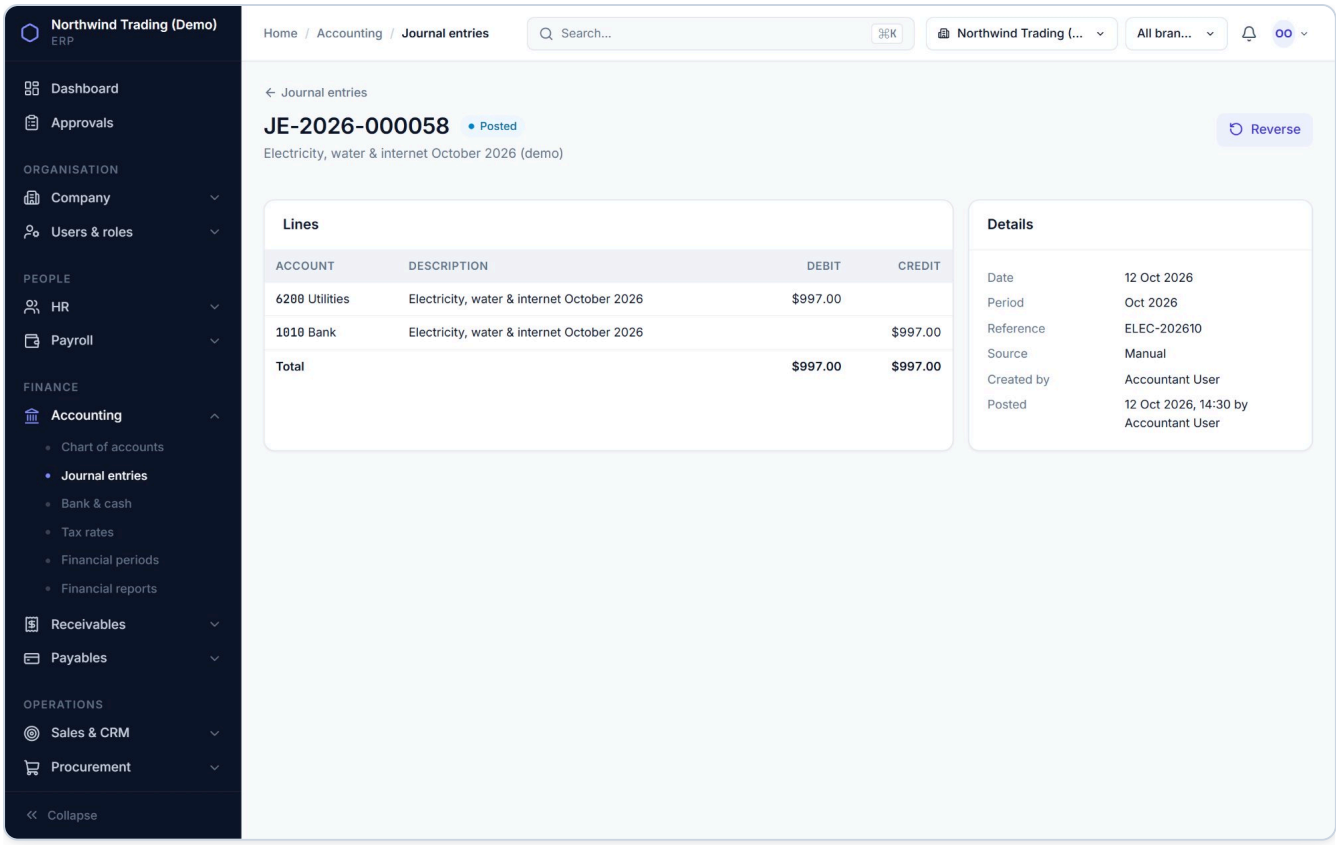

Figure 7.23 — Posted journal entry — lines, totals, period, source, kis ne banaya aur post kiya; Reverse button.

> ⚠ **DHYAN DEIN**
>
> **Posted entry edit nahi hoti.** Ghalti ho to entry khol kar **Reverse** karein — system ulti (reversing) entry post karta hai, phir sahi entry dobara banayein. Invoices, bills, payments, payroll aur stock adjustments apni journal entries khud banate hain (Source column).

Financial periods, bank, tax

- **Accounting → Financial periods**: "Postings are only allowed into open periods." Books reconcile hone ke baad periods **order mein** **Close** karein; **Add next financial year** se agla saal. Closed period mein posting reject hoti hai.
- **Accounting → Bank & cash**: bank/cash accounts — account numbers masked dikhte hain.
- **Accounting → Tax rates**: sales/purchase tax rates (maslan Standard 15%, Zero 0%) jo invoice/bill lines par lagte hain.

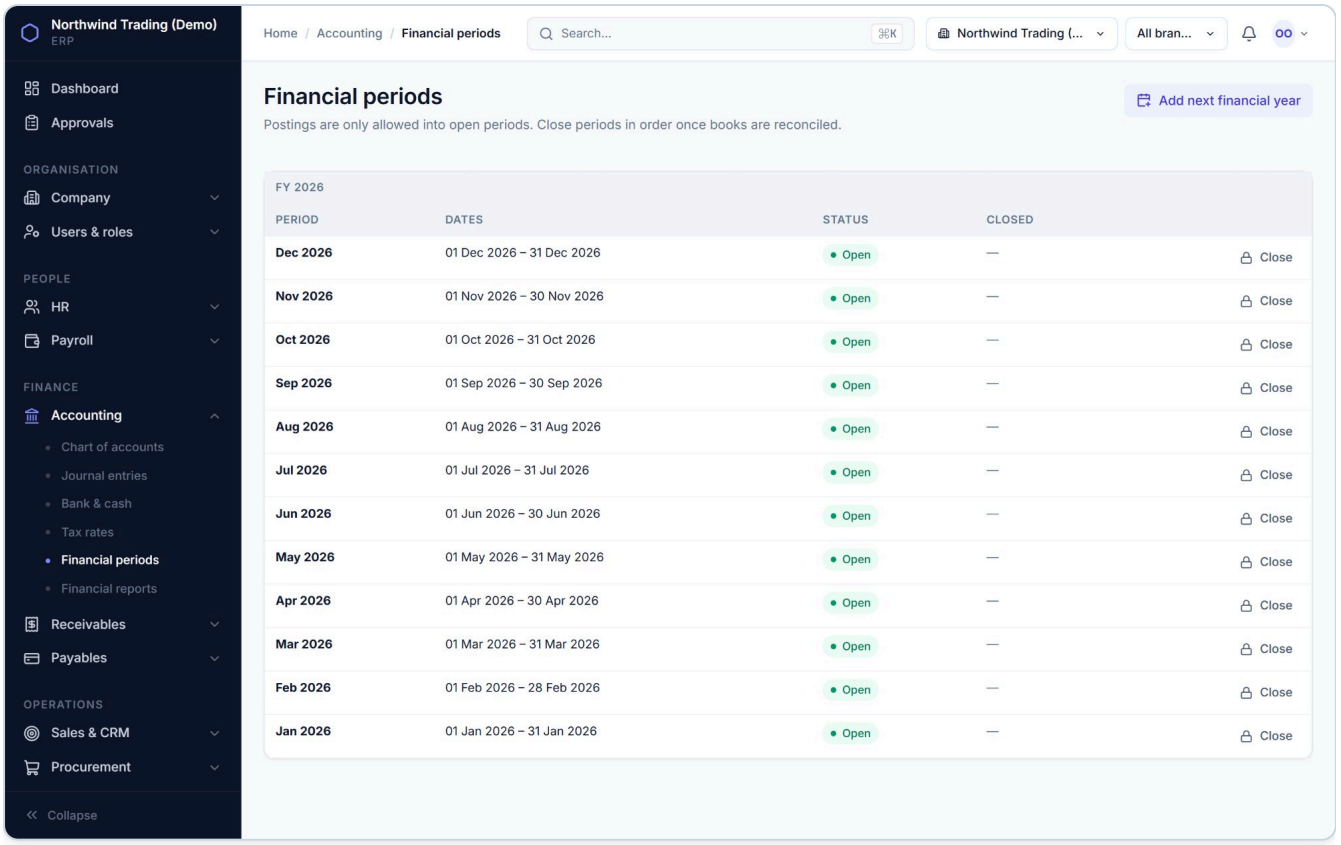

Figure 7.24 — Financial periods — har mahina Open/Closed; Close button.

Financial statements

Profit & loss, Balance sheet aur Trial balance **Insights → Reports & exports** se chalayein (7.13) — wahan se CSV/Excel/PDF export bhi hota hai. Yeh sab posted general-ledger entries se bante hain.

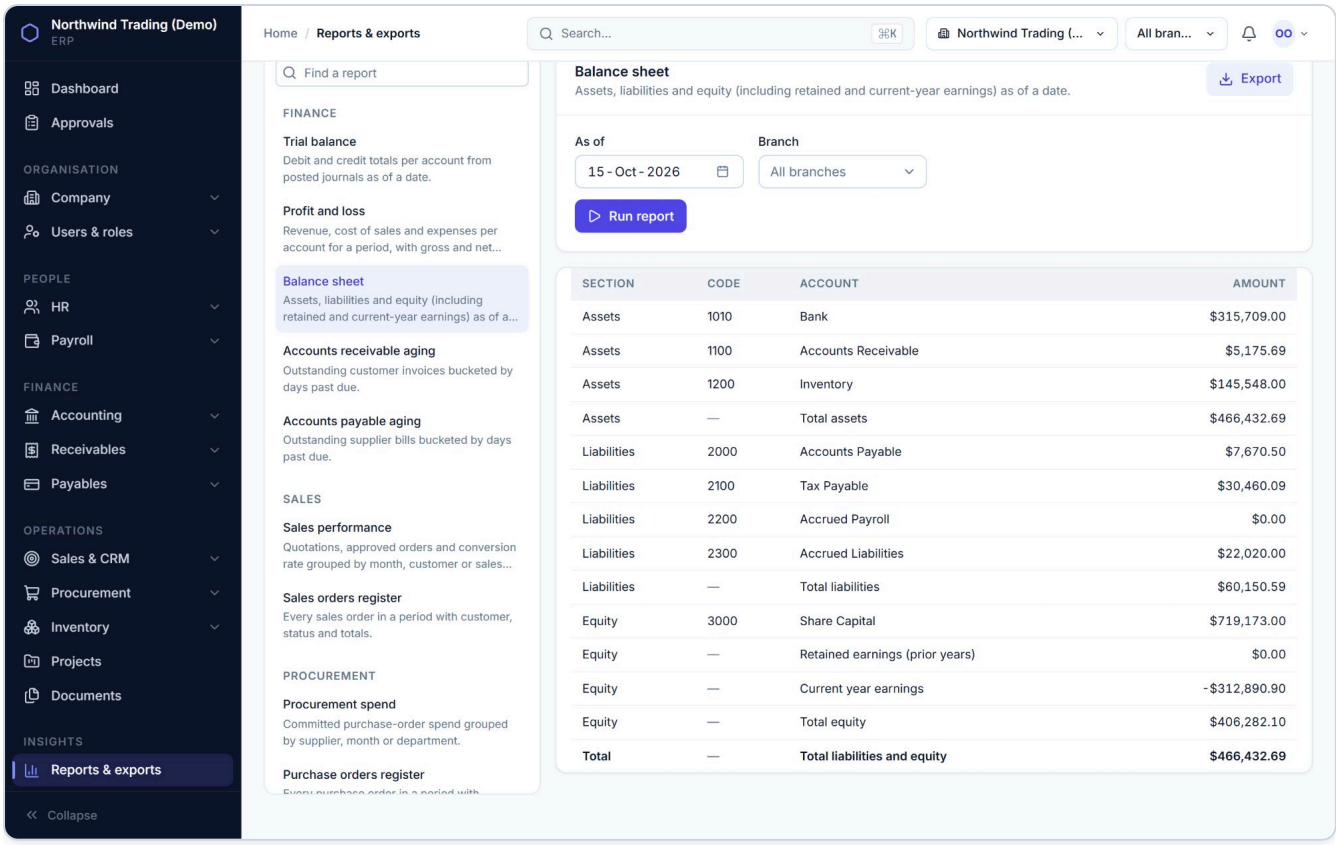

Figure 7.25 — Reports & exports → Balance sheet — assets, liabilities aur equity, as of date.

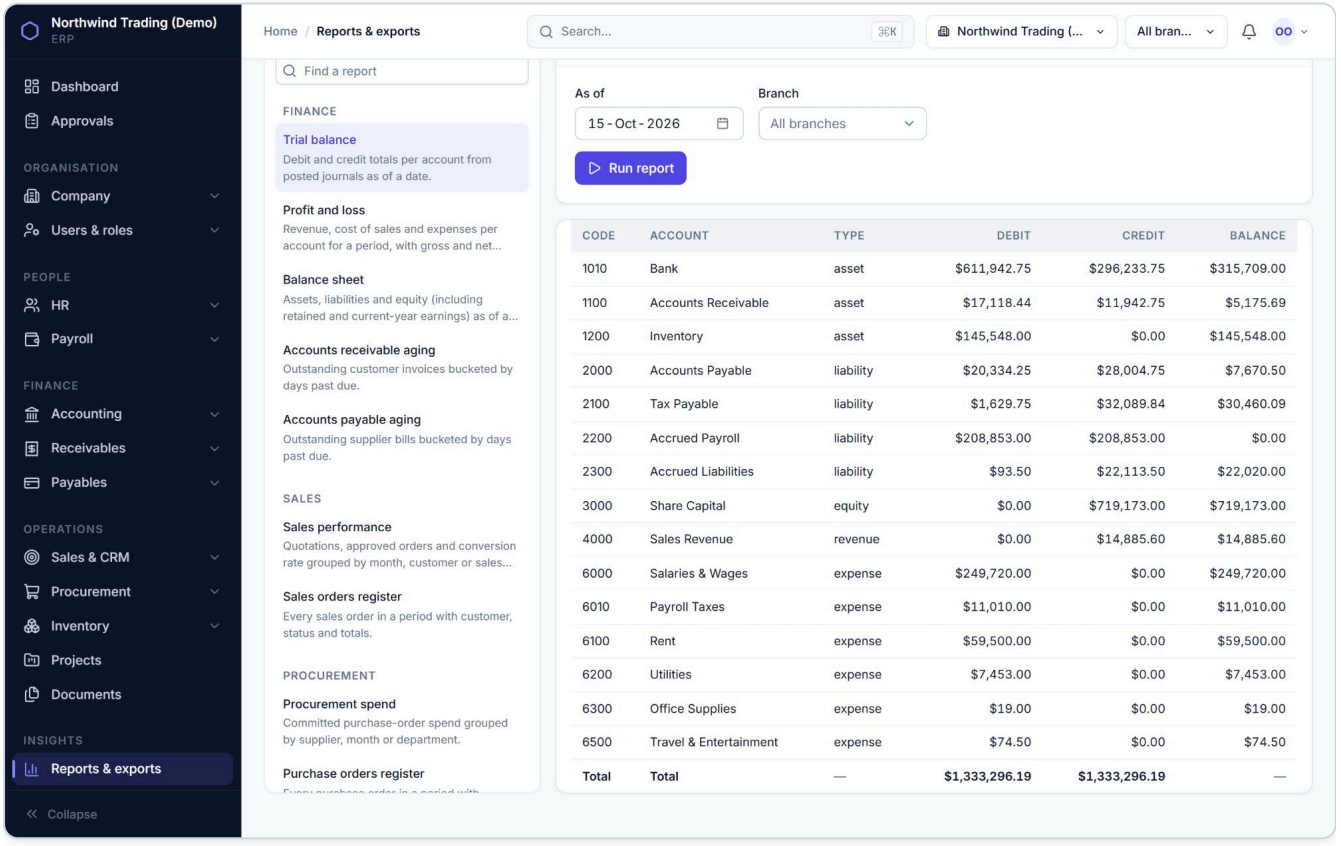

Figure 7.26 — Trial balance — har account ke debit/credit totals; dono columns barabar hone chahiye.

> **NOTE**
>
> **Accounting** → **Financial reports** page par bhi **Profit & loss**, **Balance sheet**, **Trial balance** aur **Cash flow** tabs hain, **Compare** switch (pichle period se muqabla) aur **Print** button ke sath.

7.8 Receivables & Payables

Customer invoice ka safar

Draft → **Pending approval** → **Approved** → **Partially paid** → **Paid**

1. **Receivables** → **Customer invoices** → **New invoice**: customer, dates, lines (product/description, account, qty, price, discount, tax).
2. Draft save hota hai. **Submit** se approval ke liye bhejein ("It stays editable until an approver posts it.").
3. Approver **Approve** karta hai: "Approving posts … to the general ledger (Dr receivables / Cr revenue & tax). It can then only be voided, not edited."
4. Customer paisa de to **Record receipt** — drawer mein bank account, date, method, amount aur allocation.
5. **PDF** button se invoice ka PDF.

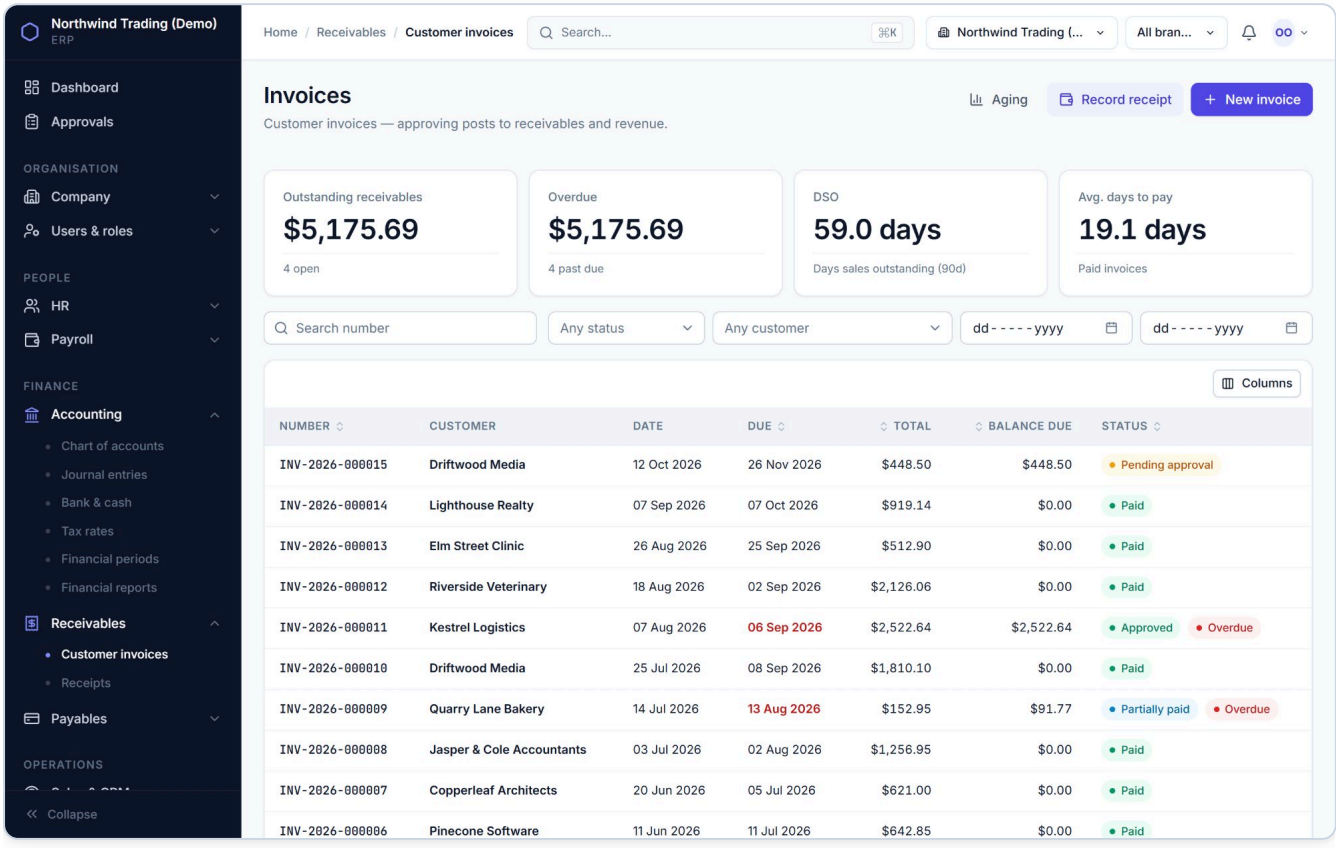

Figure 7.27 — Invoices list — outstanding receivables, overdue, DSO aur average days to pay cards.

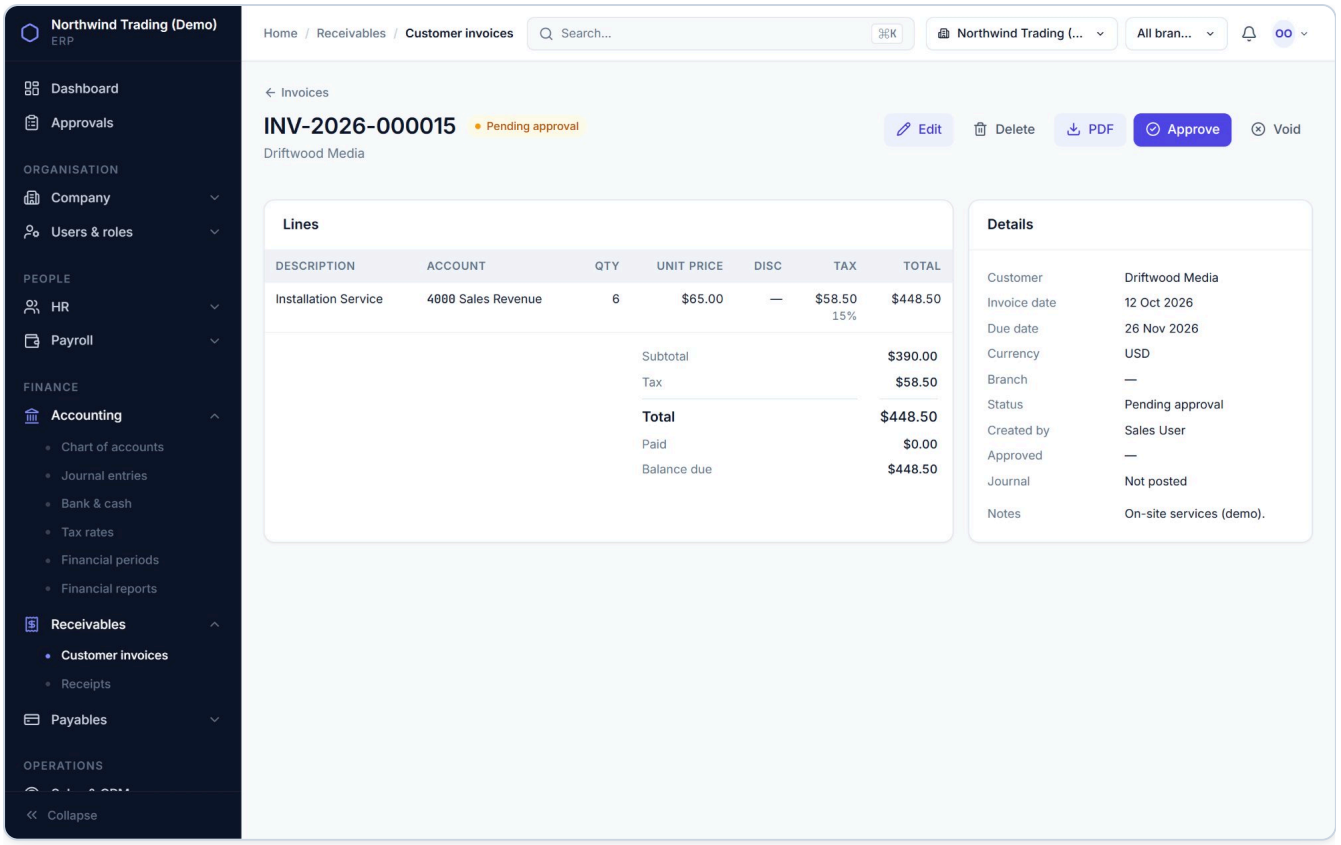

Figure 7.28 — Pending approval invoice — Edit, Delete, PDF, Approve, Void. "Journal: Not posted".

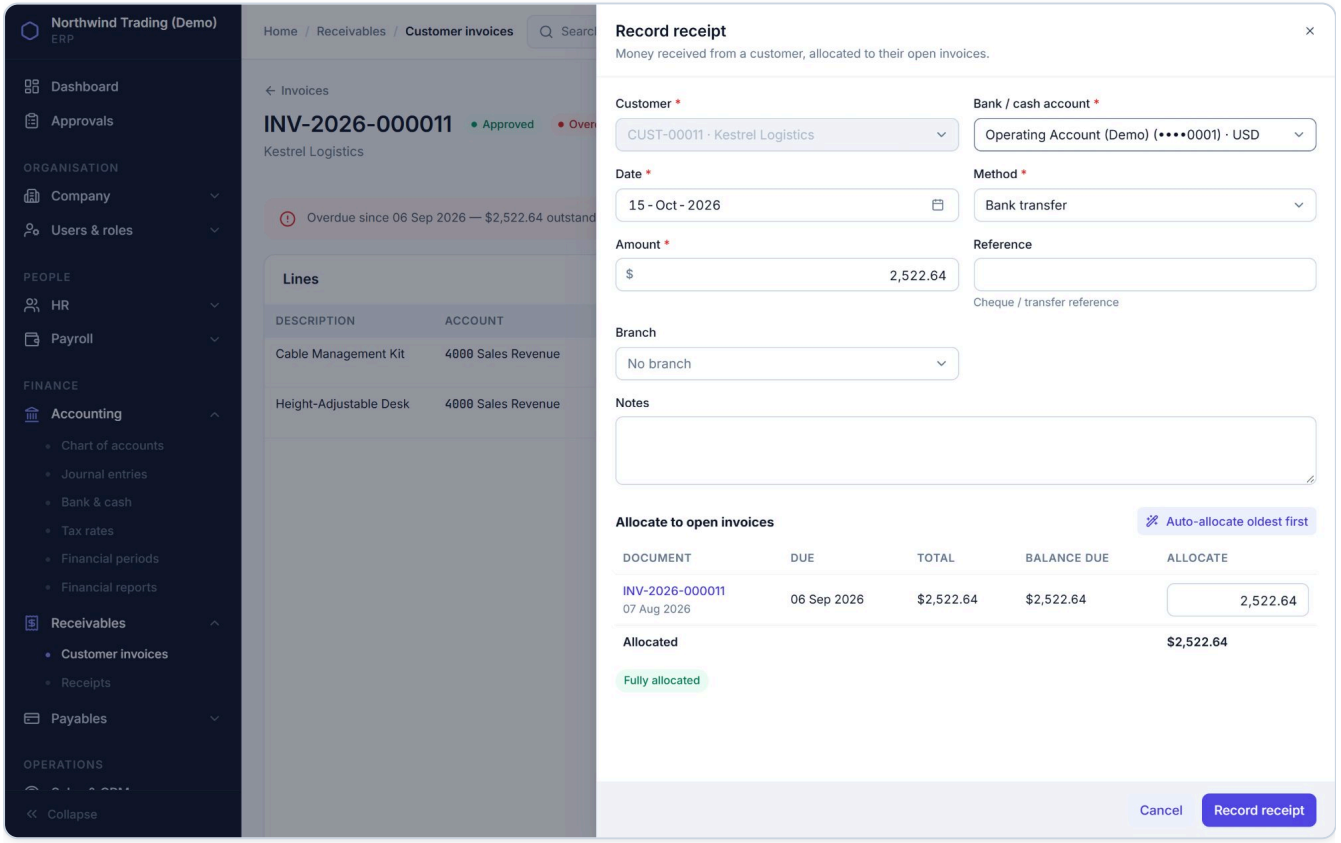

Figure 7.29 — Record receipt drawer — customer, bank/cash account, method, amount; "Allocate to open invoices" aur Auto-allocate oldest first. (Submit nahi kiya gaya.)

> ⚠ **DHYAN DEIN**
>
> **Void ka qaida:** jis invoice/bill par payments allocate hon, woh seedha void nahi hoti — "has payments allocated; void those payments first." Pehle **Receivables → Receipts** (ya Payments) mein payment void karein, phir document void karein. Approved document void karne par system reversing journal entry post karta hai.

Supplier bills, payments, expense claims

- **Payables → Supplier bills**: wohi flow — approve karne par "Dr expense & tax / Cr payables" post hota hai. **Record payment** se supplier ko adaigi.
- **Payables → Payments** aur **Receivables → Receipts**: recorded payments posted hote hain; "correct them by voiding".
- **Payables → Expense claims**: employee ke apni jeb se kharche. Approve par "Dr expenses & tax / Cr accrued liabilities" — employee ko payable.

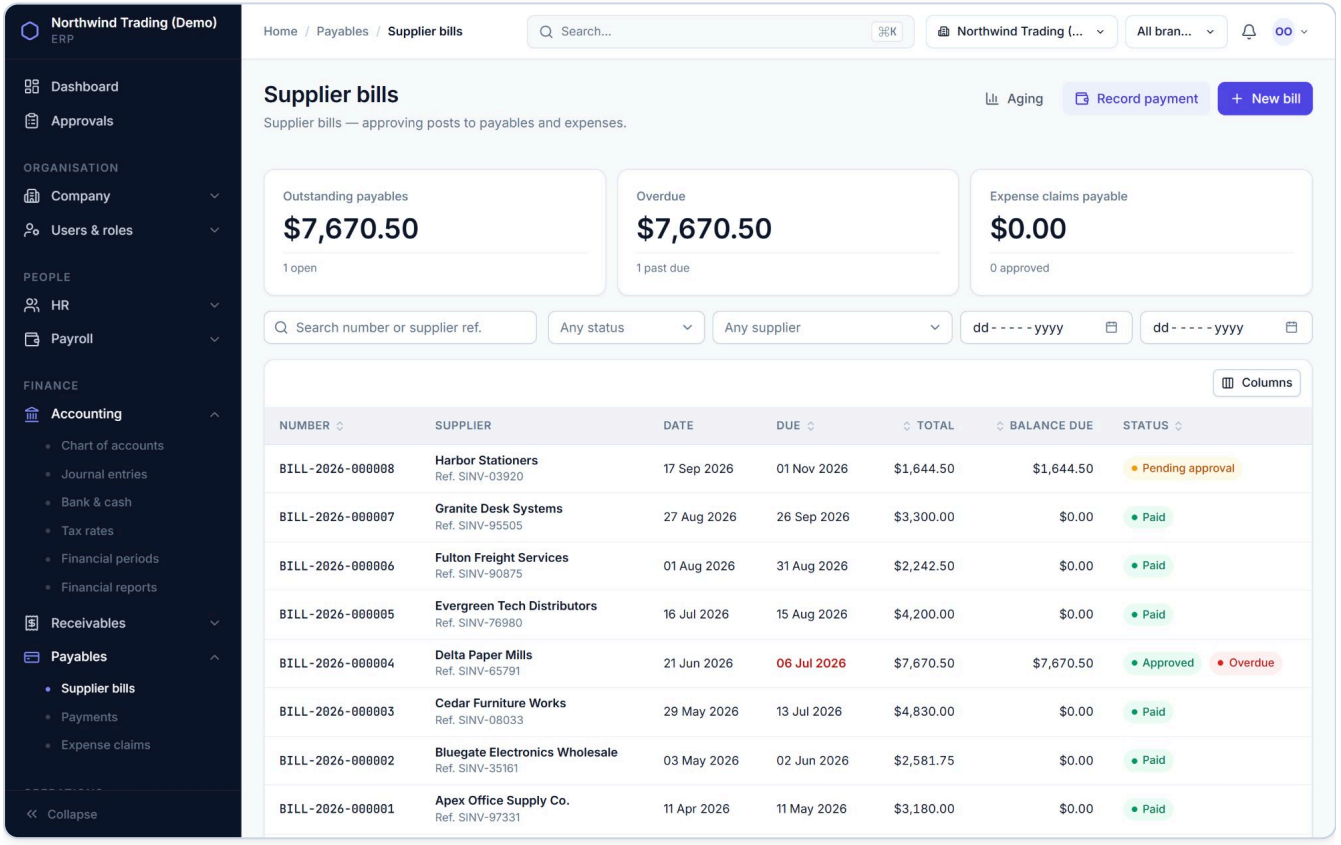

Figure 7.30 — Supplier bills — outstanding payables, overdue, expense claims payable.

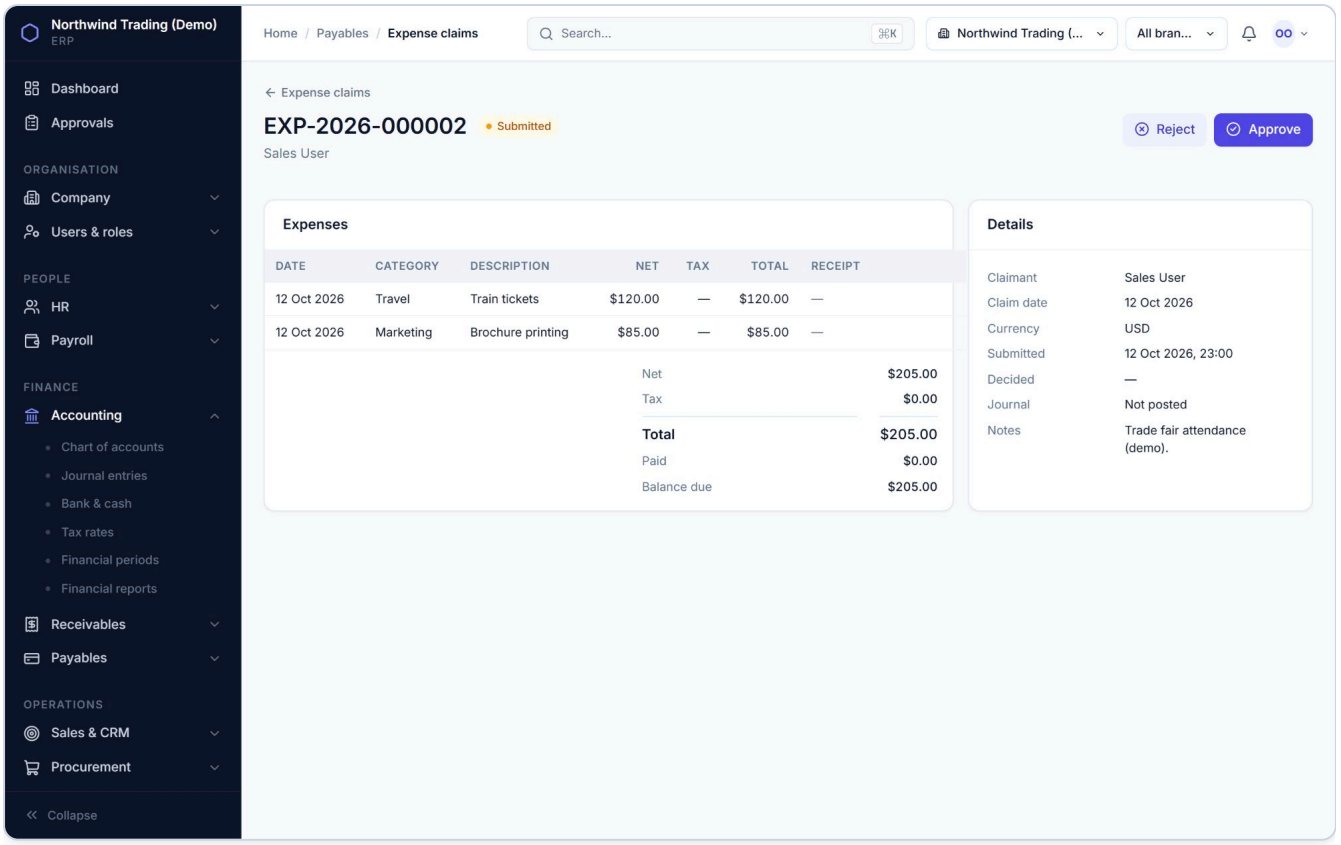

Figure 7.31 — Submitted expense claim — lines, receipts, Reject / Approve.

Aging & statements

Aging page (invoices/bills page se link) receivables/payables ko din ke hisaab se buckets mein dikhata hai: Current, 1–30, 31–60, 61–90, > 90 days, aur Customer statement tab.

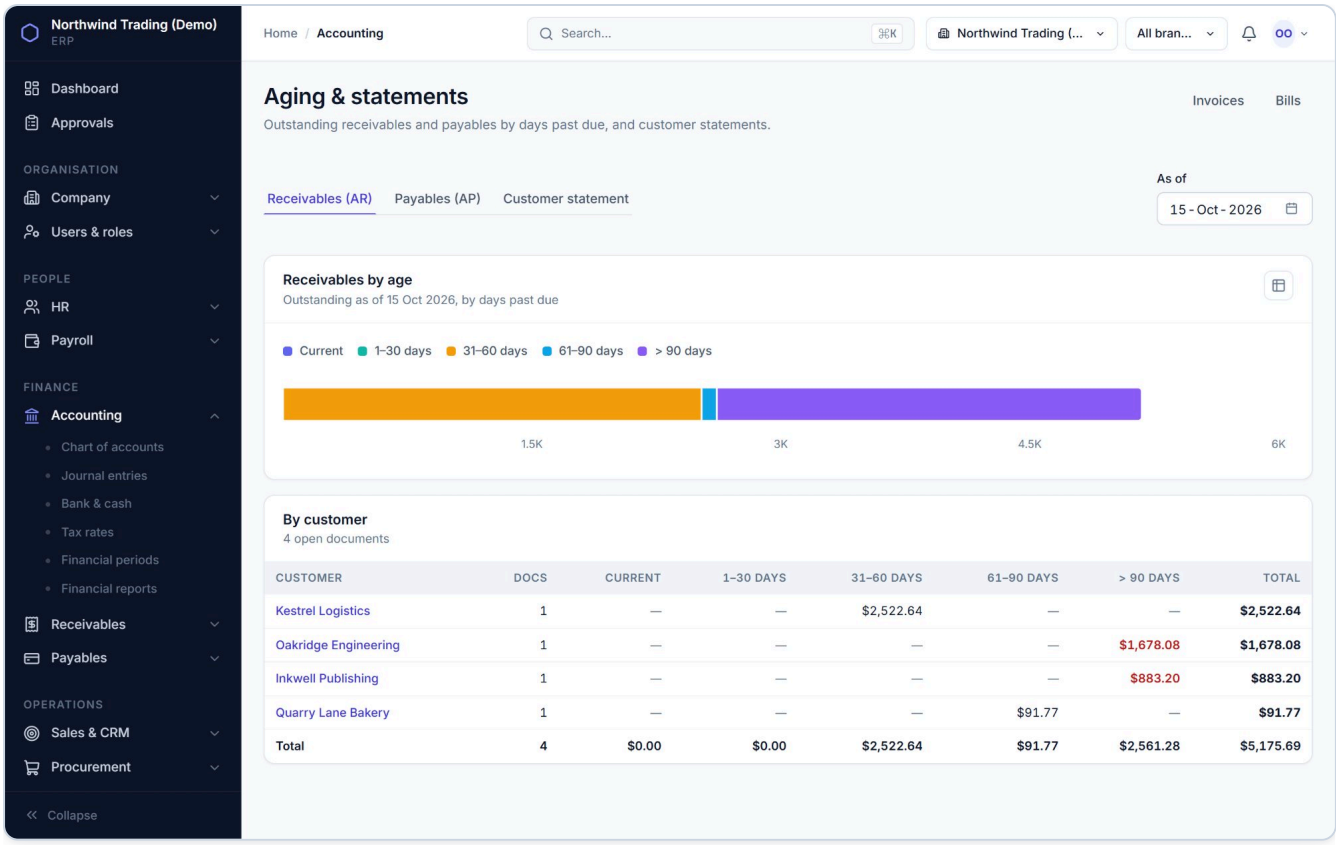

Figure 7.32 — Aging & statements — Receivables (AR), Payables (AP), Customer statement.

7.9 Sales & CRM

Lead → Opportunity (pipeline) → Quotation → Sales order → Delivery → Invoice

- **Sales & CRM → Customers**: **New customer**; detail page par **Overview**, **Contacts**, **Communications**, **Quotations**, **Orders** tabs aur AR balance. **List** / **Map** view.
- **Sales & CRM → Leads**: prospects — status New → Contacted → Qualified → Converted; **Convert lead** se customer.
- **Sales & CRM → Pipeline**: opportunities ka board (New, Qualified, Proposal, Negotiation, Won, Lost). Cards drag karein ya card ke menu se keyboard se move karein.
- **Sales & CRM → Quotations**: Draft → Sent → Accepted / Rejected / Expired → Converted. **Convert to order** se sales order.
- **Sales & CRM → Sales orders**: Draft → Confirmed → Approved → Partially delivered → Delivered. Confirm hone ke baad lines read-only. **Create delivery** se shipment.
- **Sales & CRM → Deliveries**: carrier aur tracking number haath se darj hote hain — live GPS nahi.

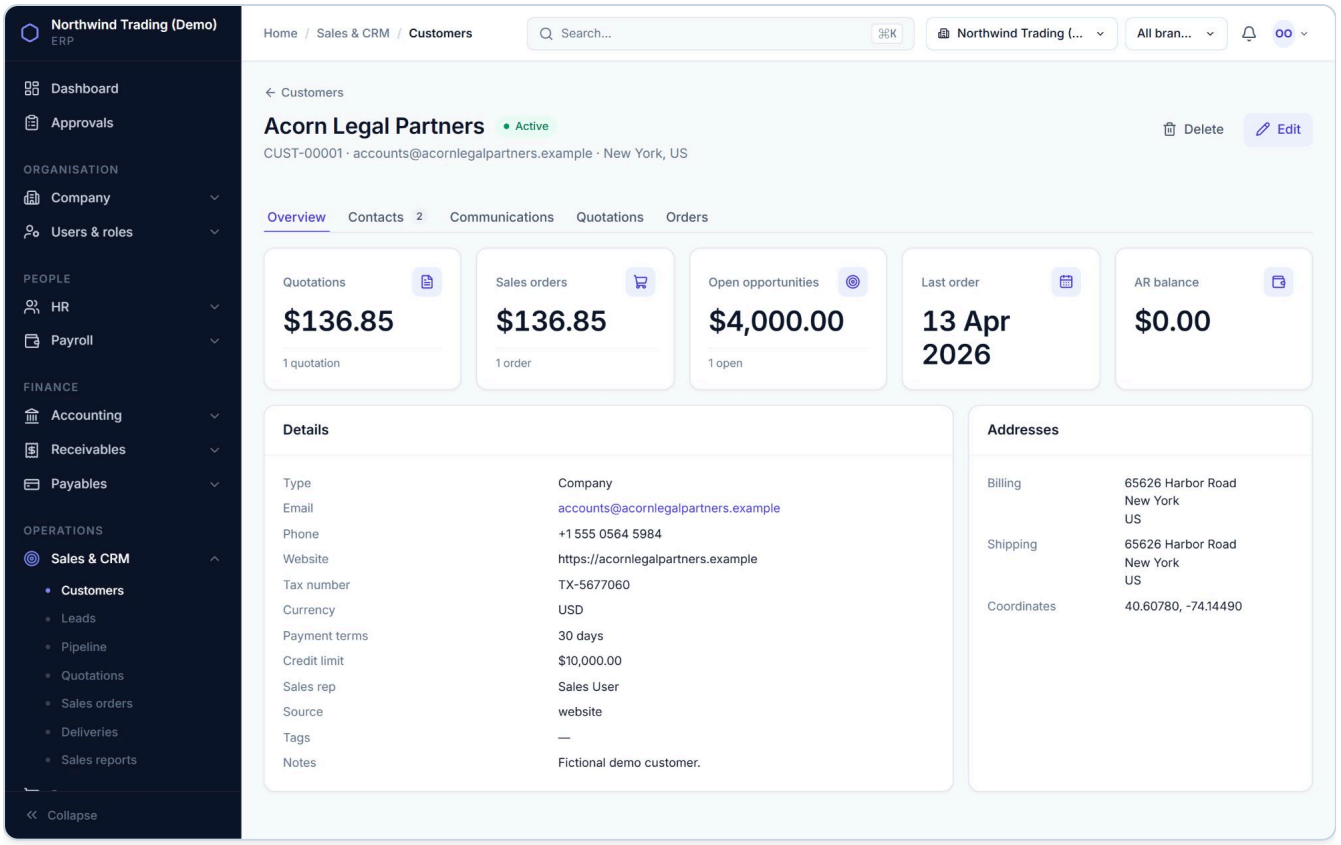

Figure 7.33 — Customer detail — quotations, orders, open opportunities, AR balance aur details.

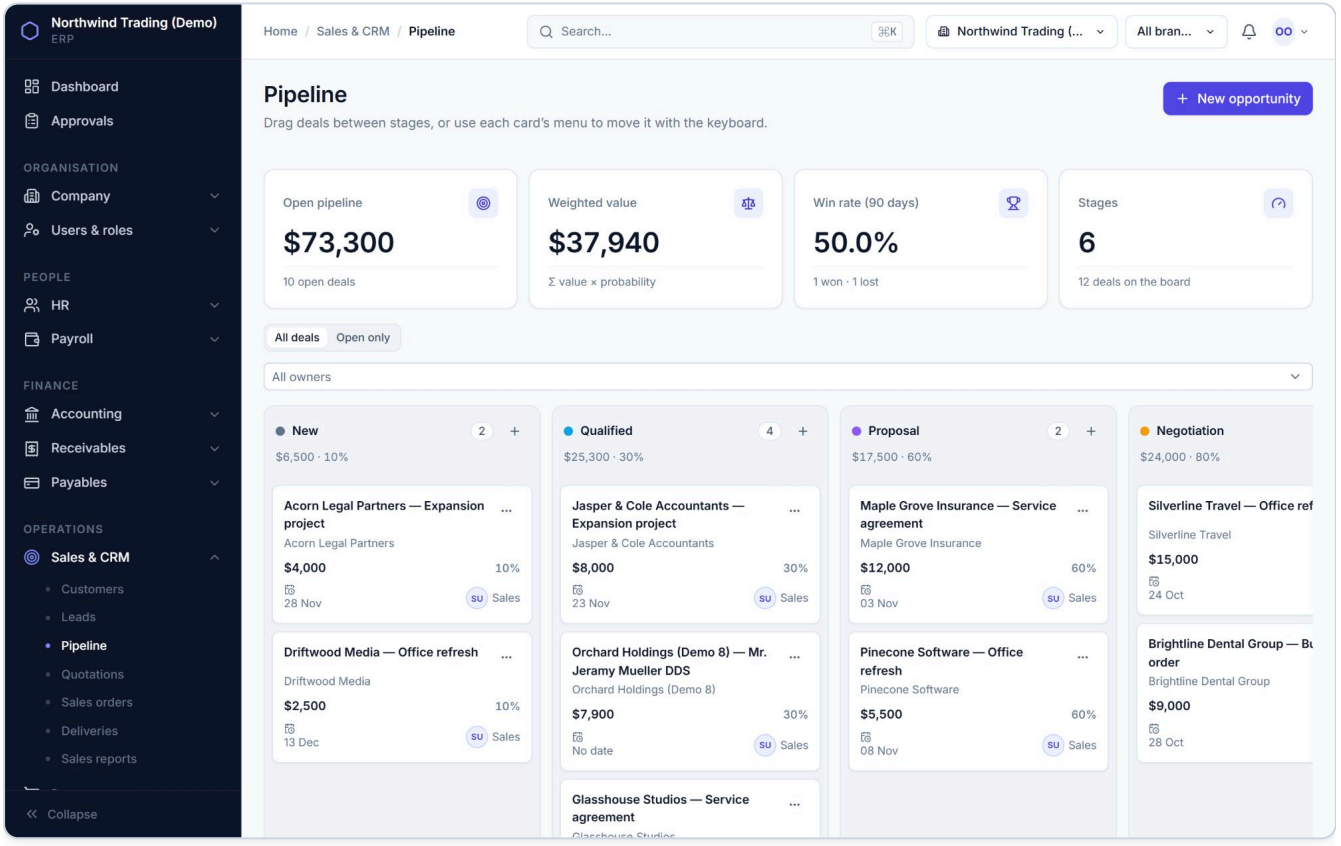

Figure 7.34 — Pipeline board — open pipeline, weighted value, win rate; stages mein deals.

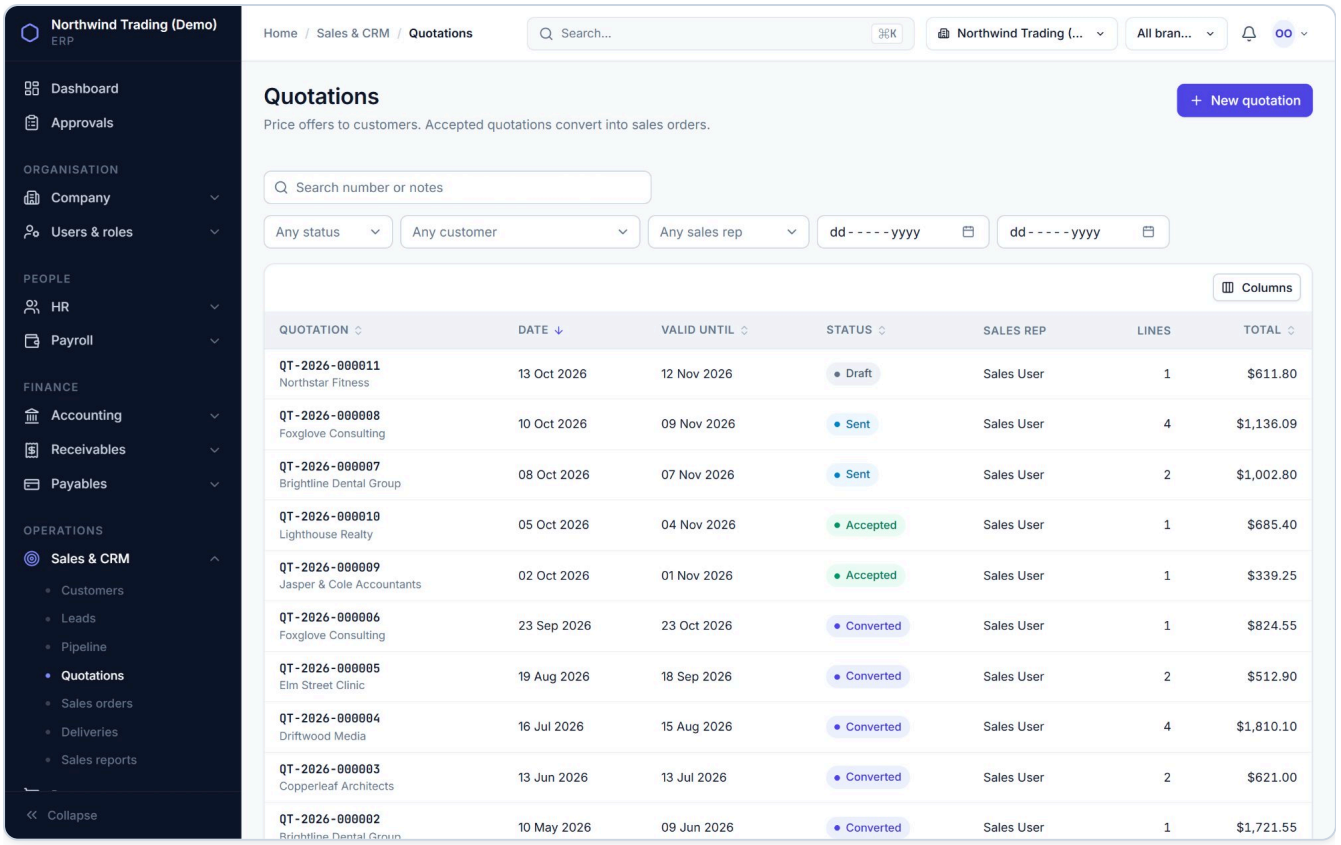

Figure 7.35 — Quotations list — status aur customer filters.

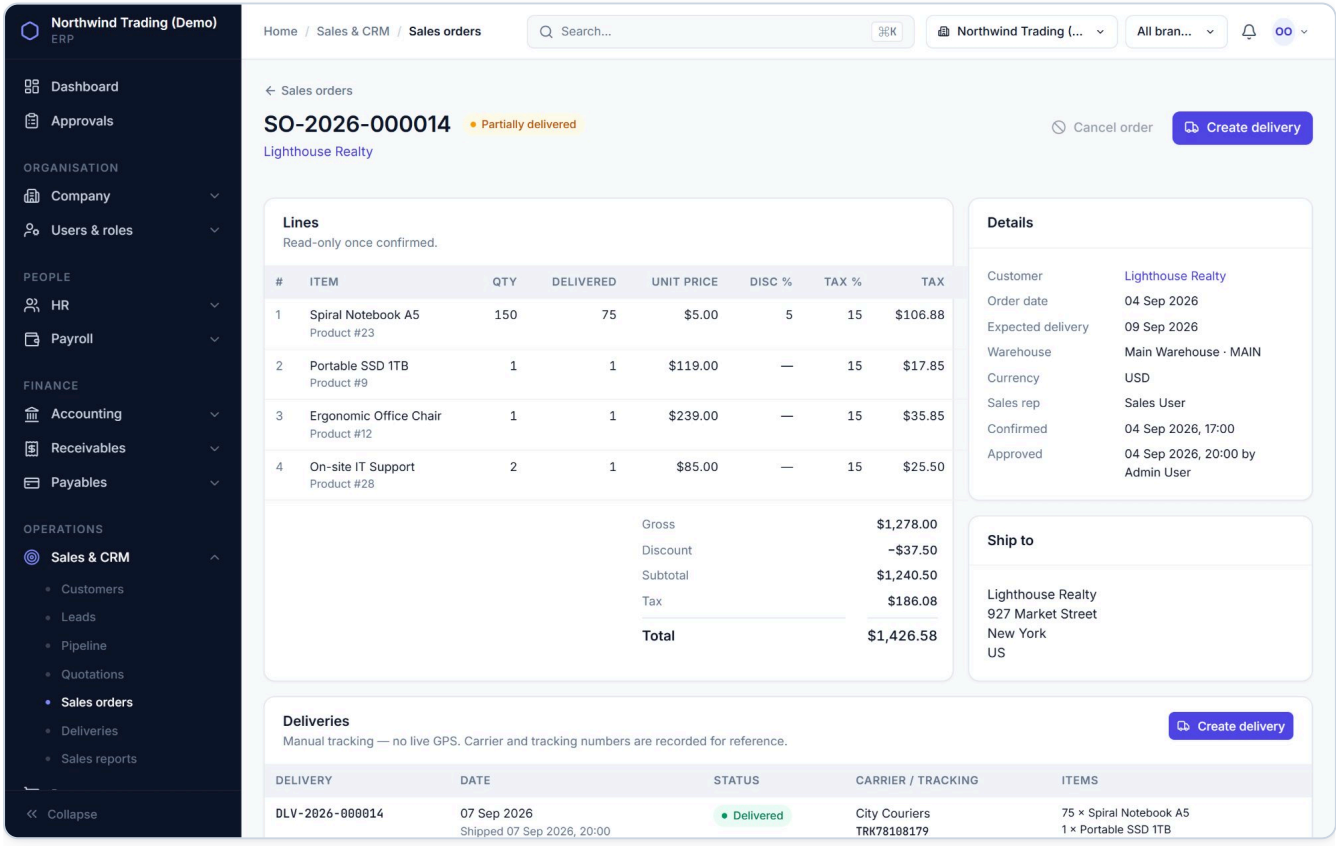

Figure 7.36 — Sales order (Partially delivered) — har line ki delivered qty, Create delivery button.

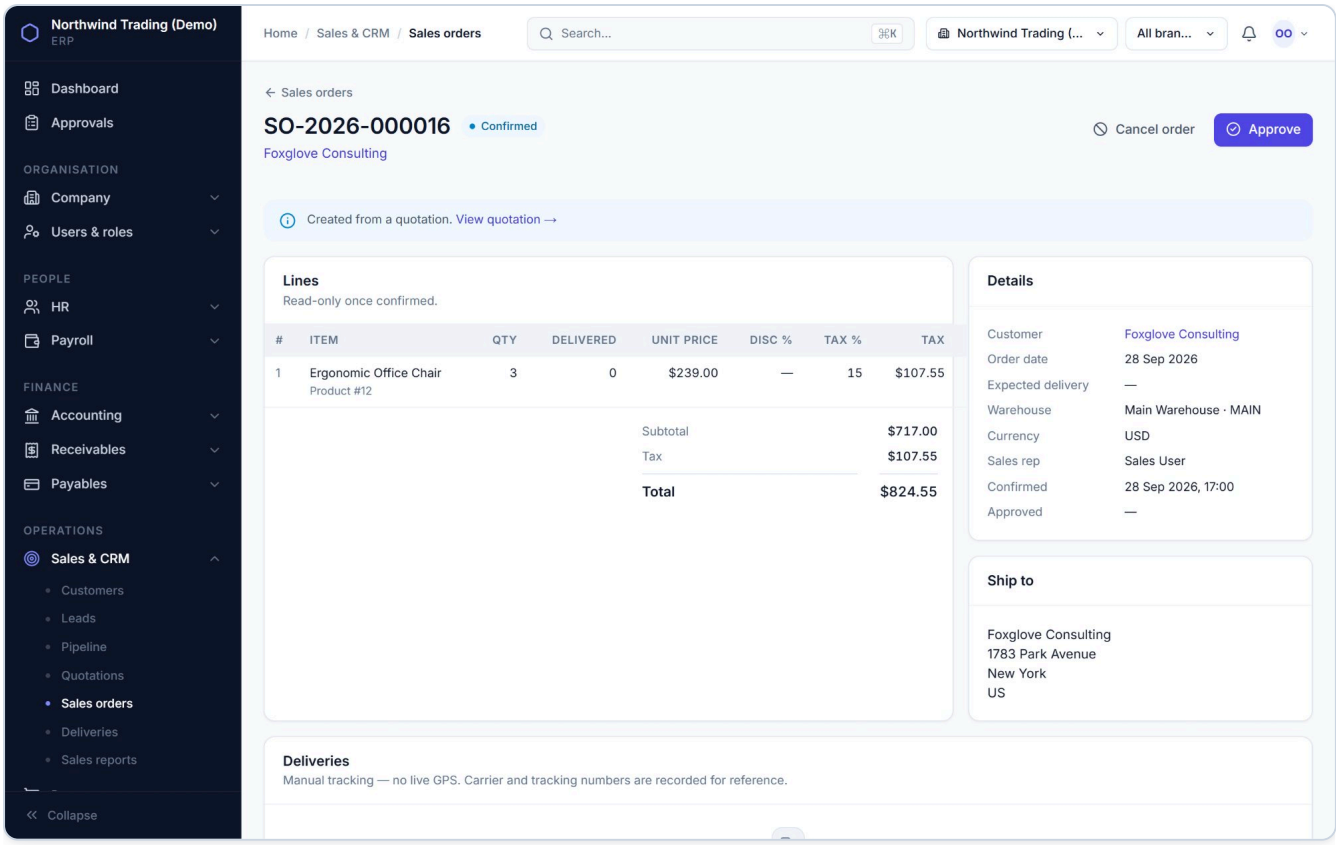

Figure 7.37 — Confirmed sales order — quotation se bana; Approve / Cancel order.

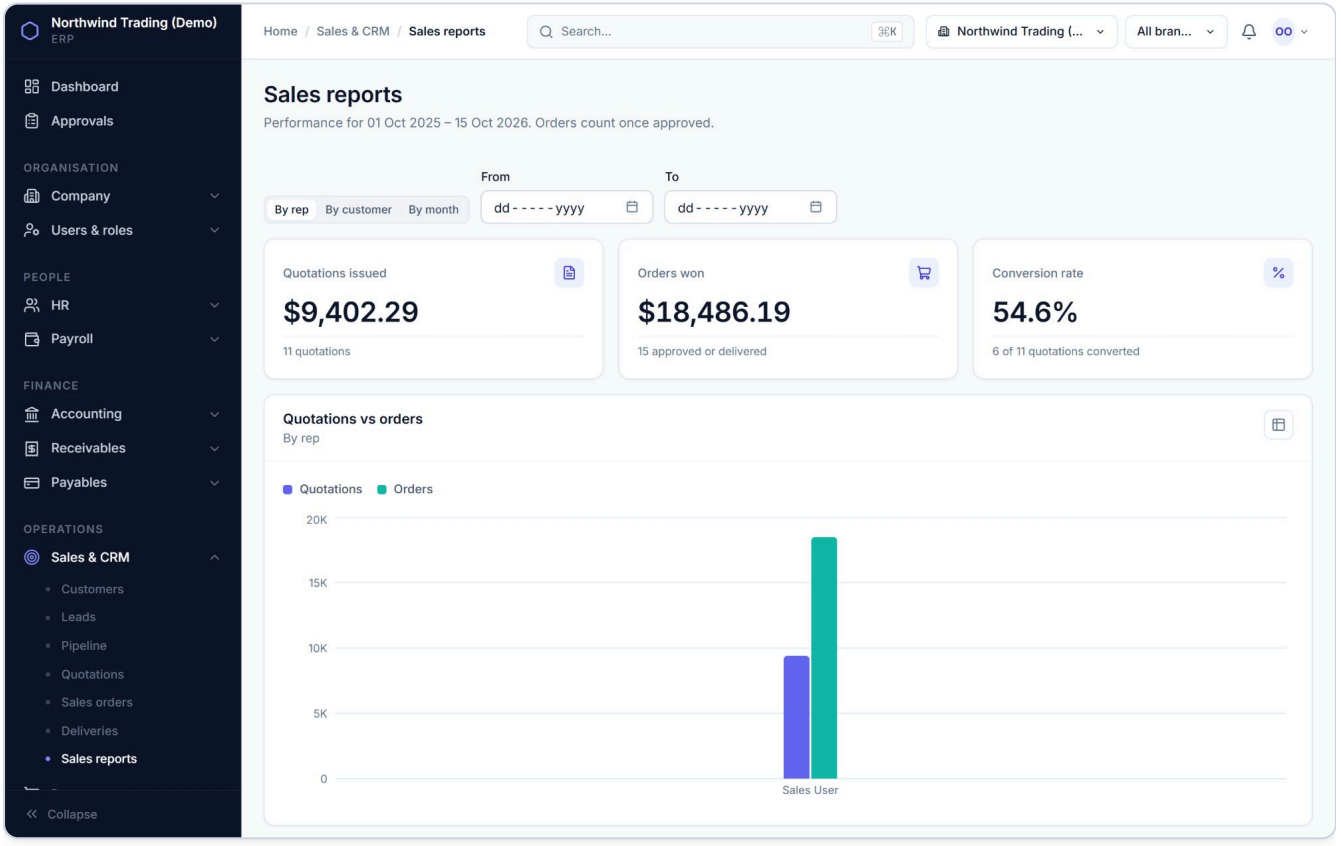

Figure 7.38 — Sales reports — quotations vs orders, conversion rate, rep/customer/month-wise.

7.10 Procurement

Requisition → Approval → Purchase order → PO approval → Goods receipt → Supplier bill

1. **Requisition:** Procurement → Requisitions → New requisition (items, qty, estimated price, needed by, priority) → Submit. Requester apni requisition approve nahi kar sakta.
2. **Approval:** manager detail page par Approve / Reject.
3. **Purchase order:** approved requisition se Create purchase order — ya Procurement → Purchase orders → New purchase order. Draft → Submit → Approve.
4. **Goods receipt:** maal aaye to PO par Receive goods (ya Procurement → Goods receipts → Receive goods): received/rejected qty, batch, expiry. Confirm karne par stock barhta hai.
5. Kharab maal: receipt ya PO se Return goods. Sab mukammal ho to PO Close.
6. **Bill:** supplier ki invoice Payables → Supplier bills mein darj karein aur approve karein.

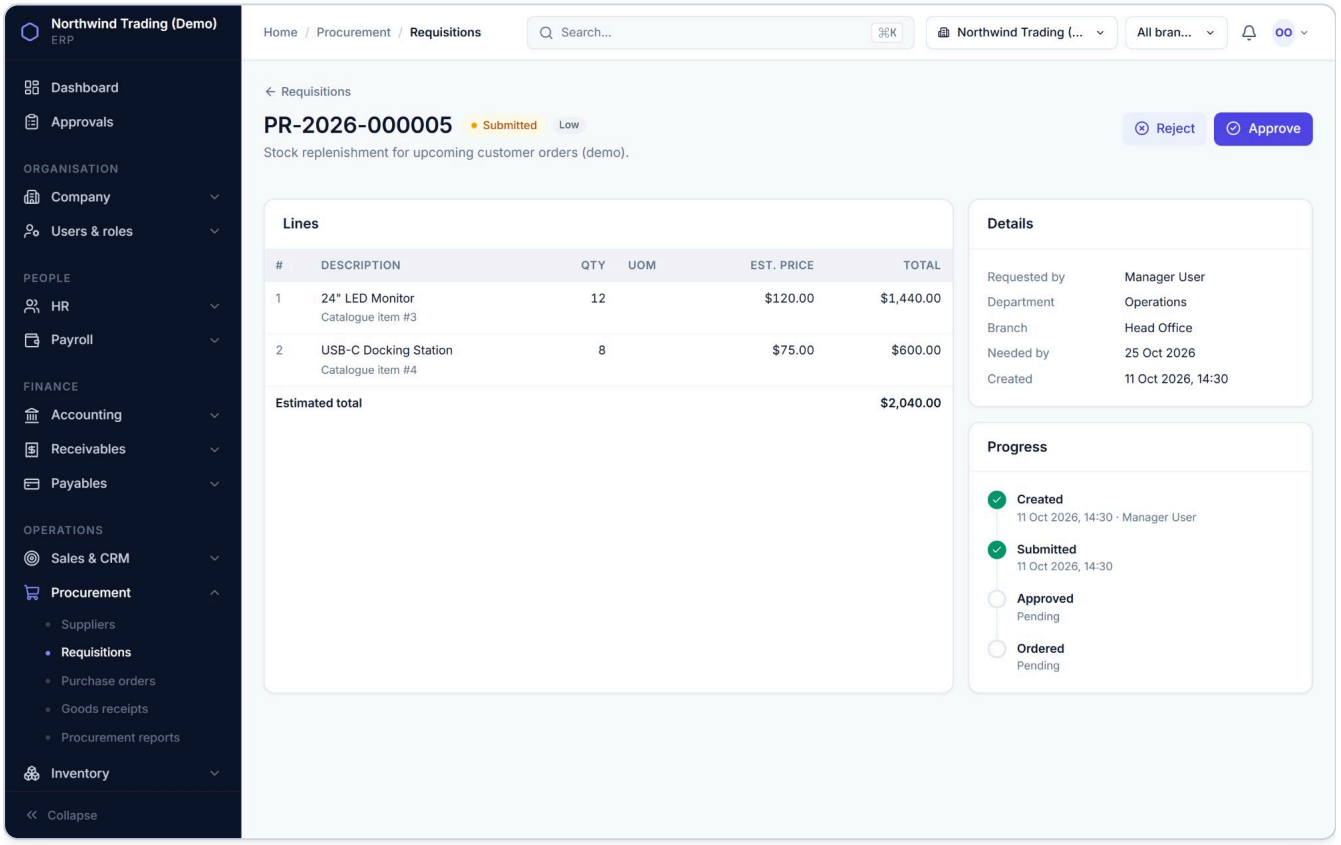

Figure 7.39 — Submitted requisition — lines, estimated total, progress; Reject / Approve.

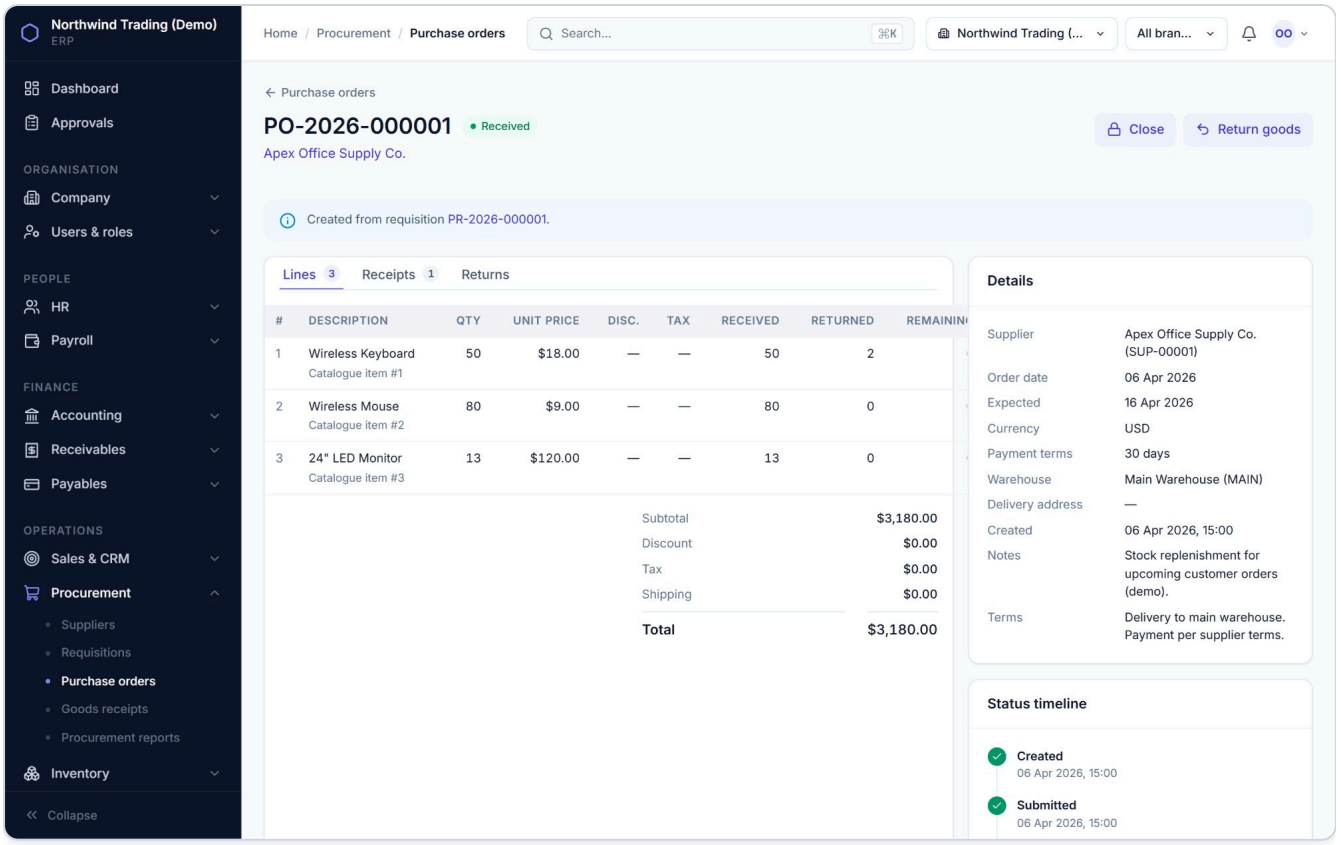

Figure 7.40 — Received purchase order — requisition se bana; received/returned/remaining qty; Close aur Return goods.

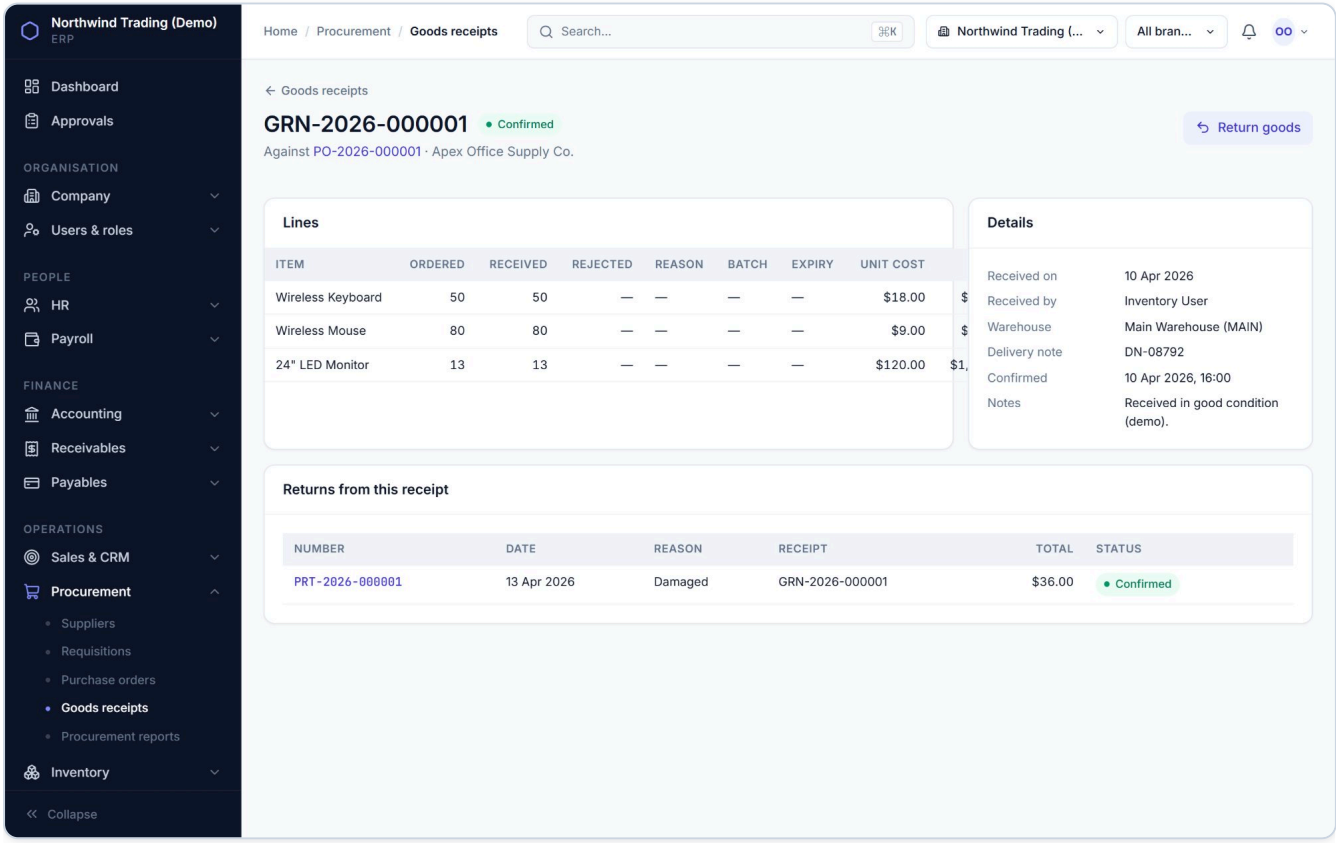

Figure 7.41 — Goods receipt — ordered vs received vs rejected, unit cost, delivery note.

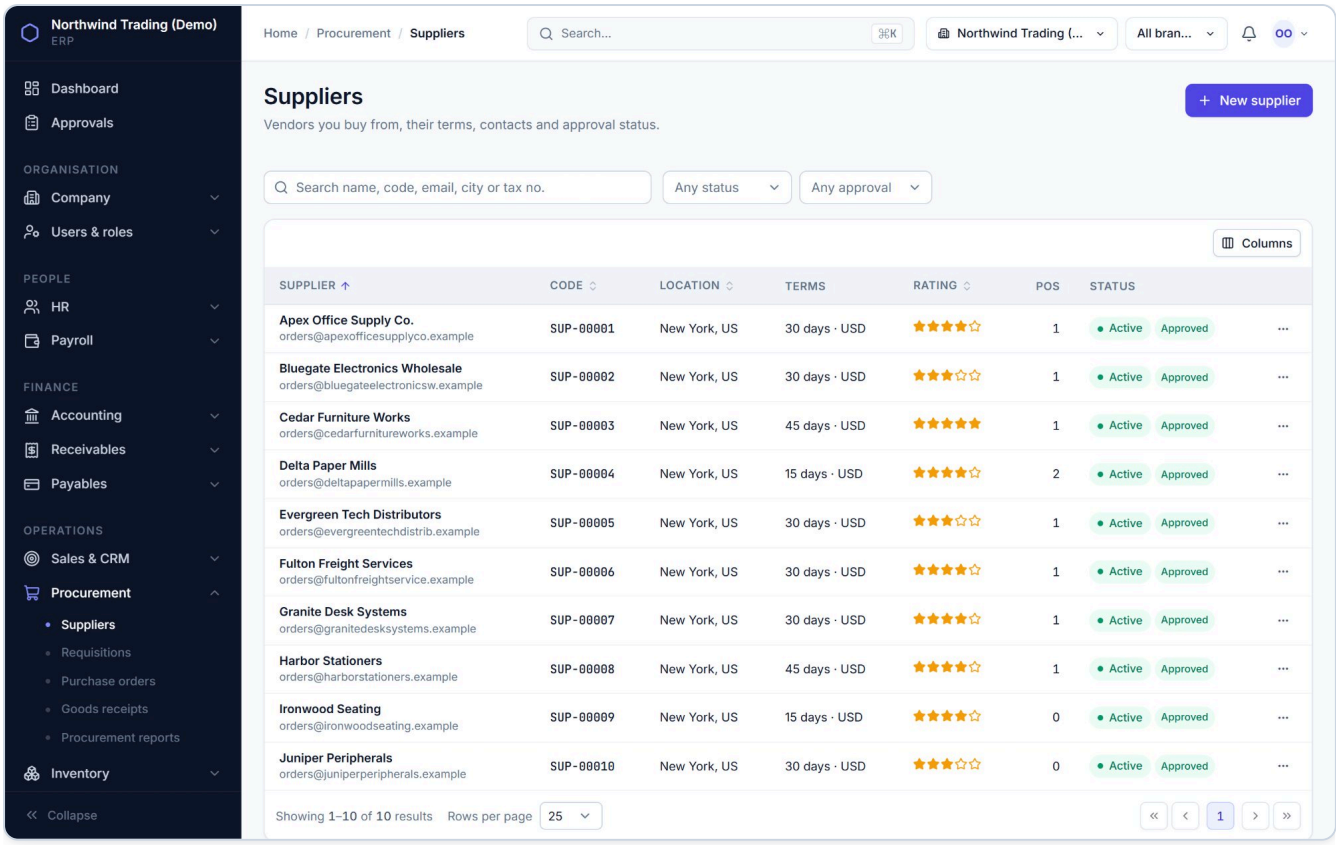

Figure 7.42 — Suppliers — terms, rating, approval status.

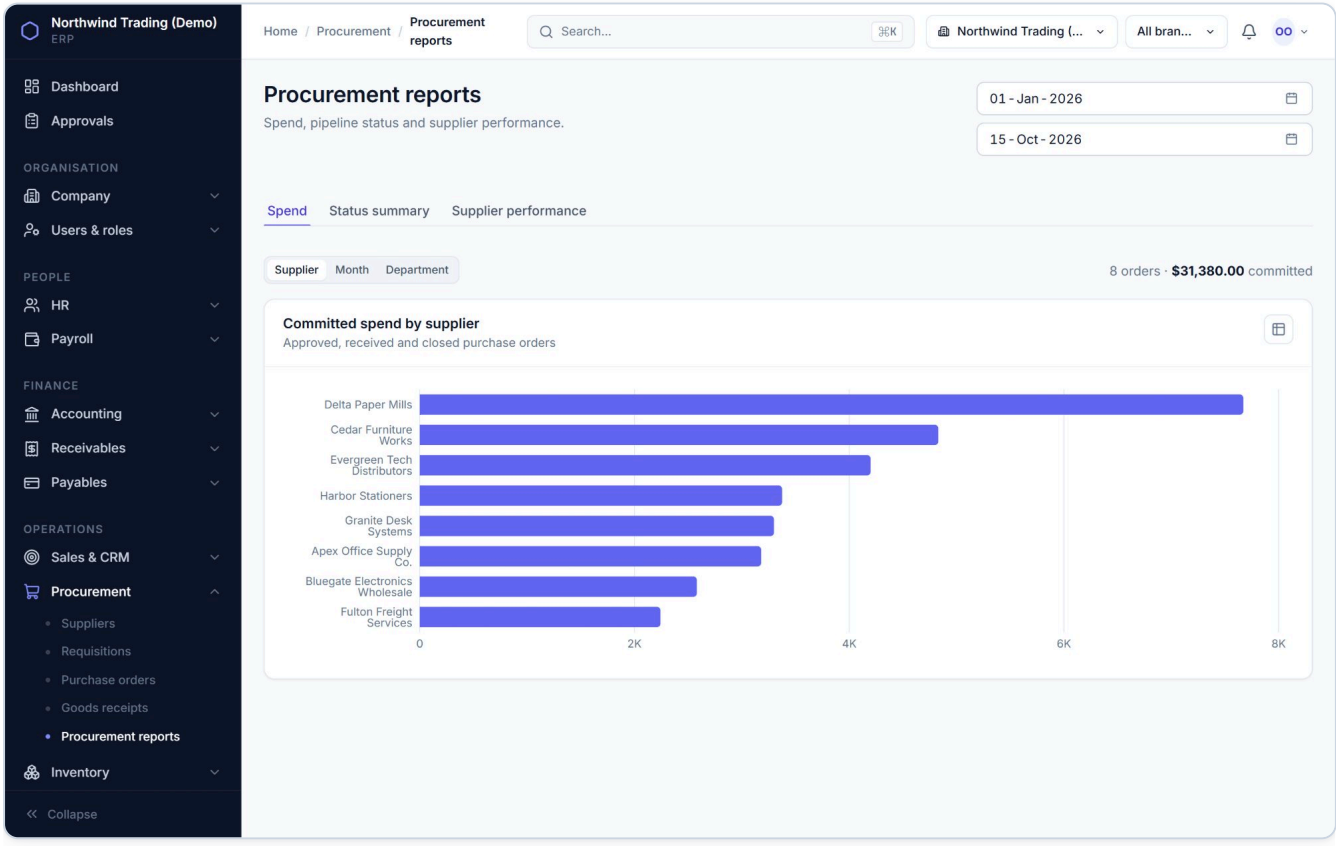

Figure 7.43 — Procurement reports — committed spend by supplier, status summary, supplier performance.

7.11 Inventory

Stock ka asal record **stock movements** hai — har receipt, issue, transfer aur adjustment aik movement banata hai jo badla nahi ja sakta ("The immutable stock ledger"). Product ki on-hand quantity isi se nikalti hai. Valuation **weighted (moving) average cost** par hoti hai.

- **Inventory → Products** : **New product** ; **Categories & units** se categories aur units of measure.
- **Inventory → Stock levels** : product × warehouse — on hand, reserved, available, reorder level, avg cost, value. **Receive** / **Issue** buttons; **Below reorder level** filter.
- **Inventory → Movements** : poori history type aur warehouse filter ke sath.
- **Inventory → Transfers** : **New transfer** → **Ship** (source se minus) → **Receive** (destination mein plus). Status Draft → In transit → Received.
- **Inventory → Adjustments** : count, damage, loss/theft, correction, expiry ke baad farq — Draft → **Post** . Posting difference stock ledger (aur GL) mein likhti hai.
- **Inventory → Warehouses** : warehouses aur un ke andar zones/aisles/racks/bins.

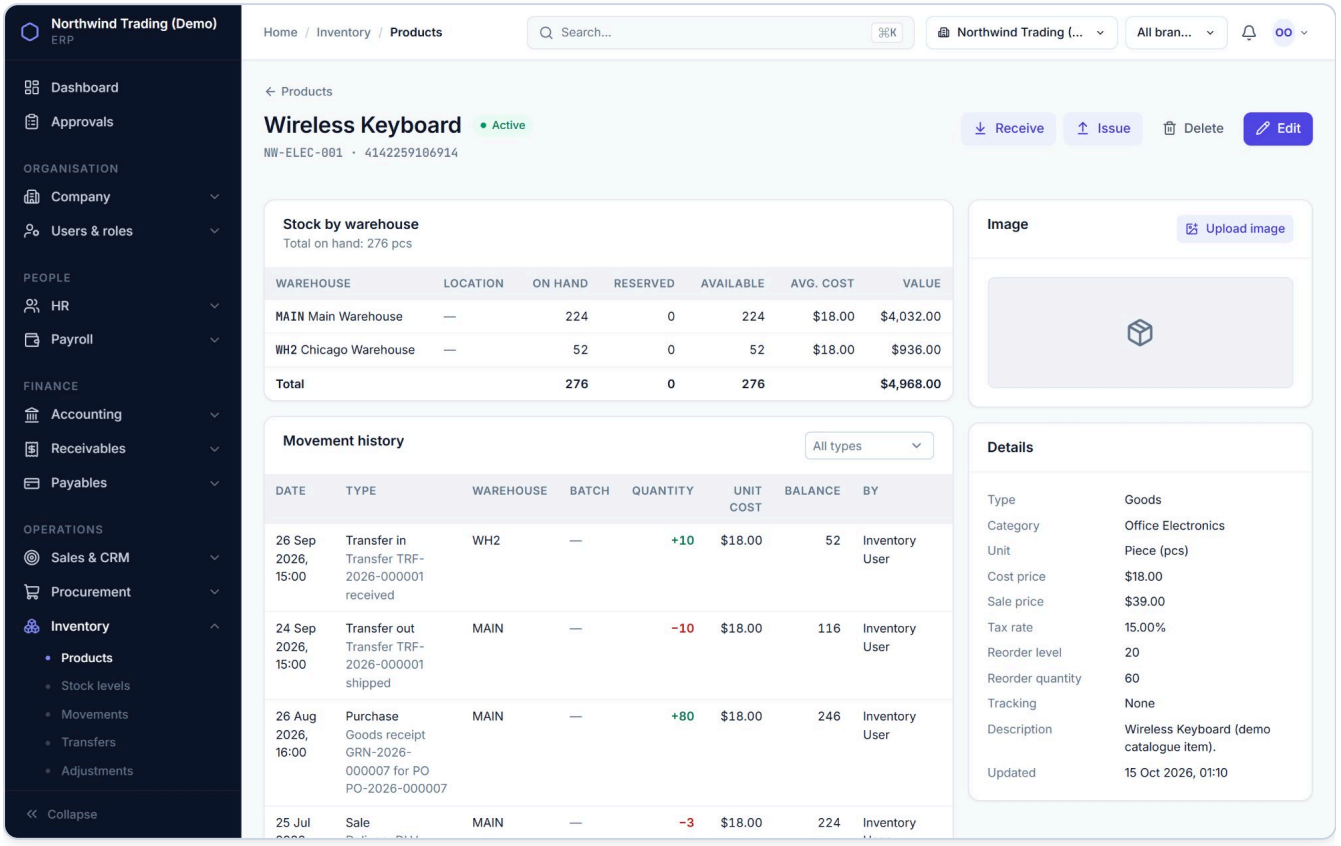

Figure 7.44 — Product detail — warehouse-wise stock, movement history, Receive / Issue.

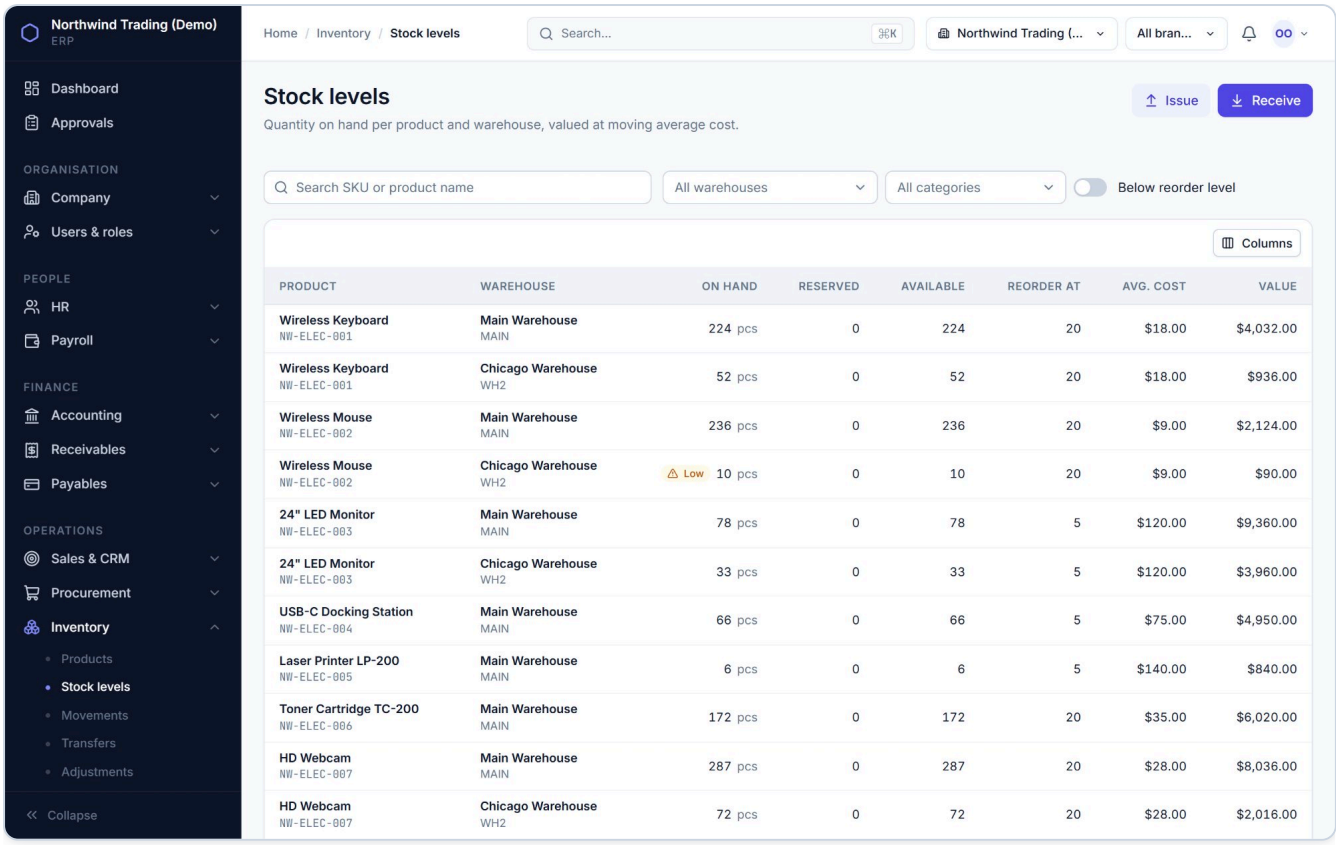

Figure 7.45 — Stock levels — on hand, reserved, available, reorder at, avg cost, value.

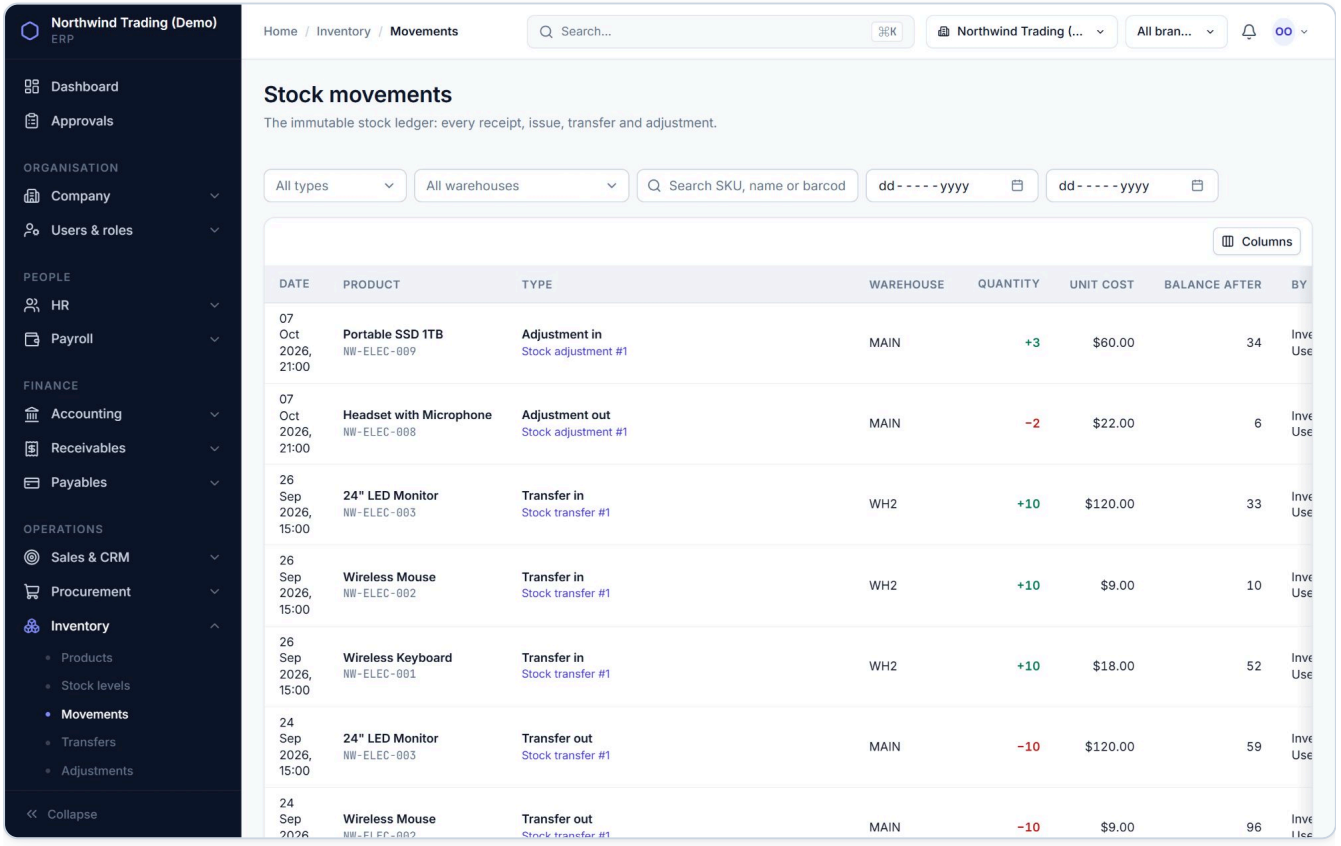

Figure 7.46 — Stock movements — immutable ledger; har movement ka type, qty, unit cost aur balance after.

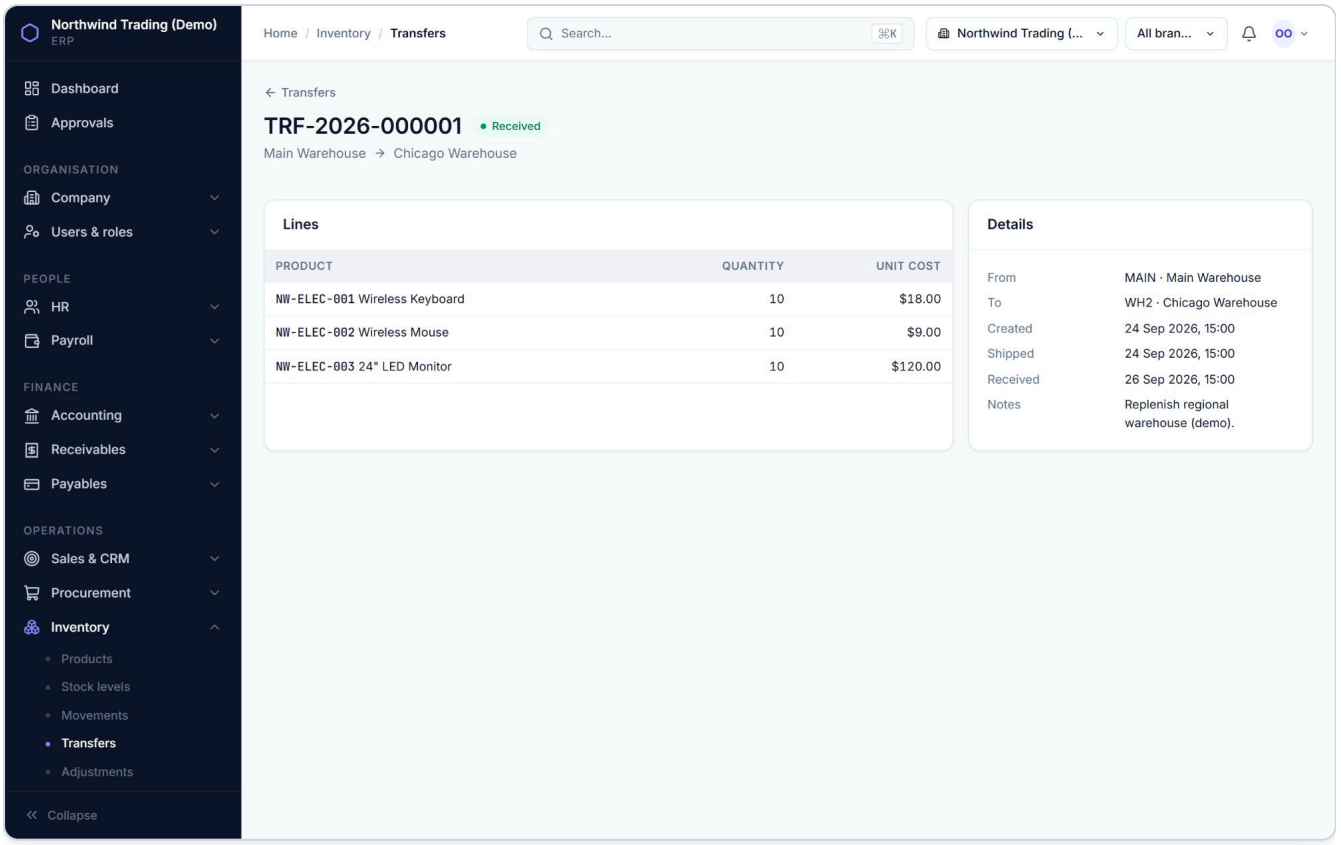

Figure 7.47 — Received transfer — Main Warehouse se Chicago Warehouse; shipped/received dates.

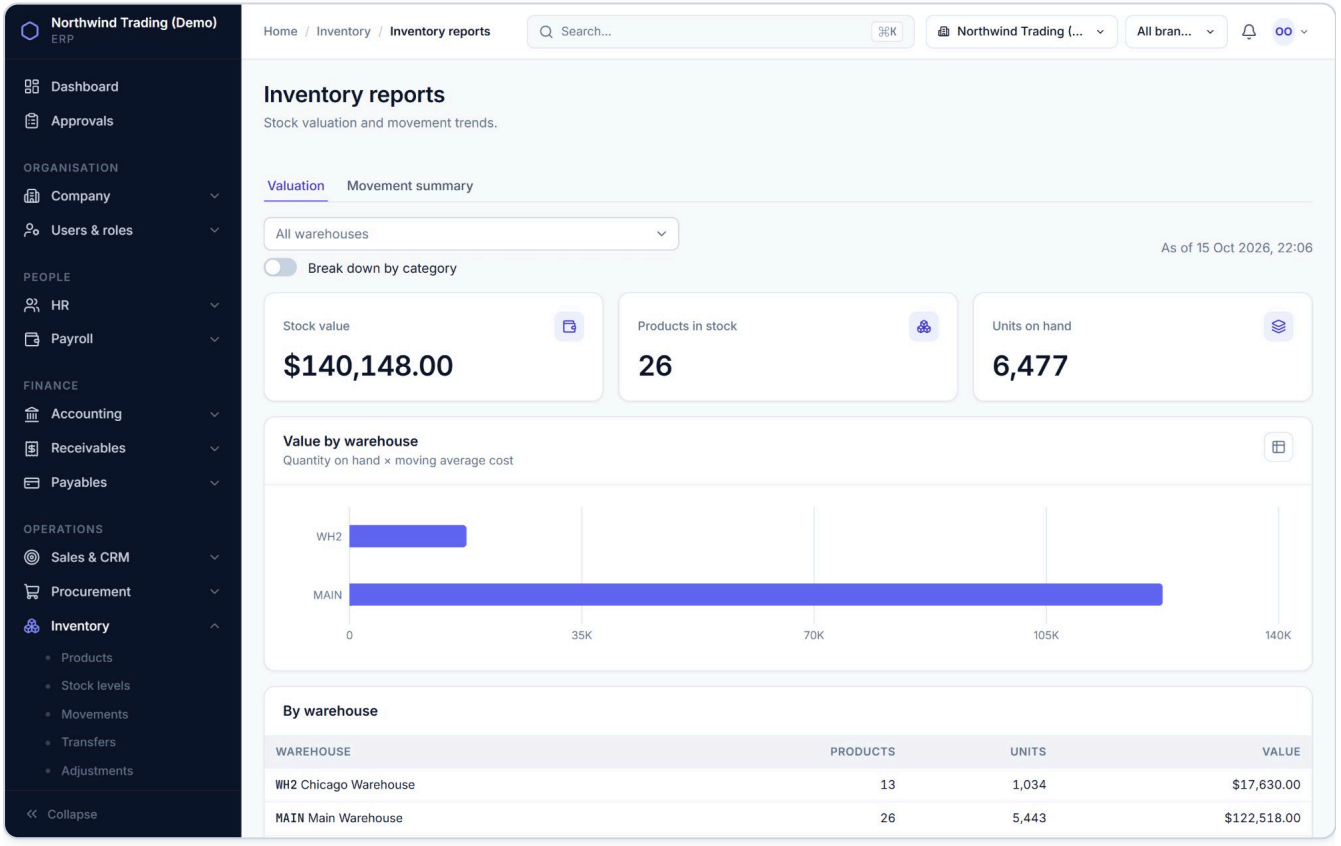

Figure 7.48 — Inventory reports — valuation by warehouse, movement summary.

> ⚠ **DHYAN DEIN**
>
> Stock ko kabhi "seedha" theek karne ki koshish na karein — hamesha Adjustment banayein taake wajah, kaun aur kab record ho. Low-stock alerts roz scheduler ke zariye inventory users ko jate hain.

7.12 Projects & Tasks

1. **Projects** kholein → **Portfolio** (saare projects, cards/list) ya **My tasks**.
2. **New project**: naam, manager, dates, budget, priority.
3. Project page tabs: **Overview**, **Board**, **List**, **Milestones**, **Members**, **Time**, **Expenses**, **Files**.
4. **New task** se task; Board par cards columns (To do, In progress, In review, Done, Blocked) mein drag karein.
5. Task par click se drawer: status, priority, assignee, milestone, dates, estimate, labels, subtasks, comments.
6. **Log time** se kaam ke ghante darj karein (har shakhs sirf apna time log karta hai).

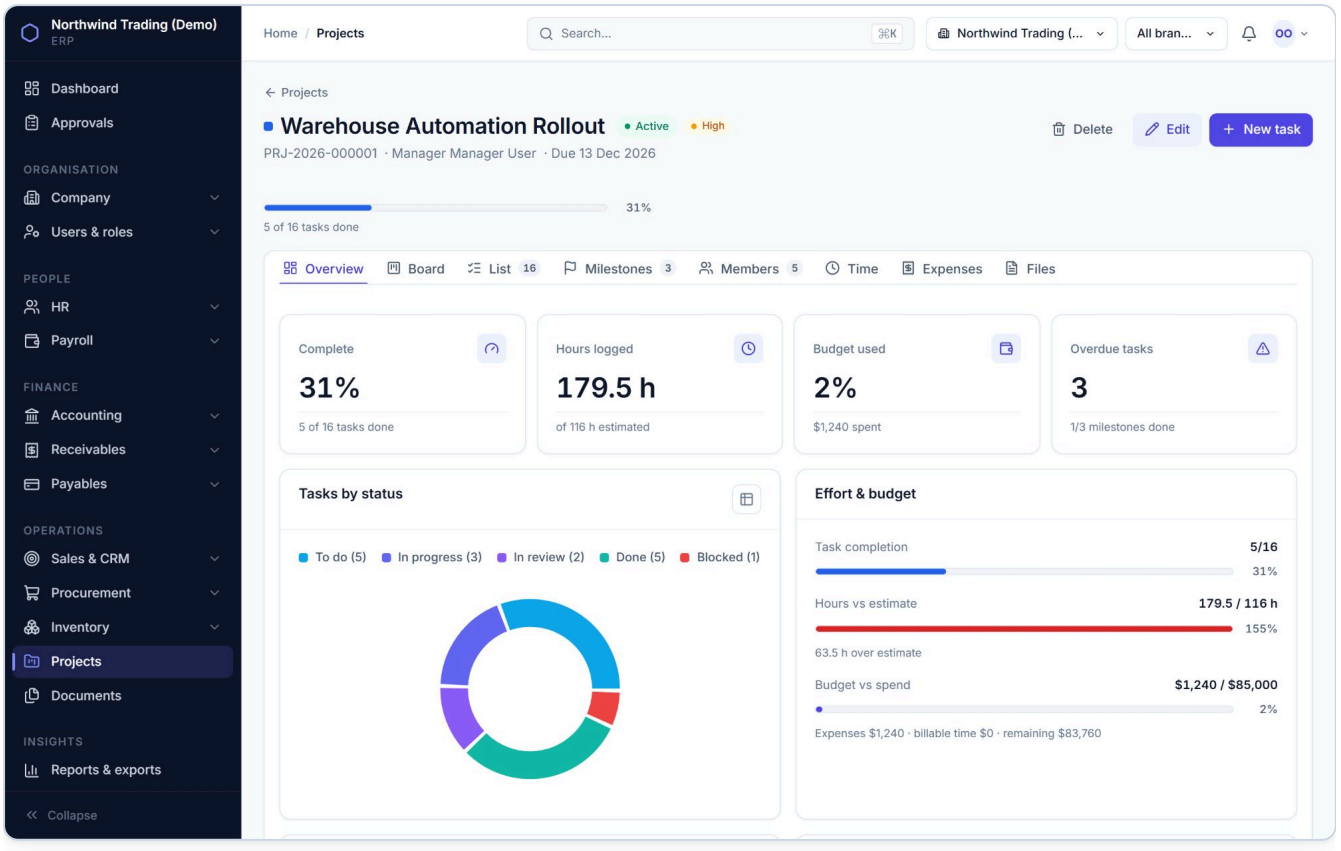

Figure 7.49 — Project overview — completion %, hours vs estimate, budget used, overdue tasks, milestones.

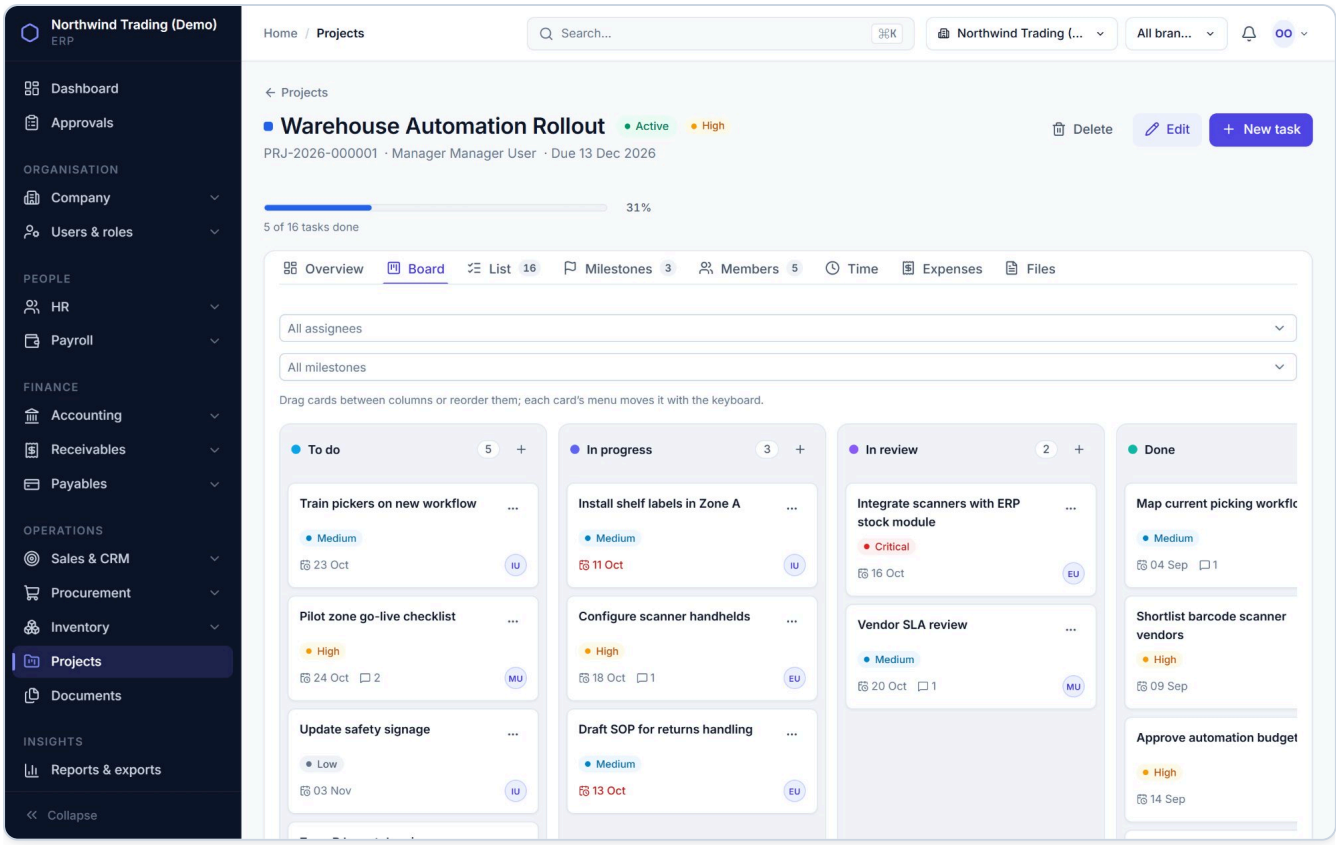

Figure 7.50 — Project board — status columns aur task cards.

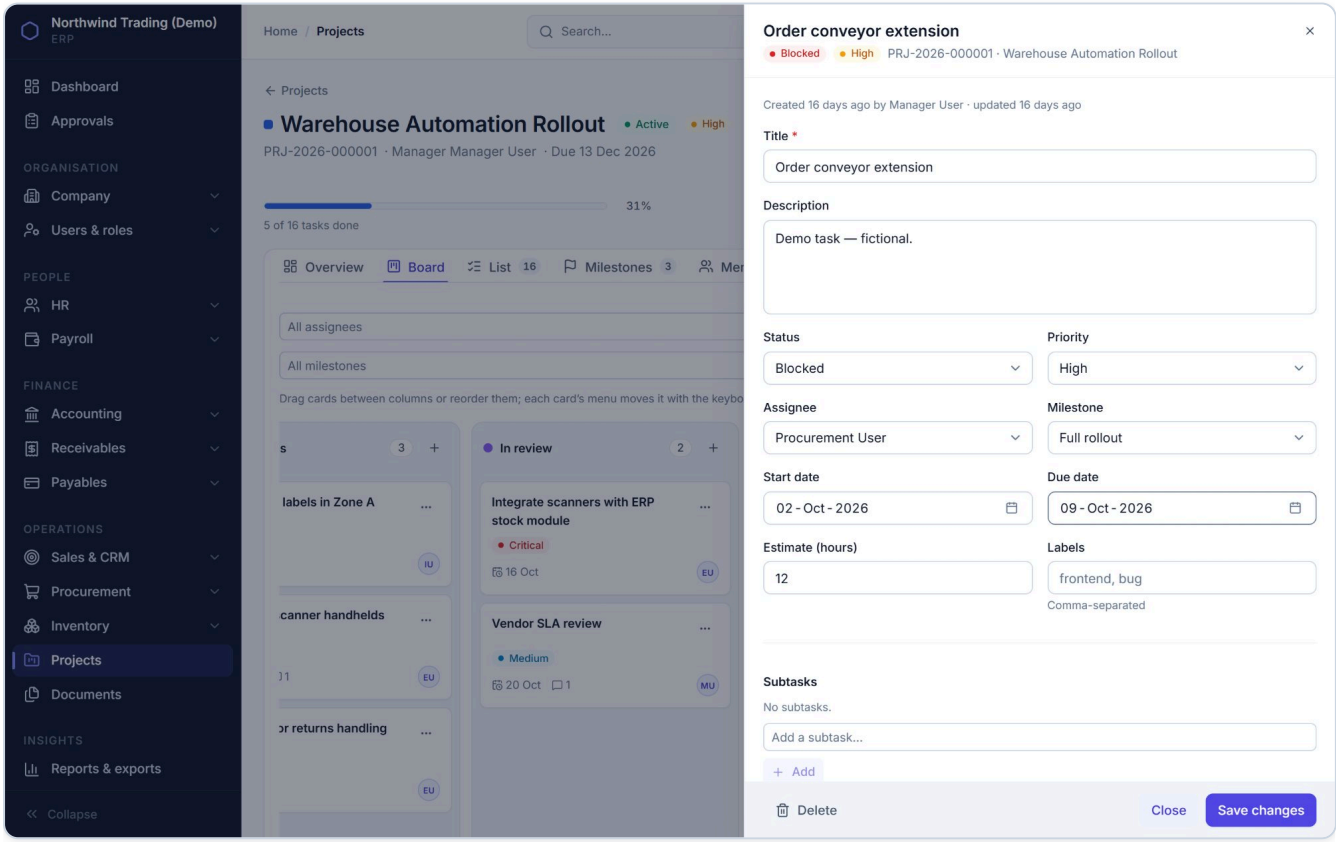

Figure 7.51 — Task drawer — title, status, priority, assignee, milestone, dates, estimate, labels, subtasks; Save changes.

7.13 Documents, Reports & Exports, Search

Documents

Documents mein company files folders (Contracts, HR, Policies...) mein rakhi jati hain — versions, visibility (Company / Restricted / Private), owner aur **retain until** date ke sath. **Upload** se file (PDF, images, Office, CSV, TXT; default max 10 MB).

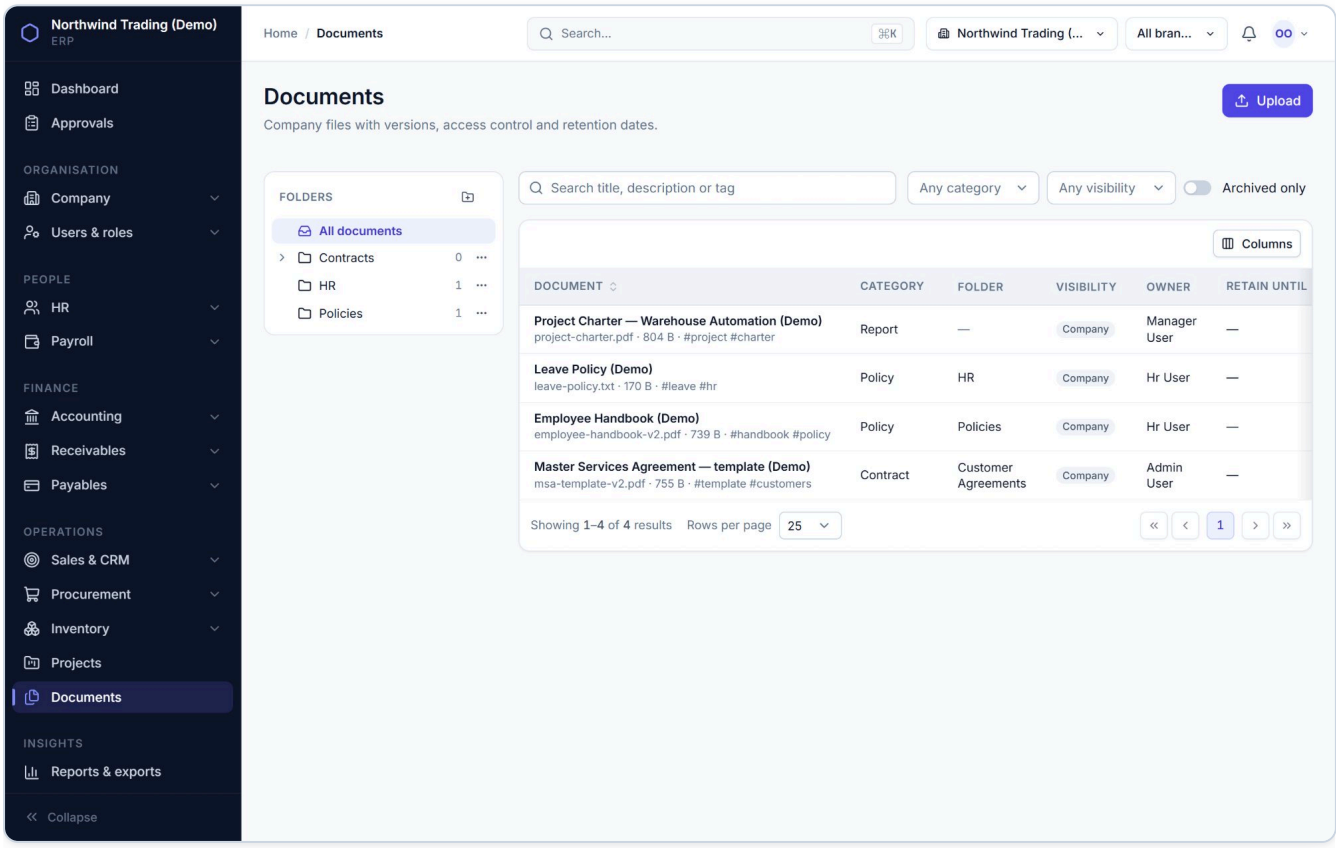

Figure 7.52 — Documents — folders, category, visibility, owner, retention date.

Reports & exports

1. **Insights → Reports & exports** kholein. Left mein reports groups mein hain: Finance, Sales, Procurement, Inventory, HR, Projects, Audit.
2. Report par click karein (maslan "Profit and loss").
3. Parameters (From/To, Branch...) set karke **Run report** — neeche preview table.
4. **Export** menu se CSV, Excel ya PDF. Chhoti export foran download; bari (2,000+ rows) background mein banti hai aur tayyar hone par notification aati hai — **My exports** mein milti hai.

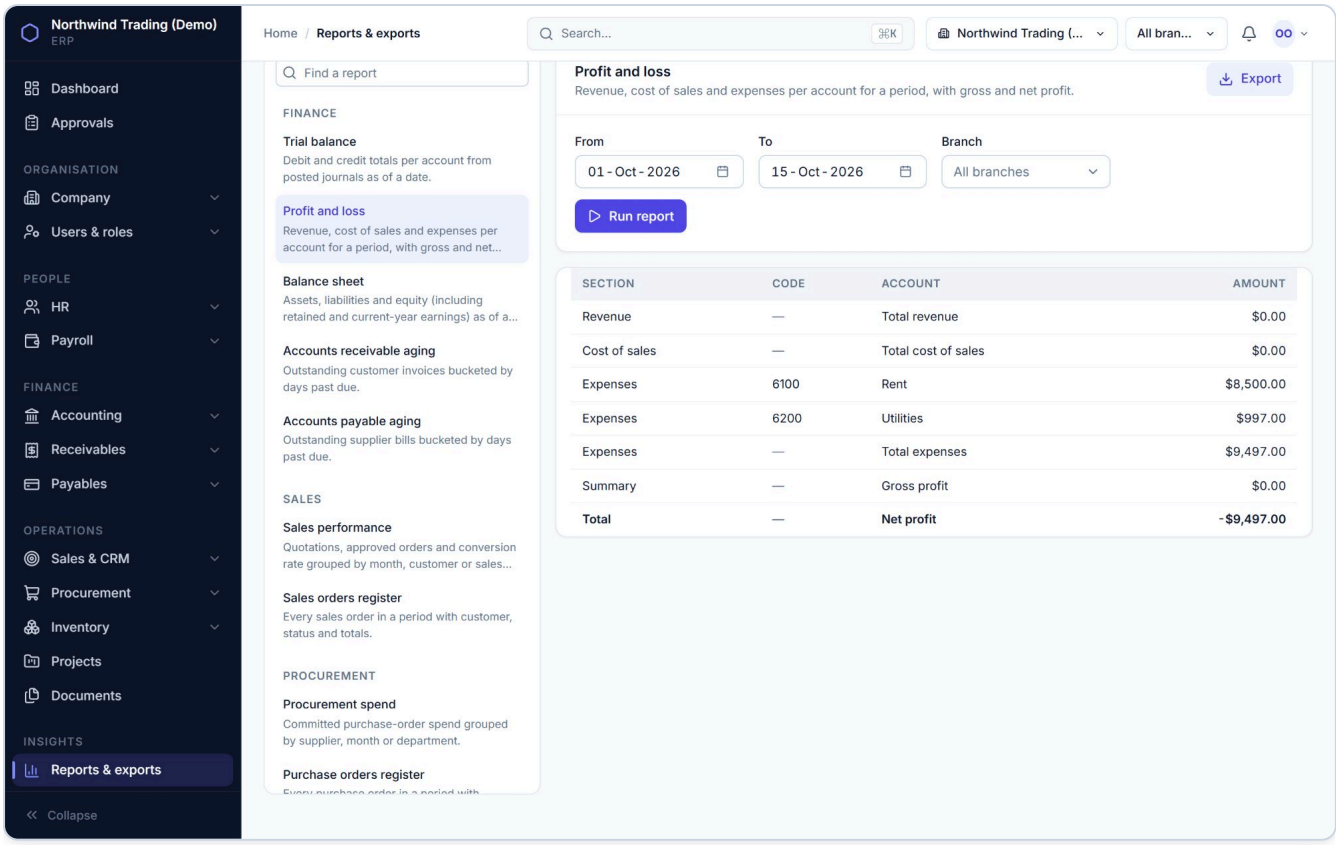

Figure 7.53 — Profit and loss report — period, branch, Run report aur Export; preview mein revenue, expenses, net profit.

SECURITY

Har export audit log mein record hota hai. Export ke liye "Export reports" permission chahiye.

Global search

Topbar ka **Search...** box (ya `/search`) employees, customers, suppliers, products, documents aur transactions mein dhoondta hai — sirf woh jo aap ki permission mein hain. Kam az kam 2 characters likhein.

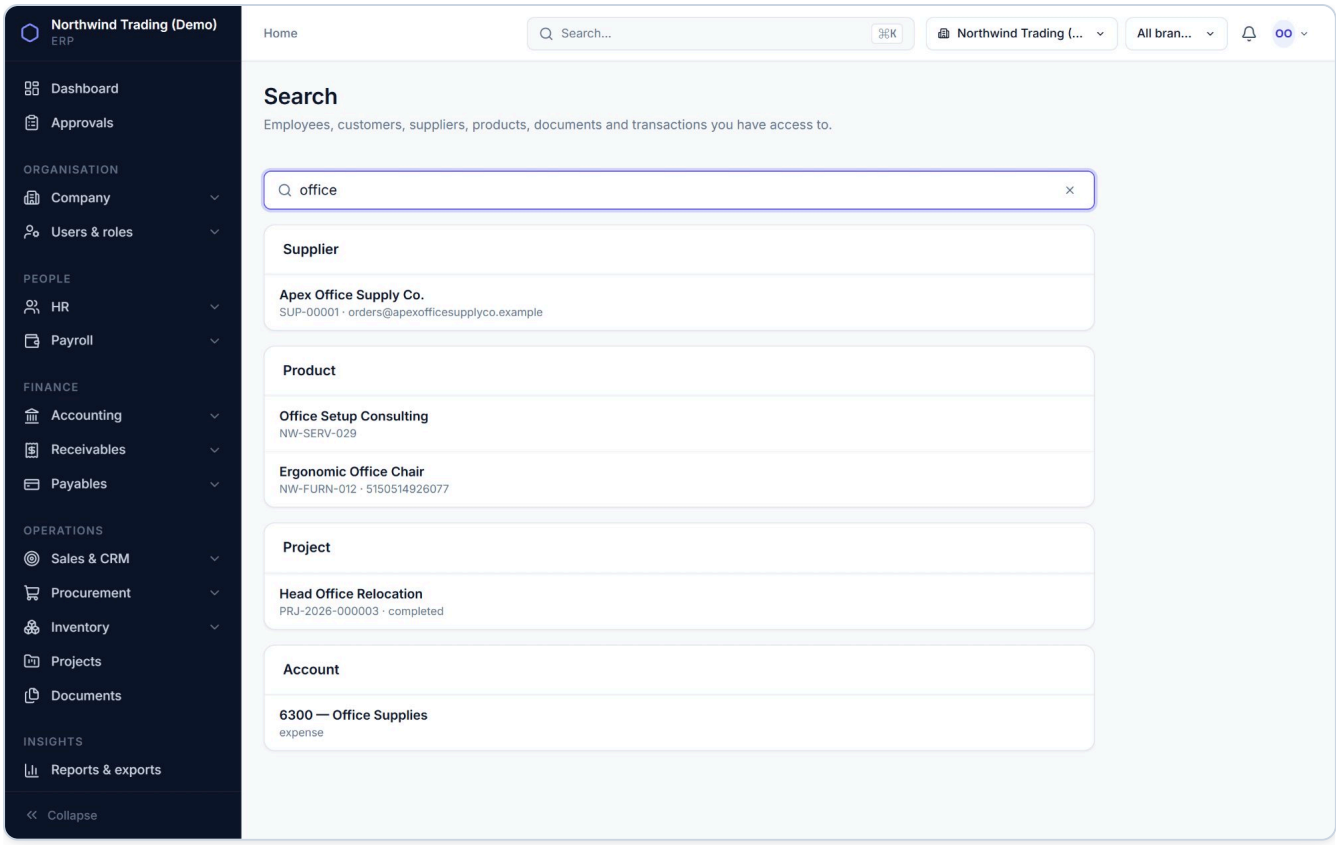

Figure 7.54 — Search results — type-wise groups.

7.14 Settings & Themes

- **My profile** : naam aur phone. Sign-in email sirf administrator badal sakta hai.
- **Security** : password change (dusre sessions sign out), two-factor (Chapter 4.4).
- **Notifications** : har event ke liye In app / Email.
- **Appearance** : Theme **Light** / **Dark** / **System** — "Saved to your account and applied on every device". **High-contrast mode** aur **Compact sidebar** .

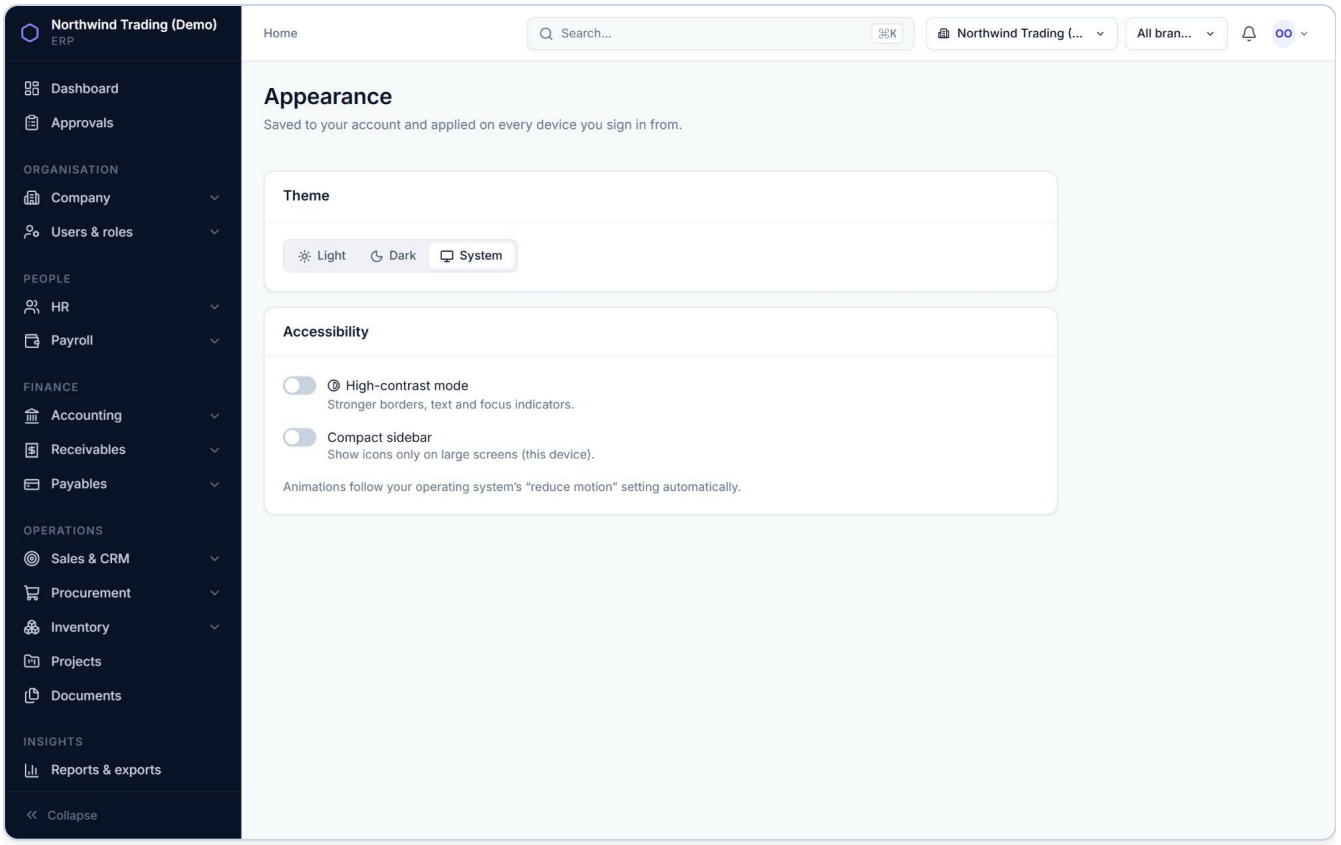

Figure 7.55 — Appearance — theme, high-contrast mode, compact sidebar.

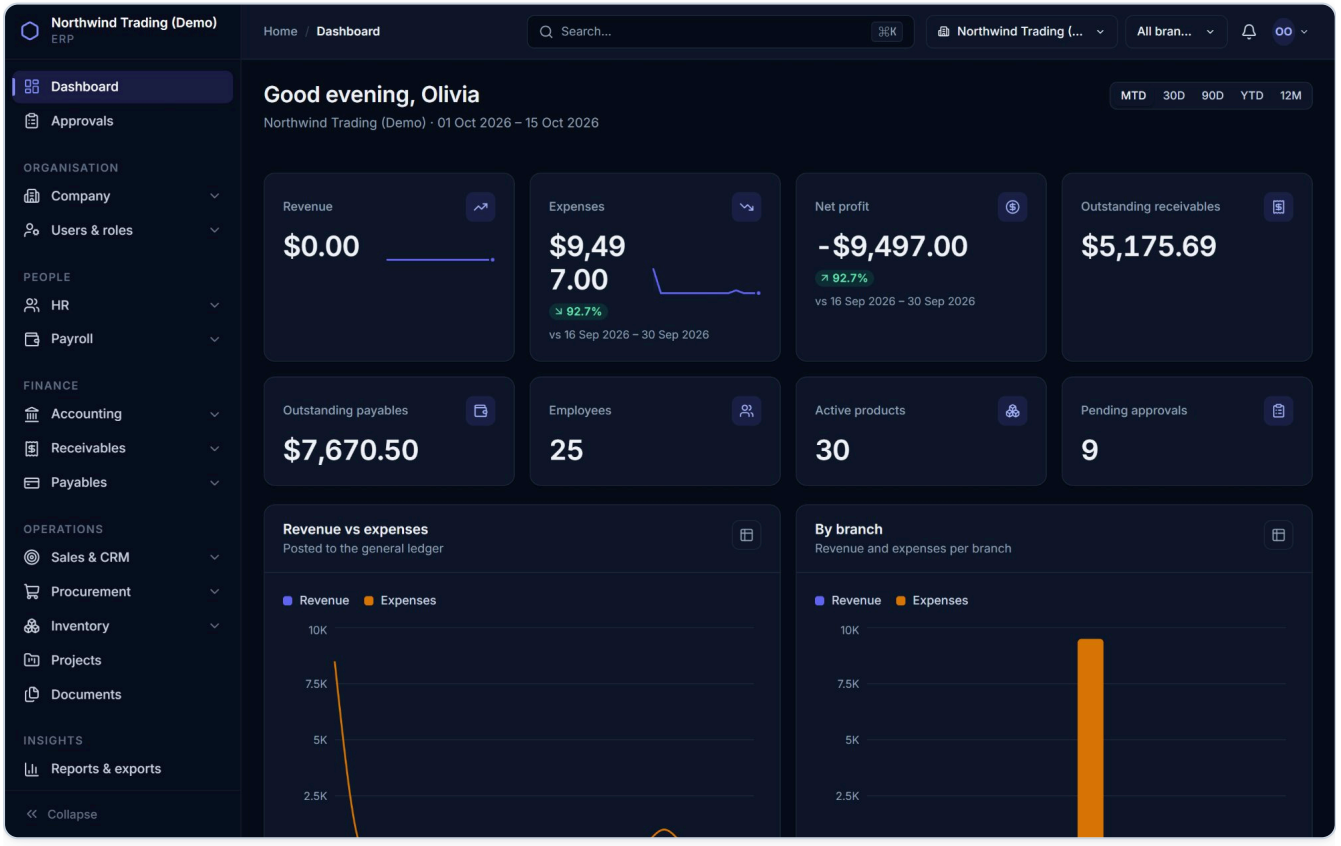

Figure 7.56 — Dark mode mein dashboard — Account menu se theme Dark.

> **TIP**
>
> Animations operating system ki "reduce motion" setting ko khud follow karti hain.

CHAPTER 8

Employee self-service

Aam employee kya dekhta hai aur apne kaam khud kaise karta hai.

8.1 Employee ko kya nazar aata hai

Employee role mein sirf self-service permissions hain: apna leave, attendance, payslips, expense claims, tasks/time aur company documents dekhna. Is liye sidebar bohat chhota hota hai.

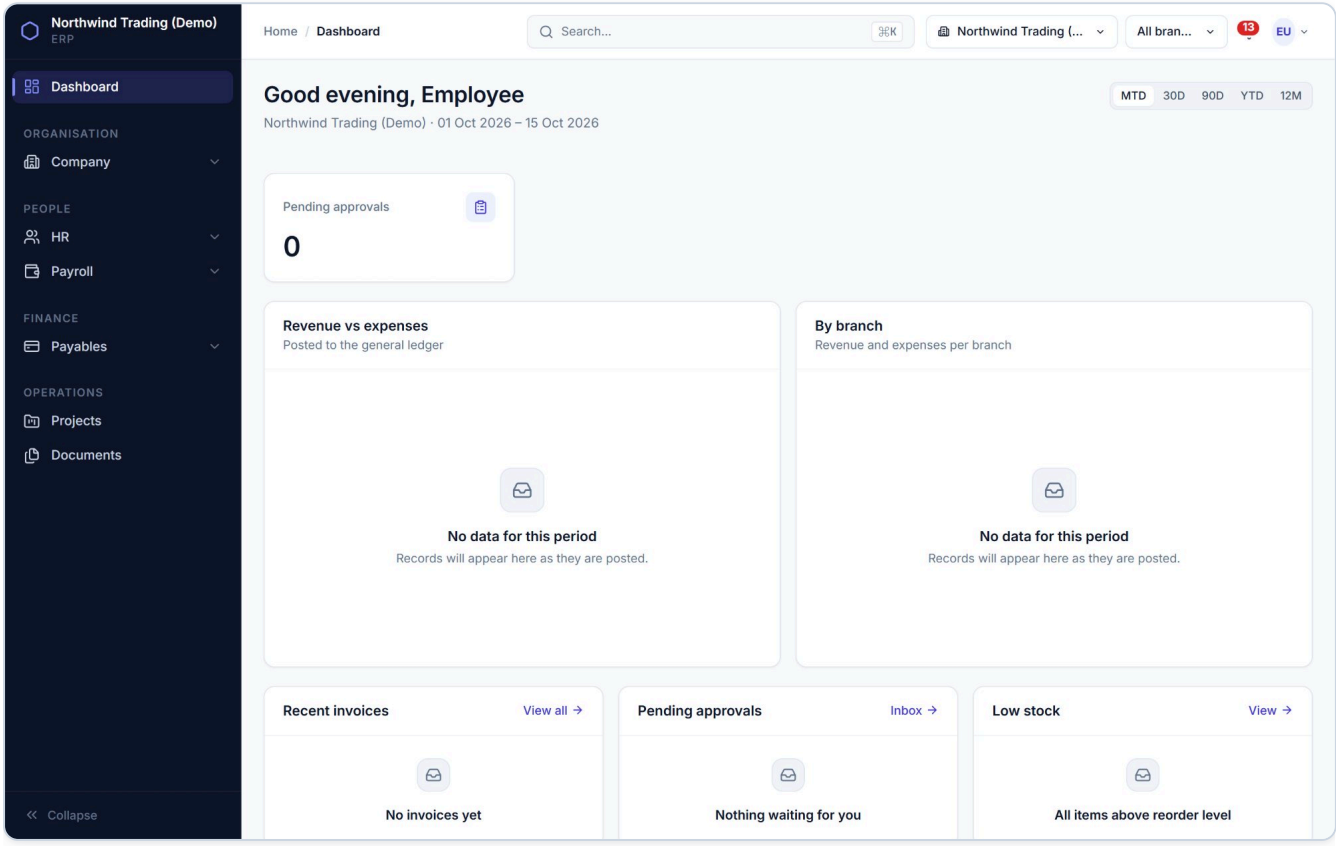

Figure 8.1 — Employee (employee@northwind.test) ka dashboard aur sidebar — sirf Dashboard, Company, HR, Payroll, Payables, Projects, Documents.

- **HR → Attendance** aur **HR → Leave** — apni attendance aur leave.
- **Payroll → My payslips** — apni salary slips.
- **Payables → Expense claims** — apne kharchon ke claims.
- **Projects** — jin projects/tasks mein woh shamil hai, aur **My tasks**.
- **Documents** — company-wide documents (jaise policies).

> **NOTE**
> Agar employee ko self-service screens par "Your account isn't linked to an employee record" nazar aaye, to HR ko kahein: **HR → Employees** → us ka record → **Edit** → **Linked user** mein us ka login select karein.

8.2 Attendance: Check in / Check out

1. Subah kaam shuru karte waqt **HR → Attendance** kholein.
2. Card par aaj ki tareekh, company ka time zone aur live ghari nazar aati hai; Status "Not checked in".
3. **Check in** dabayein. Shift ke mutabiq late hone par late minutes khud calculate hote hain.
4. Kaam khatam karke **Check out** dabayein — worked time record ho jata hai.

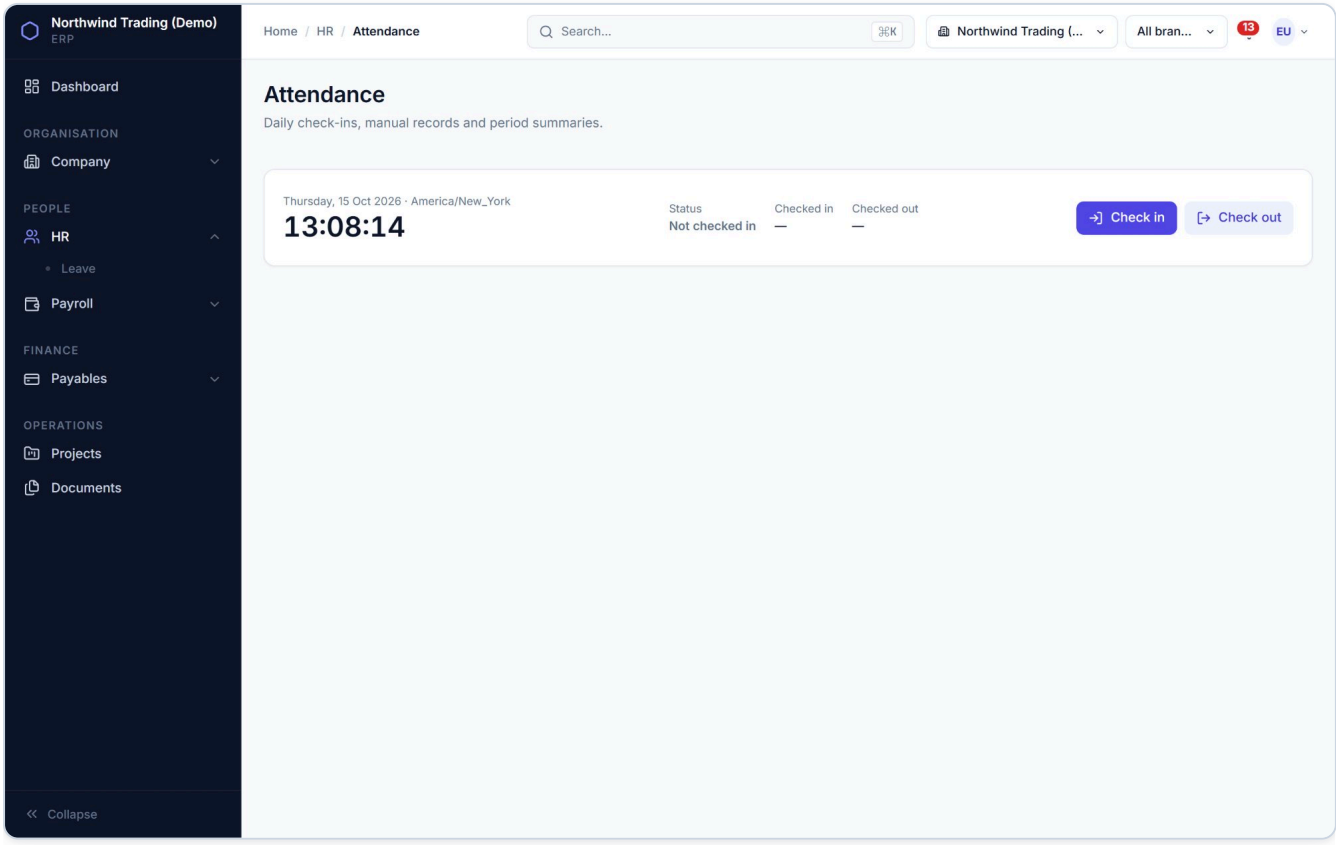

Figure 8.2 — Self check-in card — tareekh, time zone, status, Check in / Check out.

> **TIP**
> Check-in/out bhool jayein to HR ko batayein — HR "Record attendance" se manual record daal sakta hai.

8.3 Leave request karna

1. **HR → Leave** kholein — **My leave** tab mein "My balances" (har leave type ke din baqi) aur "My requests".
2. **Request leave** dabayein.
3. Leave type, start/end date aur reason likhein. Public holidays leave ke dinon ki ginti mein shamil nahi hotin.
4. **Submit request** — request "Pending" ho jati hai aur approver ko notification jati hai.
5. Faisle tak pending request **Cancel** ki ja sakti hai. Approve/reject hone par notification aati hai.

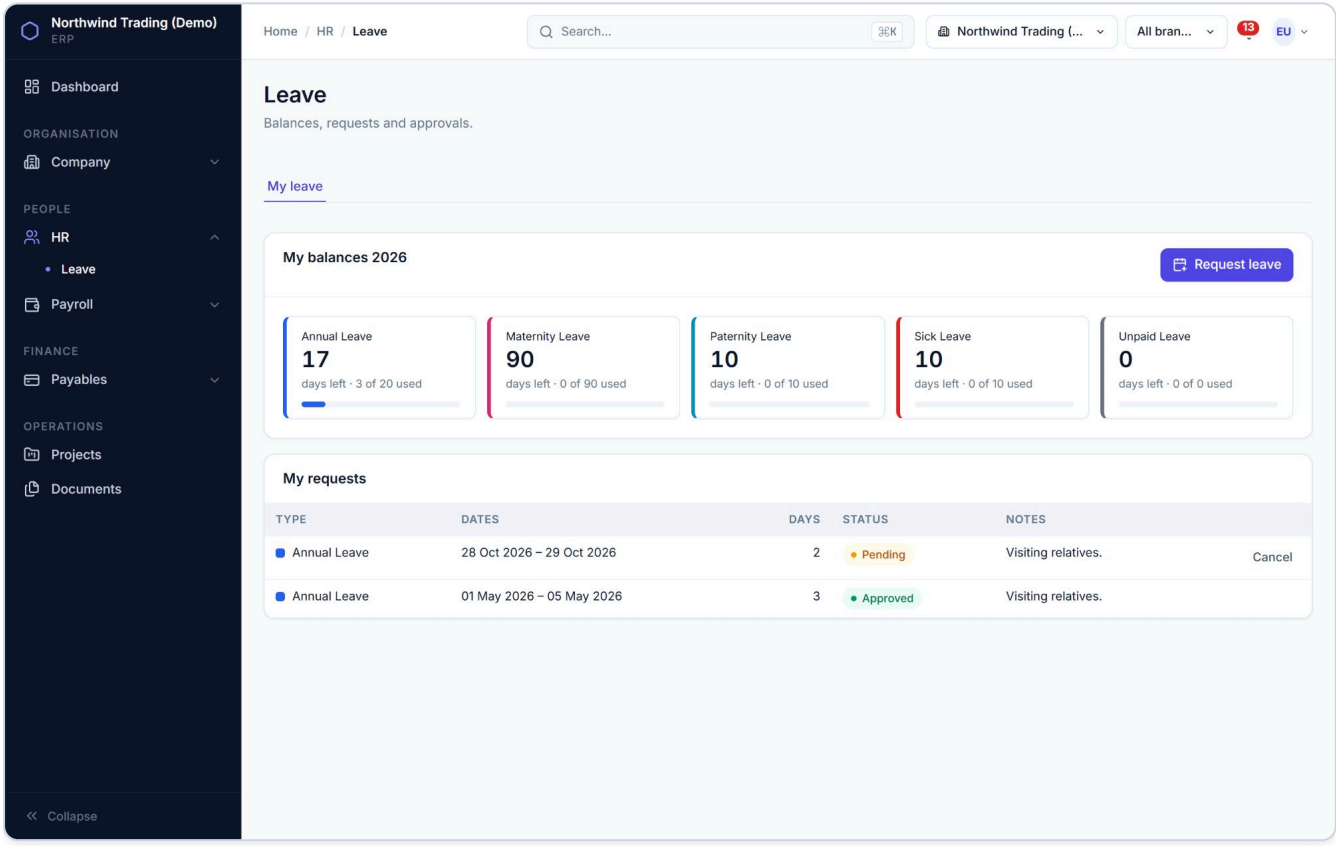

Figure 8.3 — My leave — balances (Annual 17 days left…), Request leave, aur requests ki list (Pending / Approved).

8.4 Payslips dekhna

Payroll → My payslips — har mahine ka period, pay date, gross, deductions, net pay aur status. Payslip tab nazar aati hai jab company us mahine ka payroll **finalize** kar de.

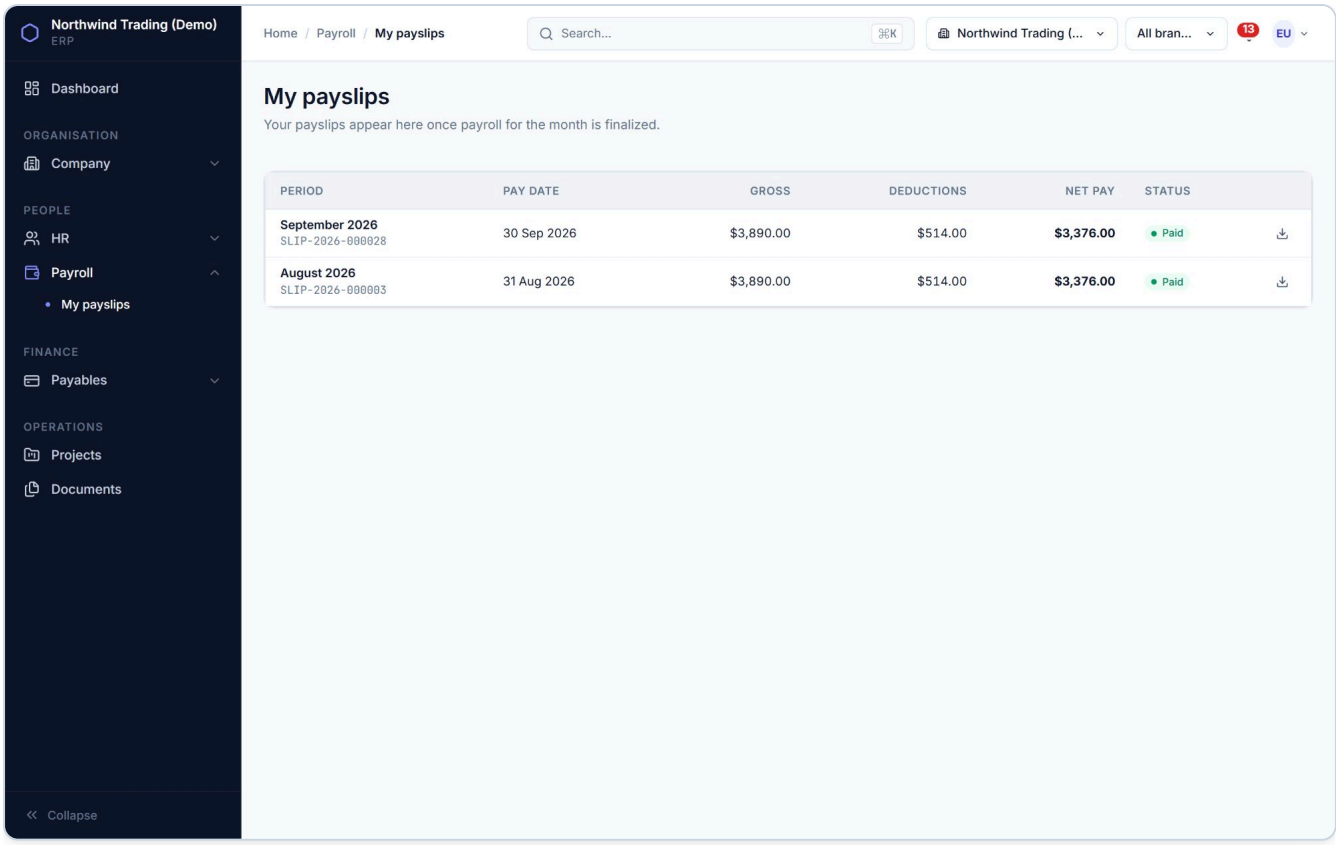

Figure 8.4 — My payslips — August aur September 2026, gross $3,890, net $3,376, Paid.

8.5 Expense claim

1. **Payables → Expense claims** → **New claim**.
2. Har kharche ki line: date, category, description, amount (aur tax).
3. Save karein (draft), phir har line par receipt (PDF ya image, max 5 MB) attach karein.
4. **Submit** — "The claim goes to an approver and can no longer be edited."
5. Reject ho to wajah nazar aati hai; claim edit karke dobara submit karein. Approve hone ke baad paisa payments ke zariye milta hai (status Paid).

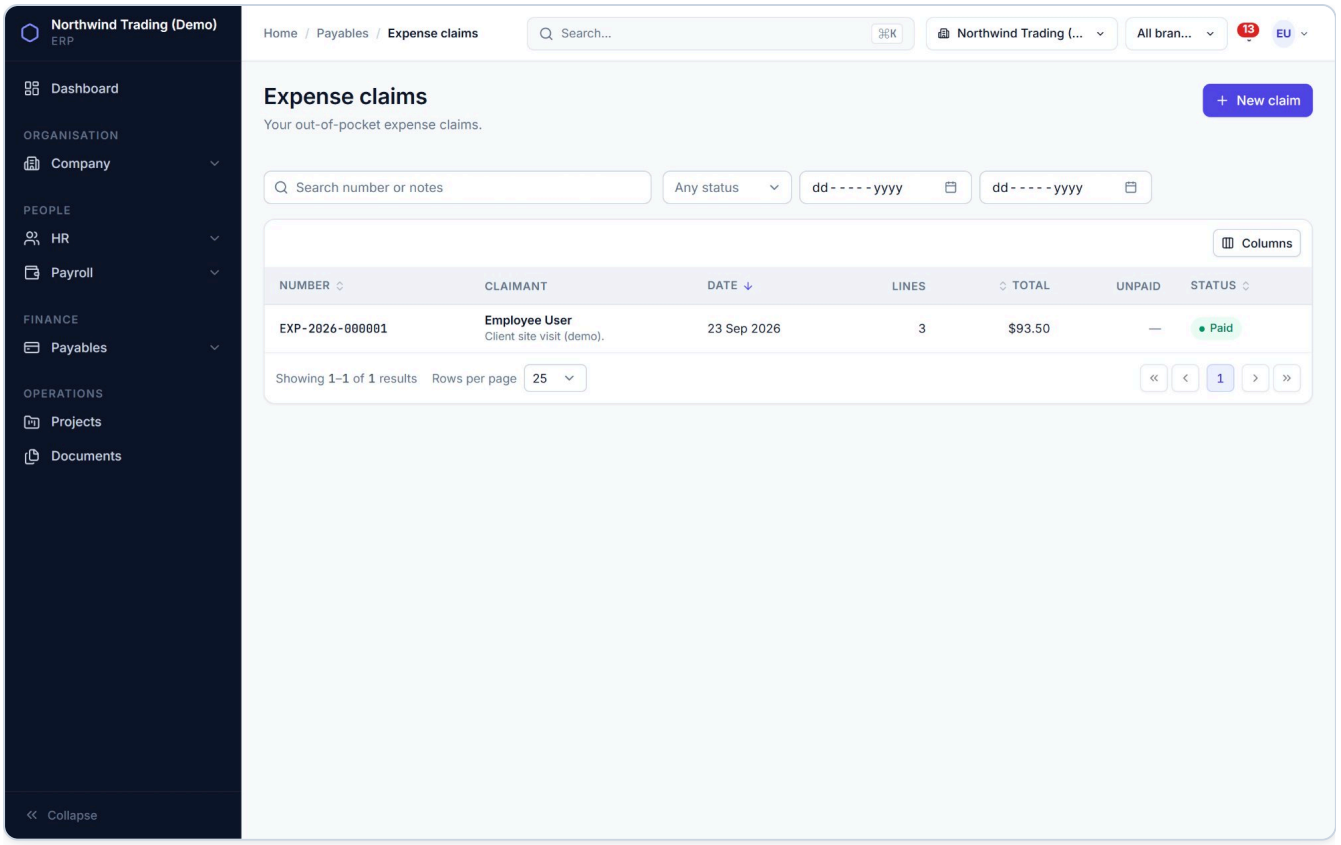

Figure 8.5 — Employee ki apni expense claims — "Your out-of-pocket expense claims."

8.6 Tasks aur time

Projects → **My tasks** mein aap ko assign kiye gaye tasks hain. Task kholein, status badlein (maslan To do → In progress → In review), comment karein, aur **Log time** se ghante darj karein.

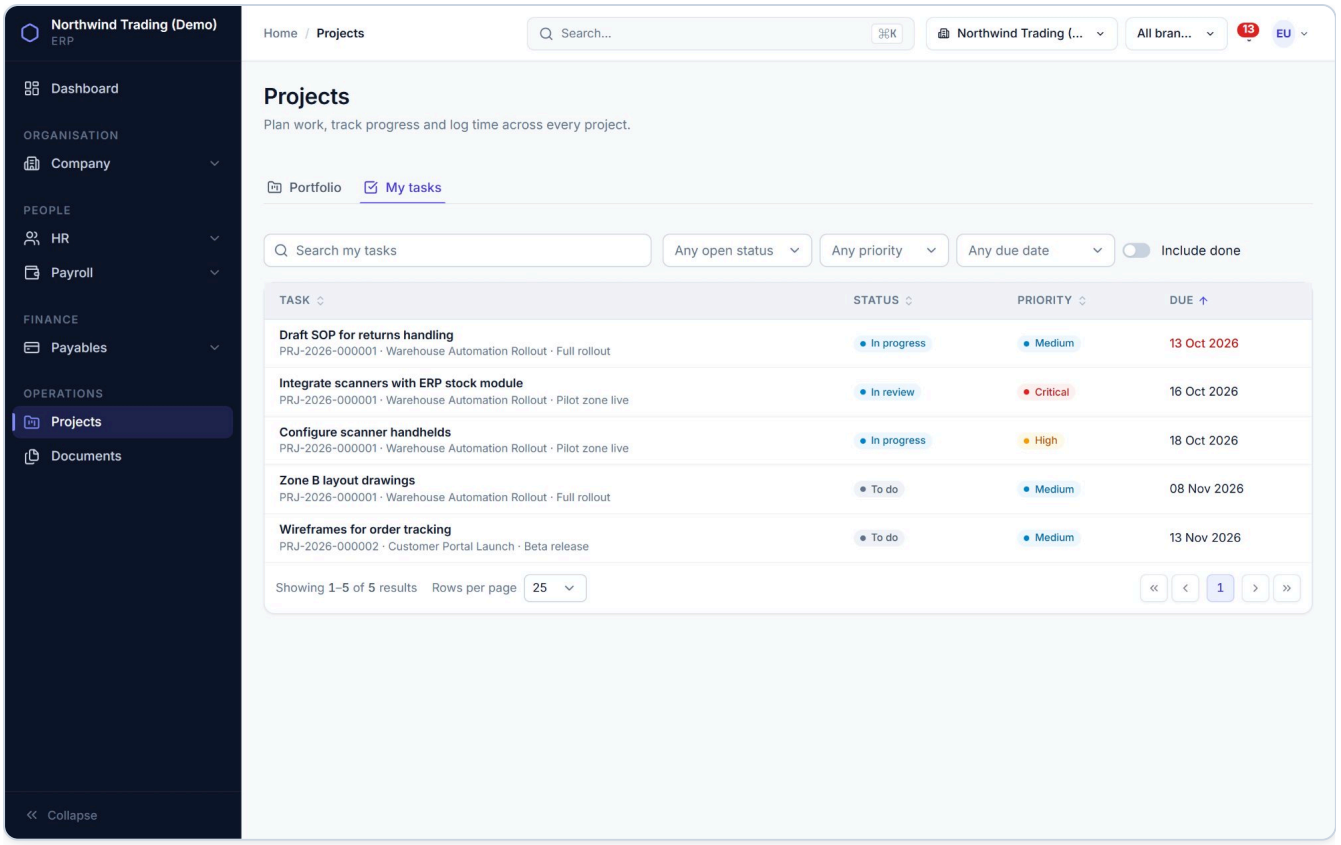

Figure 8.6 — Projects → My tasks — employee ke apne open tasks.

8.7 Jo employee nahi kar sakta

Employee doosron ka data (invoices, salaries, users…) nahi dekh sakta. Agar woh kisi aur page ka URL khud type kare, server inkaar karta hai:

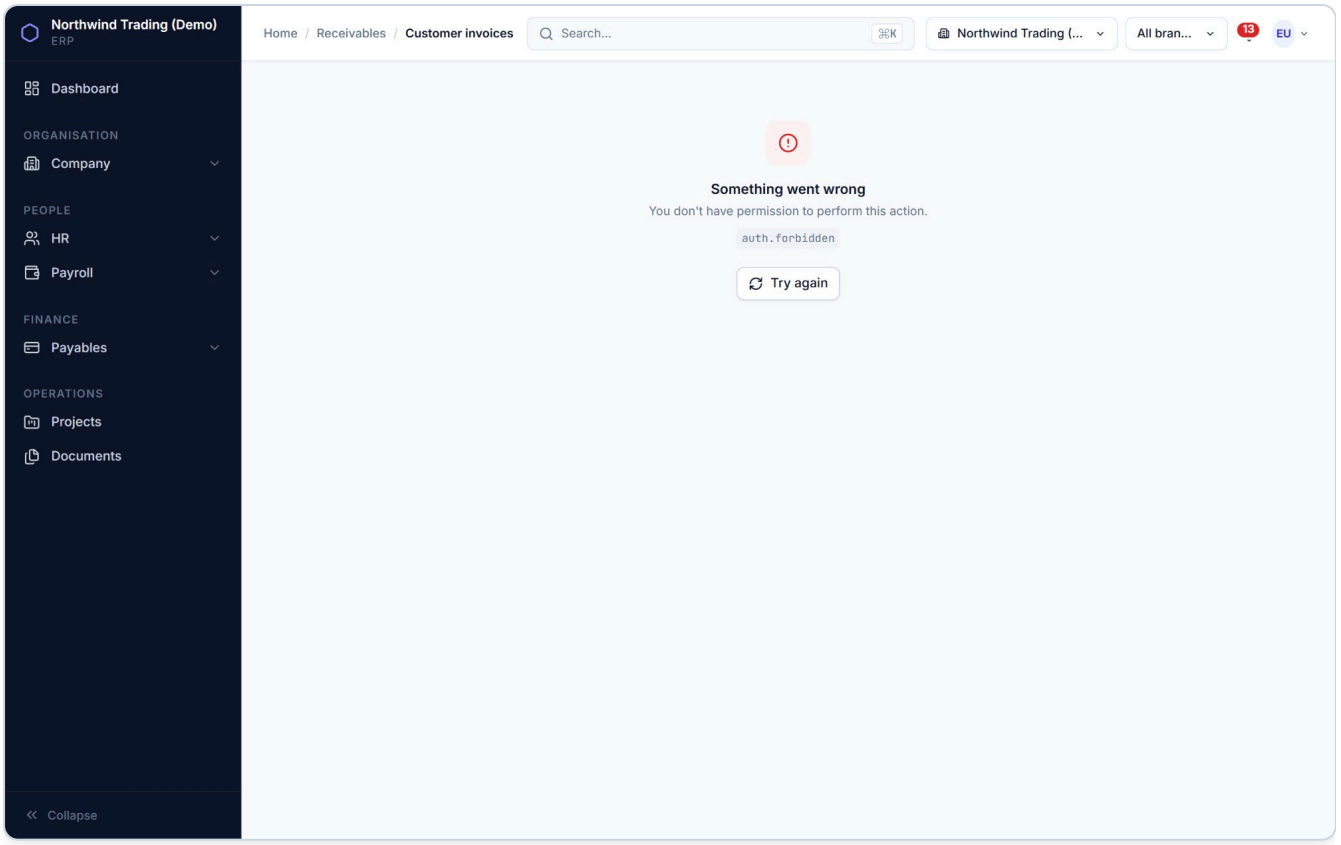

Figure 8.7 — Employee ne direct /finance/invoices/11 khola — "You don't have permission to perform this action."

> ⚠ **DHYAN DEIN**
>
> Apna password kisi ke sath share na karein — har kaam aap ke naam se audit log mein likha jata hai.

CHAPTER 9

Mobile use

Phone ke browser se ERP chalana — koi alag app install karne ki zaroorat nahi.

9.1 Phone par ERP kholna

ERP **responsive** hai: phone ke browser (Chrome, Edge, Safari) mein wohi address kholein jo computer par kholte hain. Screen chhoti hone par layout khud adjust hota hai — sidebar chhup jata hai aur menu button se khulta hai.

1. Phone ke browser mein ERP ka address kholein aur email/password se login karein (2FA on ho to authenticator app ka code).
2. Upar left mein menu button (☰, "Open menu") dabayein — sidebar drawer khulta hai.
3. Module chunein; drawer khud band ho jata hai. Band karne ke liye ✕ ("Close menu") bhi hai.
4. Lambi tables ko left/right swipe karke scroll karein.

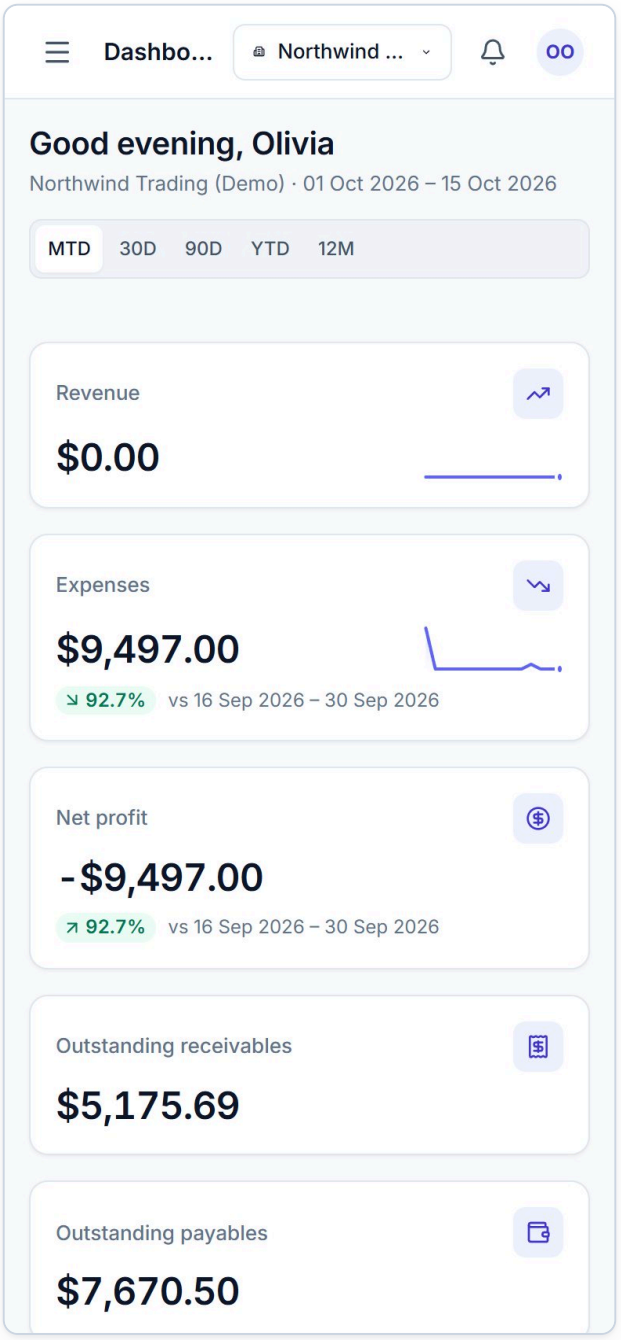

Figure 9.1 — Phone par dashboard — sidebar band, KPI cards aik column mein.

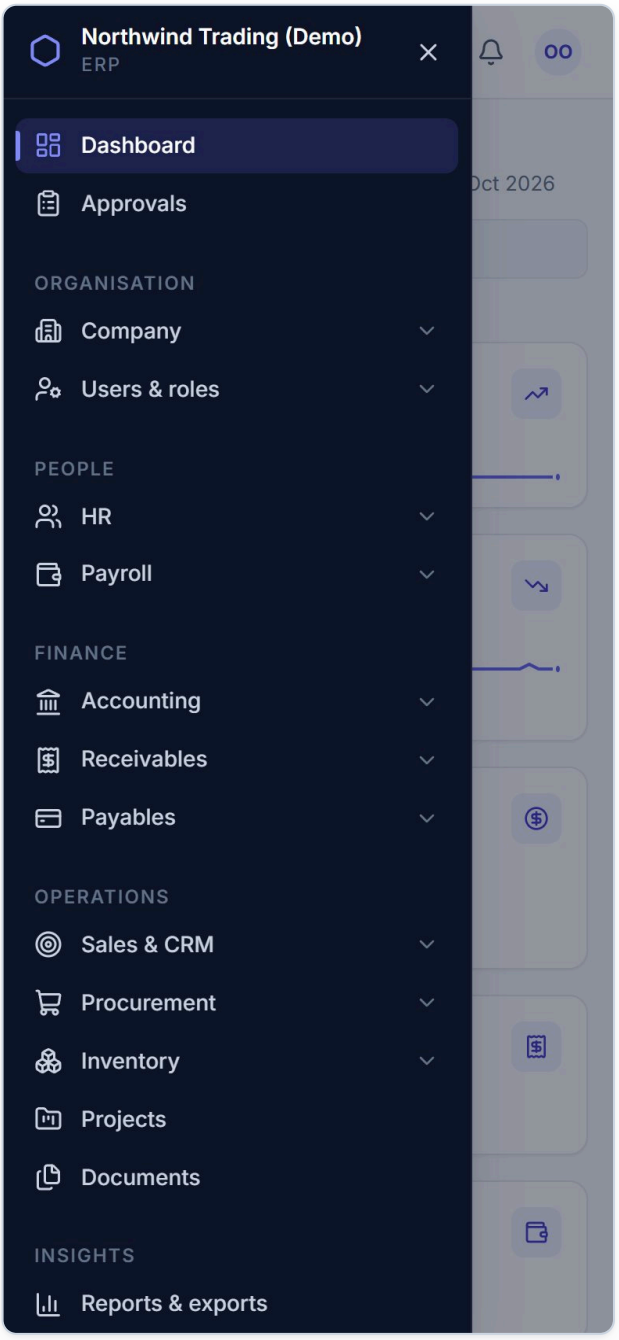

Figure 9.2 — Menu drawer khula — wohi sections aur modules.

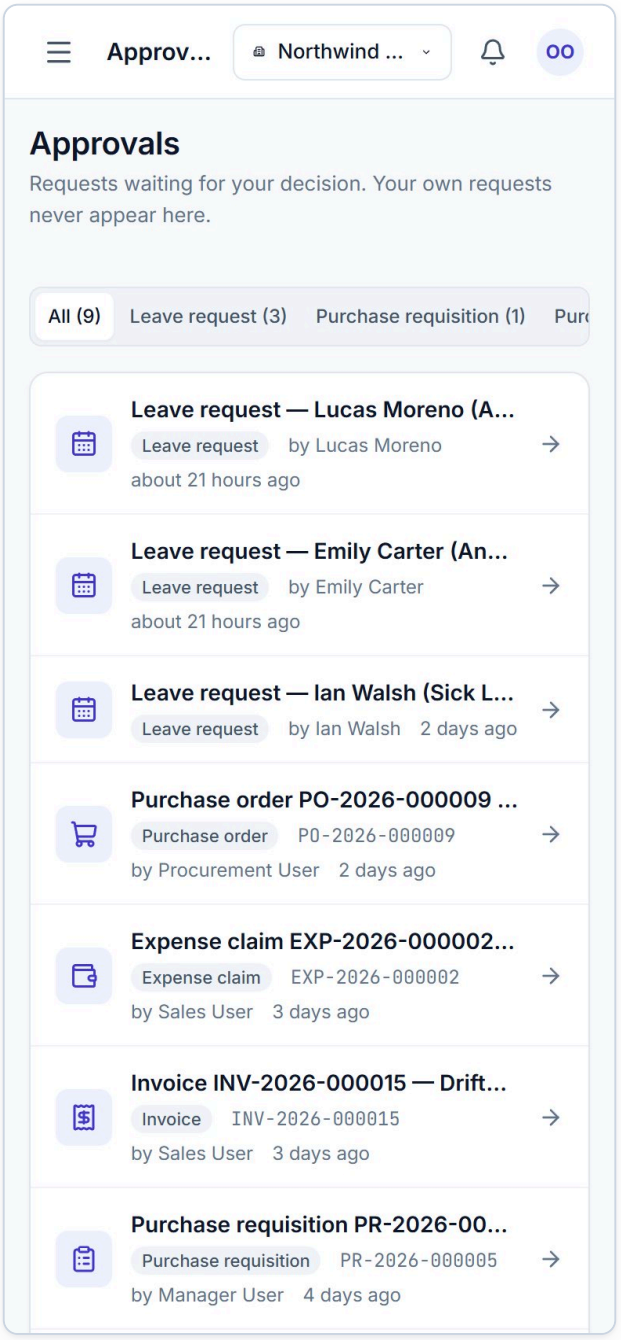

Figure 9.3 — Approvals inbox phone par — chalte phirte approve/reject.

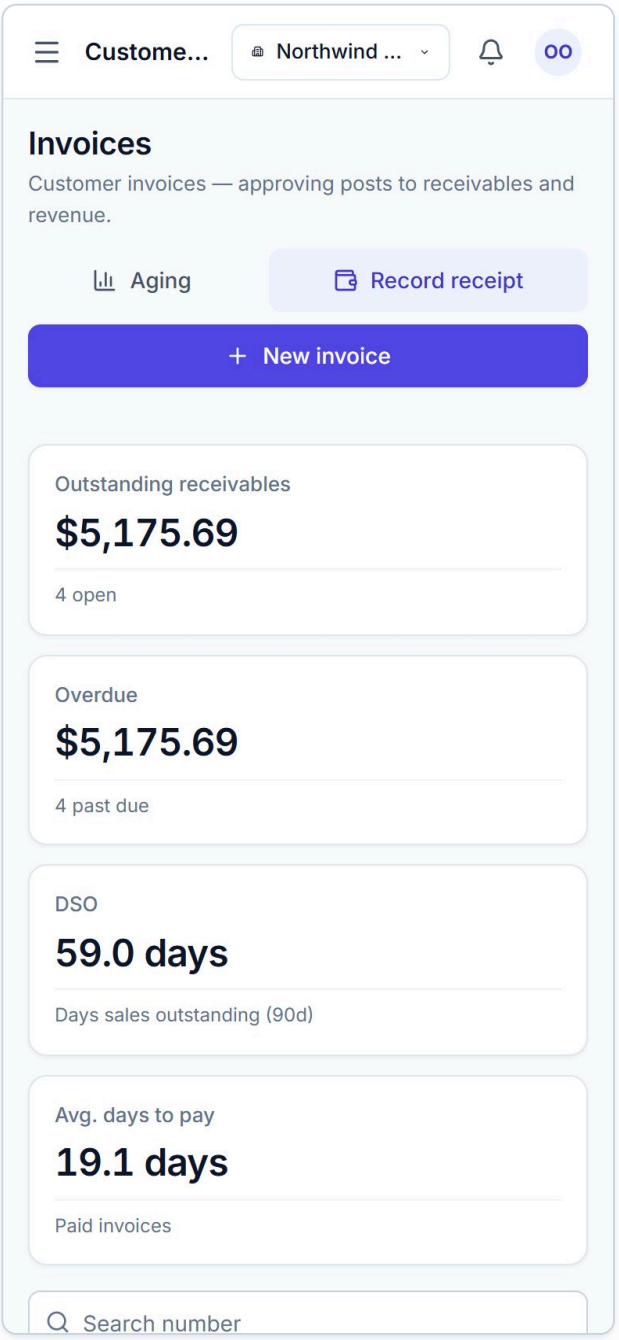

Figure 9.4 — Invoices list phone par.

9.2 Mobile tips

- Roz ke chhote kaam — approvals, leave request, check-in, payslip dekhna — phone par aasaan hain.
- Bari data entry (journal entries, invoices ki kai lines, payroll setup) ke liye computer behtar hai.
- Browser ka "Add to Home screen" option ERP ko app ki tarah icon de deta hai.
- Dark mode raat ko aankhon par kam bojh dalta hai (Account menu → **Dark**).

> **TIP**
>
> Public ya doosre ke phone par login karein to kaam ke baad **Sign out** zaroor karein, aur "Keep me signed in" tick na karein.

CHAPTER 10

Troubleshooting, FAQ & Glossary

Aam sawalat ke jawab aur ERP ki istilahat ka aasaan matlab.

10.1 Login aur account

Q **Password bhool gaya hoon. Kya karoon?**

A Login page par **Forgot password?** dabayein, email likhein aur link ka intezar karein. Security ki wajah se screen hamesha aik jaisa jawab deti hai, chahe email registered ho ya na ho. Link na aaye to spam folder dekhein ya administrator se poochein (development mein link `storage/logs/laravel.log` mein hota hai).

Q **"Too many attempts" aa raha hai.**

A Aik email se aik minute mein 5 se zyada ghalat koshishein hon to system thori der rok deta hai. 1 minute baad dobara try karein.

Q **Phone gum ho gaya, 2FA code nahi aa sakta.**

A 2FA setup ke waqt mile **recovery codes** mein se aik use karein (har code aik dafa). Codes bhi na hon to administrator se rabta karein.

Q **Main aik se zyada companies mein kaam karta hoon — dusri company kaise kholoon?**

A Topbar mein company ke naam (company switcher) par click karke dusri company chunein. Har company mein aap ka role alag ho sakta hai.

Q **Kuch der baad khud logout ho jata hoon.**

A Idle session `SESSION_LIFETIME` (default 120 minute) ke baad khatam hota hai. Password badalne par bhi baqi devices se logout hota hai. Administrator ne deactivate kiya ho to bhi foran logout hota hai.

10.2 Access aur permissions

Q **Mujhe koi menu (jaise Payroll) nazar nahi aa raha.**

A Aap ke role mein us module ki permission nahi. Company administrator se kahein ke aap ko munasib role de (Chapter 5).

Q **Page par "Something went wrong — You don't have permission to perform this action" aata hai.**

A Yeh 403 hai: aap ke paas is page/action ki permission nahi. Agar zaroorat hai to administrator se role update karwayein.

Q **"Page not found" aa raha hai lekin record mojood hai.**

A Ho sakta hai record kisi **dusri company** ka ho (current company check karein), ya aap ki branch restriction se bahar ho, ya link purana ho.

Q **Main apni leave / expense / requisition approve kyun nahi kar sakta?**

A Maker–checker rule: jo request banaye woh khud approve nahi karta. Approvals inbox mein apni requests nazar hi nahi aatin. Payroll run banane wala bhi usay approve nahi kar sakta.

Q **"Your account isn't linked to an employee record".**

A HR **HR → Employees** mein aap ka record khol kar **Edit** → **Linked user** mein aap ka login select kare. Tab check-in, leave aur payslips chalenge.

10.3 Finance, payroll aur stock

Q **Posted journal entry mein ghalti hai — edit kaise karoon?**

A Posted entries edit nahi hotin. Entry khol kar **Reverse** karein, phir sahi entry nayi banayein.

Q **Invoice void nahi ho rahi: "has payments allocated".**

A Pehle us invoice ki receipt (payment) void karein, phir invoice void karein.

Q **Journal entry save nahi ho rahi.**

A Debits aur credits barabar hone chahiyein (**Balanced** badge), har line par account ho, aur aik line mein debit aur credit dono na hon. Date wala financial period **open** hona chahiye.

Q **Payslip mein tax zero aa raha hai, "No tax table".**

A **Payroll → Salary structures** → **Tax tables** mein us period ke liye active tax table banayein, phir run **Recalculate** karein (finalize se pehle).

Q **Payroll run finalize ho gaya, ghalti nikli.**

A Finalized run badla nahi ja sakta. Accountant ko batayein — correction journal entry ya agle mahine adjustment se hogi.

Q **Stock ki quantity ghalat hai.**

A **Inventory → Movements** mein product ki history dekhein. Physical count ke baad farq ho to **Inventory → Adjustments** se adjustment bana kar post karein — wajah (Stock count, Damage...) zaroor chunein.

Q **Export "queued" hai, file nahi aayi.**

A Bari exports background mein banti hain; tayyar hone par notification aati hai aur **My exports** mein milti hai. Agar der tak na aaye to IT se queue worker (`php artisan queue:work`) check karwayein.

Q **Map par tiles nazar nahi aa rahe.**

A Map OpenStreetMap ke tiles internet se laata hai — internet connection check karein. Map par sirf woh locations aati hain jin ke coordinates saved hon.

10.4 IT / server ke masail

Table 10.1 — IT checklist

Symptom	Pehla qadam
Login page hi nahi khulta	Frontend chal raha hai? (`npm run dev` window / `http://localhost:3000`)
Har page par network error	API chal rahi hai? `http://localhost:8000/api/v1/health` kholein
Database errors	`php artisan db:check` — Hint line parhein (Chapter 2.8)
Emails nahi ja rahin	`MAIL_*` settings; dev mein `MAIL_MAILER=log` hai to emails log file mein hain
Exports / notifications atak gaye	Queue worker aur scheduler chal rahe hain? `php artisan queue:failed`

10.5 Glossary (istilahat)

Table 10.2 — English → Hinglish

English term	Aasaan matlab
Tenant / Company	ERP mein aik alag company jis ka data doosri companies se juda hai
Branch	Company ka daftar / site / shakh
Role	Permissions ka majmooa jo user ko diya jata hai
Permission	Kisi aik kaam ki ijazat (jaise invoice approve karna)
Least privilege	Sirf utni access jitni kaam ke liye zaroori
Maker–checker	Banane wala aur approve karne wala alag log
2FA (Two-factor authentication)	Password ke sath phone app ka code bhi
Audit log	Kis ne kab kya kiya — na mitne wala record
General ledger (GL)	Company ki tamam accounting entries ka markazi register
Chart of accounts	Tamam ledger accounts ki list (cash, bank, sales, rent…)
Journal entry	Aik accounting entry jis mein debit = credit
Debit / Credit	Double-entry ke do pehlu; har entry mein dono barabar
Post / Posting	Entry ko ledger mein pakka darj karna
Reverse / Reversing entry	Ghalat posted entry ko ulti entry se khatam karna
Void	Document ko mansookh karna (posting reverse ho jati hai)
Financial period	Accounting ka mahina; close hone par us mein posting band
Receivables (AR)	Customers se lena baqi
Payables (AP)	Suppliers / employees ko dena baqi
Aging	Baqaya raqam ko kitne din purani hai, us hisaab se buckets
DSO	Days sales outstanding — customers ausatan kitne din mein paisa dete hain
Invoice	Customer ko bheja gaya bill

English term	Aasaan matlab
Bill	Supplier se aaya hua bill
Receipt / Payment	Paisa aana / paisa jana
Expense claim	Employee ka apni jeb se kiya kharcha wapas lene ki darkhwast
Payroll run	Aik mahine ki tamam salaries ka hisaab
Payslip	Employee ki tankhwah ki parchi
Pay component	Salary ka hissa: allowance, deduction, employer contribution
Finalize	Payroll ko pakka karke ledger mein post karna (wapas nahi hota)
Lead / Opportunity	Mumkina customer / mumkina deal
Pipeline	Deals ke marahil ka board (New se Won/Lost tak)
Quotation	Customer ko qeemat ki pesh-kash
Sales order	Customer ka confirm order
Requisition	Andar ki darkhwast ke yeh cheez khareedni hai
Purchase order (PO)	Supplier ko diya gaya khareed ka order
Goods receipt (GRN)	Supplier se maal wusool karne ka record
Stock movement	Stock ki har aamad/raft ka na badalne wala record
Transfer	Aik warehouse se doosre mein stock bhejna
Adjustment	Ginti ya nuqsan ke baad stock theek karna
Reorder level	Woh miqdar jis se kam hone par dobara order karna chahiye
Weighted average cost	Stock ki ausat qeemat jo har khareed ke sath update hoti hai
Milestone	Project ka aham marhala
Retention	Data kitni der rakha jayega
Export	Report ko CSV / Excel / PDF file mein nikalna

APPENDIX A

Appendix — Commands, demo accounts & permission catalogue

A.1 Command cheat-sheet

Table A.1 — Commands

Kaam	Command	Kahan se
MySQL data folder banana (aik dafa)	`.\database\local\init.ps1`	Project root
MySQL start (background)	`.\database\local\start.ps1 -Background`	Project root
MySQL stop	`.\database\local\stop.ps1`	Project root
App database + user banana	`.\database\local\create-app-db.ps1`	Project root
Sab kuch start (DB + API + frontend)	`.\start-dev.ps1`	Project root
PHP packages	`composer install`	backend
App key	`php artisan key:generate`	backend
DB connection check	`php artisan db:check`	backend
DB self-test	`php artisan db:selftest`	backend
Schema + permissions	`php artisan migrate --seed`	backend
Production migration	`php artisan migrate --force`	backend
Permission catalogue (prod)	`php artisan db:seed --force`	backend
Demo data (sirf dev)	`php artisan db:seed --class=DemoSeeder`	backend
Pehla platform owner	`php artisan erp:create-admin`	backend
Pehla owner (production)	`ERP_ALLOW_ADMIN_BOOTSTRAP=true php artisan erp:create-admin --force`	backend
API server (dev)	`php artisan serve`	backend
Queue worker	`php artisan queue:work --tries=3`	backend
Scheduler (dev)	`php artisan schedule:work`	backend
Failed jobs	`php artisan queue:failed`	backend
Retention cleanup preview	`php artisan erp:prune --dry-run`	backend
Low-stock check (manual)	`php artisan inventory:check-reorder`	backend
Config cache (prod)	`php artisan config:cache route:cache event:cache`	backend
Cache saaf karna	`php artisan optimize:clear`	backend
Backend tests	`php artisan test`	backend
Frontend packages	`npm install`	frontend
Frontend dev server	`npm run dev`	frontend
Frontend production build	`npm run build`	frontend

Kaam	Command	Kahan se
Is guide ke screenshots	`npm run screenshots`	docs\build
Is guide ka PDF	`npm run guide`	docs\build

> ⚠ **DHYAN DEIN**
> Production par **kabhi nahi**: `migrate:fresh`, `migrate:reset`, `migrate:rollback`, `db:wipe`, `DemoSeeder`.

A.2 Demo accounts

DEVELOPMENT ONLY

Yeh accounts sirf demo data (`DemoSeeder`) ke sath bante hain. Is installation mein in sab ka password `password` hai (`DEMO_PASSWORD` se set). Production par yeh accounts **hargiz mojood nahi hone chahiyein**.

Table A.2 — Demo logins — password: password

Login	Role	Company
`platform@erp.test`	Platform owner (companies manage karta hai; company data nahi dekhta)	—
`owner@northwind.test`	Company Owner — sab kuch	Northwind Trading (Demo) · USD
`admin@northwind.test`	Company Administrator — sab kuch siwaye payroll processing	Northwind Trading (Demo) · USD
`hr@northwind.test`	HR Manager	Northwind Trading (Demo) · USD
`accountant@northwind.test`	Accountant	Northwind Trading (Demo) · USD
`sales@northwind.test`	Sales Manager	Northwind Trading (Demo) · USD
`inventory@northwind.test`	Inventory Manager	Northwind Trading (Demo) · USD
`procurement@northwind.test`	Procurement Officer	Northwind Trading (Demo) · USD
`manager@northwind.test`	Department Manager	Northwind Trading (Demo) · USD
`employee@northwind.test`	Employee (self-service)	Northwind Trading (Demo) · USD
`auditor@northwind.test`	Read-Only Auditor	Northwind Trading (Demo) · USD
`owner@indus.test`	Company Owner — sab kuch	Indus Manufacturing (Demo) · PKR

Login	Role	Company
`admin@indus.test`	Company Administrator — sab kuch siwaye payroll processing	Indus Manufacturing (Demo) · PKR
`hr@indus.test`	HR Manager	Indus Manufacturing (Demo) · PKR
`accountant@indus.test`	Accountant	Indus Manufacturing (Demo) · PKR
`sales@indus.test`	Sales Manager	Indus Manufacturing (Demo) · PKR
`inventory@indus.test`	Inventory Manager	Indus Manufacturing (Demo) · PKR
`procurement@indus.test`	Procurement Officer	Indus Manufacturing (Demo) · PKR
`manager@indus.test`	Department Manager	Indus Manufacturing (Demo) · PKR
`employee@indus.test`	Employee (self-service)	Indus Manufacturing (Demo) · PKR
`auditor@indus.test`	Read-Only Auditor	Indus Manufacturing (Demo) · PKR

A.3 Permission catalogue

Roles editor mein har checkbox ke neeche yeh codes nazar aate hain. Platform permissions sirf platform staff ke liye hain aur company roles mein nahi aatin.

Table A.3 — Tamam permissions

Module	Permission code	Matlab
dashboard	`dashboard.view`	View executive dashboard
company	`company.view`	View company profile
company	`company.settings.manage`	Manage company settings & branding
company	`branches.view`	View branches
company	`branches.manage`	Create/update/delete branches
company	`departments.view`	View departments
company	`departments.manage`	Manage departments
access	`users.view`	View users
access	`users.manage`	Invite, update, deactivate users
access	`roles.view`	View roles & permissions
access	`roles.manage`	Manage roles & permissions
hr	`employees.view`	View employees
hr	`employees.create`	Create employees

Module	Permission code	Matlab
hr	`employees.update`	Update employees
hr	`employees.delete`	Delete employees
hr	`employees.documents.view`	View employee documents
hr	`attendance.view`	View attendance
hr	`attendance.manage`	Record/edit attendance
hr	`leave.view`	View leave requests
hr	`leave.request`	Submit own leave requests
hr	`leave.approve`	Approve/reject leave
hr	`leave.manage`	Manage leave types & balances
hr	`shifts.manage`	Manage shifts & holidays
hr	`performance.view`	View performance reviews
hr	`performance.manage`	Manage performance reviews
payroll	`payroll.view`	View payroll
payroll	`payroll.manage`	Manage salary structures
payroll	`payroll.process`	Run & approve payroll
payroll	`payslips.view.own`	View own payslips
finance	`accounts.view`	View chart of accounts
finance	`accounts.manage`	Manage chart of accounts
finance	`journals.view`	View journal entries
finance	`journals.create`	Create journal entries
finance	`journals.post`	Post journal entries
finance	`invoices.view`	View invoices
finance	`invoices.create`	Create invoices
finance	`invoices.update`	Update draft invoices
finance	`invoices.approve`	Approve invoices
finance	`invoices.delete`	Delete draft invoices
finance	`bills.view`	View supplier bills
finance	`bills.create`	Create supplier bills
finance	`bills.approve`	Approve supplier bills
finance	`payments.view`	View payments & receipts
finance	`payments.create`	Record payments & receipts
finance	`expenses.view`	View expense claims
finance	`expenses.submit`	Submit expense claims
finance	`expenses.approve`	Approve expense claims
finance	`bank.manage`	Manage cash & bank accounts

Module	Permission code	Matlab
finance	`tax.manage`	Manage tax configuration
finance	`periods.manage`	Manage financial periods
finance	`reports.financial.view`	View financial reports
sales	`customers.view`	View customers
sales	`customers.manage`	Manage customers
sales	`leads.view`	View leads & opportunities
sales	`leads.manage`	Manage leads & opportunities
sales	`quotations.view`	View quotations
sales	`quotations.manage`	Manage quotations
sales	`sales_orders.view`	View sales orders
sales	`sales_orders.manage`	Manage sales orders
sales	`sales_orders.approve`	Approve sales orders
procurement	`suppliers.view`	View suppliers
procurement	`suppliers.manage`	Manage suppliers
procurement	`requisitions.view`	View purchase requisitions
procurement	`requisitions.create`	Create purchase requisitions
procurement	`requisitions.approve`	Approve purchase requisitions
procurement	`purchase_orders.view`	View purchase orders
procurement	`purchase_orders.manage`	Manage purchase orders
procurement	`purchase_orders.approve`	Approve purchase orders
procurement	`goods_receipts.manage`	Record goods received
inventory	`products.view`	View products
inventory	`products.manage`	Manage products & categories
inventory	`warehouses.view`	View warehouses
inventory	`warehouses.manage`	Manage warehouses & locations
inventory	`inventory.view`	View stock levels & movements
inventory	`inventory.receive`	Receive stock
inventory	`inventory.issue`	Issue stock
inventory	`inventory.transfer`	Transfer stock
inventory	`inventory.adjust`	Adjust stock
projects	`projects.view`	View projects
projects	`projects.manage`	Manage projects
projects	`tasks.view`	View tasks
projects	`tasks.manage`	Manage tasks
projects	`time_entries.manage`	Log time

Module	Permission code	Matlab
documents	`documents.view`	View documents
documents	`documents.manage`	Upload/manage documents
system	`approvals.view`	View approval inbox
system	`reports.view`	View reports
system	`reports.export`	Export reports
system	`audit.view`	View audit logs
platform	`platform.companies.manage`	Manage all companies
platform	`platform.users.manage`	Manage platform users
platform	`platform.system.view`	View system health & database overview